

CRF Complete Edition v17.17 — Part M

ZC94–106: The Mass Sector, End to End

Mass-hierarchy registry · running bridge · tower obstruction and center law · spectral closure · derivation boundary · accumulation integral · operation–group law · closure discriminant · first numerical simulator · the first-rung effect, excluded and split

The charged-fermion spectrum reduced to one running and one grading, parameter-free down to two constants — then the residual located, bounded, and named

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If you are just arriving — Part M in plain language

CRF tries to derive the structure of physics from a single starting idea called Distinction (δ). This Part is about *mass*: why the charged fermions (electron, muon, tau; the up- and down-type quarks) have the masses they do, in the pattern they do. The arc does three things in turn. **First** (ZC94–98) it shows the whole charged-fermion spectrum is *one running and one grading*: a single rate that “runs” with scale, plus a three-family splitting by a fixed step of $\pm d_s/2$, together reproduce the mass ladder. **Second** (ZC99–102) it pins down how far this is parameter-free — the spectrum reduces to *two* constants (κ_R and L_0), and the structure that fixes them is traced to which operation “closes” (mass accumulates by addition, selecting $\mathbb{Z}_{15}$; force couples by a Clifford product, selecting a different count). **Third** (ZC103–106) it builds the chain’s first numerical simulator, confirms the running law with zero free parameters, then isolates the one place it disagrees with data — the “first rung” (generation 1→2) — excludes seven explanations for that residual, and splits it into an old piece (inherited from ZC16) and a genuinely new one.

Minimum vocabulary: δ (*Distinction*) — the one starting idea; *running* — a quantity that changes smoothly with scale, here the log-linear drift of a mass ratio; *tower / grading* — the three-generation ladder and the fixed $\pm d_s/2$ step that separates families; κ_R, L_0 — the two constants the spectrum reduces to; *first-rung effect* — the residual deviation at the lowest generation step, where the smooth law meets a boundary; *gap / GAP-...* — an open question the chain has not yet closed. Full symbol list: Master Notation Key (front matter).

These results are pieces of a map under construction, not a claim of final truth: each carries the conditions under which it would fail (see front matter, “Map, not reality”).

Part LX

The Mass Sector, End to End: Running, Grading, Closure, and the First-Rung Effect (ZC94–106, v17.17)

0.1 ZC94 — The Mass-Hierarchy Prediction Registry

Abstract

CRF already derives the count structure of the Standard Model fermions exactly: fifteen Weyl fermions per generation is the unique minimal content (FIND-AM03), and the three observed generations are three copies of it. CRF also derives a parameter-free *ratio* between generation masses — $m_H \propto (15/|H|)^5$, EQ-MC-03 — in which every quantity descends from the single primitive δ . What CRF does *not* yet have is a match: the ratio predicts 243 and 3125 for the second and third charged-lepton generations where experiment reads 207 and 3477, a +17% and –10% residual (GAP-MC-01).

This paper opens Part M to confront that residual as a programme, exactly as ZC64 opened Part I’s prediction programme. It consolidates the mass sector into a single falsifiability-oriented registry, classifies its predictions by tier, and names the one open item that would close the gap.

The central observation is that the residual is not noise. It is *signed*: the second generation is over-predicted, the third under-predicted, and the sign turns over monotonically with the compression ratio. A static ratio cannot produce a sign flip across generations; a scale-dependent correction that grows with compression can. This is the same situation the electroweak constants were in before ZC70 — exact R_0 -layer values that became data-matching only once carried across a *running bridge* from the resolution-current layer to the maximum-resolution scale. That bridge, for the electroweak sector, was first asserted and then derived (GAP-ZC64-01 $\rightarrow$ COR-ZC70-01).

FIND-ZC94-01 records that the GAP-MC-01 residual is signed and bounded — a fingerprint of a missing running correction. **FIND-ZC94-02** reads the two mass-sector scars as one signpost: the falsified logarithmic *map* (PRED-ZC15-01) and the surviving parameter-free constraint-cost ratio (EQ-MC-03) together say the spectrum is not a closed-form map but a constraint-cost form correct up to scale. **CONJ-ZC94-01** conjectures that the residual closes under a per-sector mass running bridge built from the ZC70 machinery, and attaches a falsification signature. **GAP-ZC94-01** registers that bridge, $f_{\text{mass}}(R_0 \rightarrow R_{\text{max}})$, as the primary frontier of Part M. Nothing here is solved; it is framed, registered, and made falsifiable.

0.2 Background: What the Mass Sector Already Has

If you are just arriving — Part M in plain language

Why do the electron, muon, and tau — three copies of the same particle — weigh so wildly differently? CRF answers the *why* already: existing under constraint costs possibility, and that cost *is* mass (so the constraint-free ground state is massless). It also answers the *rough how much*: a clean, no-knobs formula that says each heavier generation is the compression ratio raised to the fifth power. That formula lands within ten to seventeen percent of the measured masses — impressive for zero fitting, but not exact. This paper does not try to fix the formula. It does something more disciplined: it collects every mass-sector claim into one place, sorts them by how they can be tested, and identifies the single missing piece — a “running bridge” that carries the formula’s static value up to the scale where masses are actually measured. The same kind of bridge already turned other CRF near-misses into exact hits. Part M is the programme to build it for mass; ZC94 is its registry and its falsification contract.

Numerical Summary — the live mass-sector frontier

Generation	CRF ($15/ H $) ⁵	Observed	Residual	Status
Gen 1 ($ H =15$)	$1^5 = 1$	1 (base)	—	exact (baseline)
Gen 2 ($ H =5$)	$3^5 = 243$	207	+17%	GAP-MC-01
Gen 3 ($ H =3$)	$5^5 = 3125$	3477	−10%	GAP-MC-01

The residual is *signed* (over, then under) and turns over with the compression ratio — the structural fingerprint this paper registers (FIND-ZC94-01). All inputs (15 , $|H|$, $n_m=5$) are fixed by the δ -chain; there is no free parameter.

0.2.1 Three things are already settled

The mass sector is not a blank page. Three results stand as pillars, and Part M builds on them rather than reopening them.

First, the *count*. THM-GEN-01 (ZC15) fixes the admissible generation subgroups $|H| \in \{3, 5, 15\}$ from the subgroup lattice of $\mathbb{Z}_{15}$, and FIND-AM03 (ZC09) establishes that fifteen Weyl fermions is the unique minimal content of one generation, with the three observed generations being three copies. There is no freedom here and no open question; the number three is derived.

Second, the *origin*. COR-MC-03 (ZC16) proves, as a theorem rather than a metaphor, that conditioned stability necessitates mass while the unconditioned ground state is massless: mass *is* the accumulated constraint cost of existing under S_{dep} . The proposition that “the price paid for the existence of reality is the loss of possibility” — first stated in natural language and then formalised in ZC16 — is the conceptual ground of the whole sector. Part M does not revisit why mass exists; it asks only how much.

Third, the *ratio*. THM-MC-01 (ZC16, EQ-MC-03) gives the parameter-free generation

mass ratio

$$m_H \propto \left(\frac{15}{|H|} \right)^{n_m} = \left(\frac{15}{|H|} \right)^5,$$

where the exponent $n_m = 5$ is the matter-mode count and every other quantity descends from the δ -chain. This is the formula whose residual defines the frontier.

0.2.2 And one thing is open

The ratio is close but not exact: +17% at the second generation, −10% at the third (GAP-MC-01). For a derivation with no adjustable parameter a residual of this size is neither a success to be overstated nor a failure to be hidden; it is a precise, documented gap. The question Part M exists to answer is whether that gap is missing physics or a missing *bridge*.

0.3 DEF-ZC94-01: Mass-Sector Tier Classification

ZC64 sorted CRF’s predictions into four tiers by how they can be tested. The mass sector needs the same discipline, specialised to its own structure.

DEF-ZC94-01: Mass-Sector Prediction Tiers (M1–M4)

A mass-sector prediction is classified by what is required to confront it with experiment:

- **Tier M1 — structurally exact, directly checked.** The count structure: $|H| \in \{3, 5, 15\}$ (THM-GEN-01), 15 Weyl fermions / generation (FIND-AM03), three generations. No bridge needed; matches observation exactly.
- **Tier M2 — exact in principle, simulation-checkable.** The constraint-cost identification (DEF-MC-03): rest mass = e^{M_H} . Structurally exact; its numerical content is the R_0 -layer ratio.
- **Tier M3 — requires the mass running bridge.** The *values* of the generation mass ratios against data (GAP-MC-01). The R_0 -layer formula gives 243 and 3125; matching 207 and 3477 requires carrying these across $f_{\text{mass}}(R_0 \rightarrow R_{\text{max}})$. This is the tier that defines Part M.
- **Tier M4 — cosmological / long-range.** Any running of mass parameters with cosmic scale, by analogy with the $G(z)$ Tier-4 prediction of Part I. Registered for completeness; not the focus of ZC94.

The reclassification is the point: **GAP-MC-01 moves from “an open numerical gap” to “a Tier-M3 item awaiting a specific, namable bridge.”** That is the same move ZC64 made for the electroweak constants, which were Tier-3 until ZC70 derived their bridge and promoted them to Tier-1.

0.4 FIND-ZC94-01: The Residual Is Signed

The decisive feature of GAP-MC-01 is one a reader can check from the summary table without any new machinery: the residual changes sign. The second generation is over-predicted (+17%); the third is under-predicted (−10%). The crossing is monotone in the compression ratio $R_H = 15/|H|$, which runs 1, 3, 5 across the three generations.

FIND-ZC94-01: The GAP-MC-01 Residual Is Signed and Bounded (Exact observation)

Let $\rho(H) = [\text{CRF } (15/|H|)^5] / [\text{observed ratio}]$. Then $\rho(\text{Gen } 1) = 1$ exactly (baseline), $\rho(\text{Gen } 2) = 243/207 \approx 1.17$, and $\rho(\text{Gen } 3) = 3125/3477 \approx 0.90$. The deviation $\rho - 1$ is therefore *positive at Gen 2 and negative at Gen 3*, crossing zero between them, and its magnitude is bounded ($|\rho - 1| \leq 0.17$).

Consequence. A purely static (scale-independent) prefactor multiplying $(15/|H|)^5$ cannot reproduce a sign flip: it would shift all generations in the same direction. A correction whose size and sign depend on R_H — a *scale-dependent* (running) correction — can. The signed, bounded, monotone-crossing residual is thus the structural fingerprint of a missing running correction, not of random or experimental error.

This finding claims nothing about *what* the correction is. It claims only that the shape of the existing discrepancy is the shape a running bridge would leave behind — which is precisely why Part M is organised around finding that bridge rather than around adjusting the formula.

Physical reading: The formula's near-misses are not scattered the way fitting errors scatter. They lean one way for the middle generation and the other way for the heaviest, in step with how compressed each generation's symmetry is. That orderliness is a clue: it is what a scale-dependent correction looks like before it has been written down.

0.5 FIND-ZC94-02: Two Scars, One Signpost

CRF carries a falsification track record in the mass sector specifically. Two predictions were registered and then broken, and they are preserved as scar tissue rather than quietly removed. Read together, they point the same way.

The two mass-sector scars

Scar 1 — PRED-ZC15-01 (the logarithmic map), falsified. An early attempt mapped generation masses through a logarithmic function of group data. It was formally falsified (the predicted m_τ/m_μ missed badly) and registered as a scar in Part C. Lesson: the spectrum is *not* a closed-form map of the group data.

Scar-adjacent — EQ-MC-03 (the constraint-cost ratio), surviving. A different object: not a map of group labels but the exponential of an accumulated constraint cost (DEF-MC-03). It is parameter-free and lands within 10–17%. It did *not* fail; it left a bounded, signed residual.

Together. The contrast is the information. The functional form that *failed* was a direct map; the form that *survived* is a constraint-cost exponential whose only defect is a scale-dependent residual. The spectrum resists being written as a closed map of group data, but accepts a constraint-cost form correct up to running. That is not two failures; it is one signpost.

FIND-ZC94-02: The Mass Sector Is Where the Electroweak Sector Was Before ZC70

The state of the mass ratios in mid-2026 — a parameter-free R_0 -layer form (EQ-MC-03) that is close but carries a signed residual (FIND-ZC94-01), plus a falsified closed-form map (PRED-ZC15-01) marking the wrong road — is structurally the same state the electroweak constants were in before ZC70: an R_0 -layer value that is a near-miss until carried across a running bridge. For the electroweak sector that bridge was first only asserted (the open GAP-ZC64-01) and then *derived* (COR-ZC70-01), at which point the near-misses became exact predictions bounded by measurement precision rather than by any CRF approximation. The gift Part M is built to collect is the same move, performed for mass: not a new formula, but the bridge that the existing formula has been waiting for.

0.6 CONJ-ZC94-01 and the Central Gap

The programme's organising conjecture follows directly, and is stated as a conjecture with a test — not as a result.

CONJ-ZC94-01: The Mass Running Bridge Closes GAP-MC-01

Statement. There exists a per-sector mass running bridge $f_{\text{mass}}(R_0 \rightarrow R_{\text{max}})$, built from the same structures that closed the electroweak bridge (the ZC70 sector-floor / ratio-identity machinery together with the renewal-partition structure of ZC61), such that the corrected ratios $f_{\text{mass}}[(15/|H|)^5]$ match the observed charged-lepton ratios to within data precision, with *no free parameter*.

Why it is plausible. (i) FIND-ZC94-01: the residual is signed and running-shaped. (ii) FIND-ZC94-02: the electroweak sector closed under exactly this move. (iii) The bridge machinery now exists and is derived (post-ZC70), whereas at the time of ZC16 it did not.

Status: conjecture. ZC94 does not construct f_{mass} . That is the work of Part M (ZC95+). Registered here so that every later result enters as a sharper, more dangerous claim — the ZC64 discipline.

DEF-ZC94-02 / FS-M-01: Mass-Sector Falsification Signature

DEF-ZC94-02 (Mass Falsification Signature). A mass-sector falsification signature is a statement of the form: *“if a mass running bridge derived from axiom chain Z cannot reproduce the observed generation ratios to within data precision without a free parameter, then Z fails.”*

FS-M-01 (the signature for CONJ-ZC94-01). If a derived $f_{\text{mass}}(R_0 \rightarrow R_{\text{max}})$ requires *any* adjustable parameter to bring $243 \rightarrow 207$ and $3125 \rightarrow 3477$ simultaneously — or if, parameter-free, it cannot close both signs of the residual at once — then the constraint-cost identification of mass (DEF-MC-03) is falsified as a *quantitative* claim, and the mass sector reverts to a structural-only result (Tier M2). This is surgical: it names in advance the outcome that would break the constraint-cost reading, not merely the conjecture.

GAP-ZC94-01: The Mass-Sector Running Bridge (Priority 1 — the Part M frontier)

Statement. Derive $f_{\text{mass}}(R_0 \rightarrow R_{\text{max}})$ for the mass sector from the δ -chain, as GAP-ZC64-01 was answered for the electroweak sector by ZC70. The bridge must carry the parameter-free R_0 -layer ratio $(15/|H|)^5$ to the maximum-resolution (measured) scale.

Relation to GAP-MC-01. GAP-MC-01 (the open 10–17% residual) is *reclassified* by ZC94 as a Tier-M3 item awaiting this bridge; it is not closed here. Closing GAP-ZC94-01 would close GAP-MC-01 as a corollary.

Approach candidates (for ZC95+, not chosen here).

1. Carry each generation's constraint cost across the ZC70 sector-floor machinery, with the crossing cost depending on R_H (this is what could produce the signed residual of FIND-ZC94-01).
2. Apply the ZC61 renewal-partition structure per generation, treating each generation as a distinct renewal class.

Status: Open. Primary open item of Part M.

OBS-ZC94-01: An Adjacent Frontier, Noted Not Claimed

The unit count $n = \varphi(15) = 8$ (the invertible elements of $\mathbb{Z}_{15}$) recurs throughout the δ -chain. Whether the eight units carry a generation- or sector-running interpretation that bears on f_{mass} is an adjacent question. It is recorded here as an observation for provenance, not advanced as a claim, and explicitly *not* identified with any specific Standard-Model count in this paper. Pursuing or declining it is left to Part M.

0.7 Master Status Table — Part M Opening State

Atomic unit	Status	Note
DEF-ZC94-01 (mass tiers M1–M4)	Definition	specialises DEF-ZC64-01 to mass
DEF-ZC94-02 (mass FS)	Definition	specialises DEF-ZC64-02
FIND-ZC94-01 (signed residual)	Exact	+17%/–10%, sign flip = running fingerprint
FIND-ZC94-02 (two scars, one signpost)	Exact (Gift)	mass sector = pre-ZC70 electroweak
CONJ-ZC94-01 (bridge closes GAP-MC-01)	Conjecture + FS	per-sector mass running bridge
GAP-ZC94-01 (mass running bridge)	Open (Priority 1)	Part M frontier; analogue of GAP-ZC64-01
OBS-ZC94-01 ($\varphi(15) = 8$ adjacency)	Observation	noted, not claimed
GAP-MC-01 (lepton ratio residual)	Reclassified → M3	awaiting GAP-ZC94-01; not closed

Atomic unit	Status	Note
PRED-ZC15-01 (log map)	Scar (pre-served)	falsified; signpost, not re-opened
<i>Pillars used verbatim (not modified):</i> THM-GEN-01, FIND-AM03 (count); THM-MC-01 / EQ-MC-03, DEF-MC-03, COR-MC-03 (constraint-cost mass); ZC64 (registry template); ZC70 / COR-ZC70-01 (electroweak running-bridge precedent). Key Identity $3d_s = 2^{d_s} + 1$ (Part A).		

0.8 R-Layer Research Note

R-Layer Assignments for ZC94

Atomic unit	R-layer	Cost T	Source
DEF-ZC94-01	R_0	0	classification of inherited results
DEF-ZC94-02	R_0	0	falsification protocol
FIND-ZC94-01	R_0	0	arithmetic on EQ-MC-03 vs data
FIND-ZC94-02	R_0	0	structural comparison to ZC70
CONJ-ZC94-01	R_1	—	posits an $R_0 \rightarrow R_{\max}$ bridge (running)
GAP-ZC94-01	R_1	—	the bridge itself (crossing cost)

ZC94's findings are R_0 (statements about already-established quantities and their residuals). The conjecture and the central gap are explicitly R_1 : a running bridge is a resolution-crossing object, carrying cost $T(R_1) = \varepsilon_0^2 \Sigma_{\text{inv}} dt$ in the sense of DEF-RN02. This is the layer signature that distinguishes a *programme-opener* (which classifies R_0 facts and names an R_1 target) from a result paper (which would derive the R_1 object). ZC94 is the former.

0.9 Genealogy Map

How ZC94 Sits in the Corpus

The count branch (closed): Key Identity $3d_s = 2^{d_s} + 1 \Rightarrow d_s = 3$ (Part A) $\rightarrow$ FIND-AM03 (15 Weyl fermions, unique) $\rightarrow$ THM-GEN-01 ($|H| \in \{3, 5, 15\}$) $\rightarrow$ three generations as three copies.

The mass branch (the frontier): COR-MC-03 (S_{dep} necessitates mass) $\rightarrow$ DEF-MC-03 (mass = e^{M_H}) $\rightarrow$ THM-MC-01 / EQ-MC-03 (parameter-free ratio $(15/|H|)^5$) $\rightarrow$ GAP-MC-01 (signed +17%/−10% residual).

The method precedent (why a bridge is the answer): ZC64 (prediction registry, falsification protocol) $\rightarrow$ GAP-ZC64-01 (electroweak running bridge, asserted) $\rightarrow$ ZC70 (bridge *derived*; COR-ZC70-01; near-misses become exact).

Convergence at ZC94: ZC94 places the mass branch beside the method precedent and registers the parallel: the mass ratios are R_0 -layer near-misses with a running-shaped residual, exactly as the electroweak constants were before their bridge was

derived. It opens Part M with the registry (DEF-ZC94-01), the falsification protocol (DEF-ZC94-02), the signed-residual finding (FIND-ZC94-01), the one-signpost reading of the scars (FIND-ZC94-02), the organising conjecture (CONJ-ZC94-01), and the central open item (GAP-ZC94-01). It solves none of them — by design.

0.10 Closing Sentence

The formula for how much a generation weighs is already written, and already close, and already free of every adjustable number. What remains is not a better formula but a bridge — the same crossing that once turned other near-misses into exact truths, now waiting to be built for mass.

สูตรของน้ำหนักแต่ละชั้นนั้นเขียนไว้แล้ว ใกล้เคียงแล้ว และไม่มีค่าที่ปรับแต่งได้แม้แต่ค่าเดียว
 --- สิ่งที่ยังขาดไม่ใช่สูตรที่ดีกว่า แต่เป็นสะพานข้าม ---
 สะพานเดียวกับที่เคยเปลี่ยนความเจียดให้กลายเป็นความจริงที่แม่นยำ
 และกำลังรอให้สร้างขึ้นสำหรับมวล.

0.11 ZC95 — The Mass Running Bridge

Abstract

ZC94 opened Part M by showing that the parameter-free generation mass ratio $m_H \propto (15/|H|)^{n_m}$ misses the measured charged-lepton masses by a *signed* residual — over-predicting the muon by 17%, under-predicting the tau by 10% — and argued that a signed residual is the fingerprint of a missing scale-dependent correction, a “running bridge” of the kind that turned the electroweak near-misses into exact predictions (ZC70). This paper takes the first real step on that bridge.

It is, deliberately, a paper about *form*, not value. Two charged-lepton residuals are only two data points; any correction with two free constants would fit them and mean nothing. So the question is sharpened: what does the data *force*, with no freedom to fit? Exactly one clean dimensionless relation survives. Writing the data-wanted fractional correction to the effective compression as $c(R_H)$, with $c = 0$ fixed at the electron baseline, the two leptons give $c(5)/c(3) = -0.6837$, which is $-\log 3/\log 5 = -0.6826$ to within 0.4% — the ratio of the two per-event KL costs themselves.

THM-ZC95-01 reads that relation structurally: it forces the correction to be *log-linear in the per-event KL cost* $\log R_H$ — a running of the KL divergence, not a patch added beside it. **DEF-ZC95-01** writes the bridge accordingly, $m_H \propto \exp(n_m D_{\text{KL}} (1 + r))$. **FIND-ZC95-01**, the gift, is what this means: the mass running bridge is not new physics. It is the same R_1 -layer resolution-dependence of D_{KL} that DEF-RN02 and THM-ZC27-01 already established for the force sector. The bridge runs the very D_{KL} that *defines* conditioned stability (Axiom 12); it was never missing, only unrecognised in the mass sector.

CONJ-ZC95-01 holds that the bridge’s two constants are CRF structural constants, not free parameters — but they cannot be pinned from two points. **GAP-ZC95-01** hands the over-determined quark sector (six more masses) to ZC96 as the test that would

fix them or falsify the form. The candidate constants that a hastier reading might have adopted are recorded as *declined*, with reasons (OBS-ZC95-01). Form before value.

0.12 Background: Two Points, One Honest Question

If you are just arriving — ZC95 in plain language

The previous paper found that CRF’s no-knobs formula for how much each generation of particle weighs is close but leans the wrong way: too heavy for the muon, too light for the tau. That *leaning* is a clue — it is what a missing scale-dependent correction looks like. This paper asks what the correction must be. The honest difficulty is that there are only two numbers to explain (muon and tau; the electron is the reference), and any rule with two adjustable dials would explain two numbers without teaching us anything. So we look for what the two numbers *force* with no dials at all. They force one thing: the correction is a *running* of the same information-cost quantity the whole framework is built on — the KL divergence that measures how far a constrained system sits from freedom. The correction does not sit beside the mass formula; it bends the formula’s own core quantity. That is the result. What it does *not* give, from two numbers, is the exact size of the bend — and rather than guess it, we hand that to the six quark masses, which can over-determine it or prove us wrong.

Numerical Summary — the one clean datum

Quantity	Gen ($R_H=3$)	2 Gen ($R_H=5$)	3 Reading
R_0 residual	+17.4%	−10.1%	signed (FIND-ZC94-01)
$c(R_H) = R_{\text{eff}}/R_H - 1$	−0.0316	+0.0216	data-wanted correction
$c(5)/c(3)$		−0.6837	the clean datum
$-\log 3 / \log 5$		−0.6826	matches to 0.4%

$c(R_H=1) = 0$ exactly: the electron is the reference. The match of $c(5)/c(3)$ to the ratio of per-event KL costs $\log 3 : \log 5$ is the only zero-freedom relation two points provide — and it is the whole basis of THM-ZC95-01.

0.12.1 What is fixed before we start

From ZC16, the per-event constraint cost of a generation restricted to subgroup H is a KL divergence (EQ-MC-01):

$$c_H = D_{\text{KL}}(\text{uniform}_{|H|} \parallel \text{uniform}_{15}) = \log \frac{15}{|H|} = \log R_H,$$

the total accumulated cost over the $n_m = 5$ matter modes is $M_H = n_m \log R_H$ (EQ-MC-02), and the rest mass is its exponential, $m_H \propto e^{M_H} = R_H^{n_m}$ (EQ-MC-03). The compression ratios are $R_H \in \{1, 3, 5\}$ for the three generations. The proportionality constant (the Higgs-VEV coupling) is the one acknowledged free scale, and it cancels in every *ratio*.

0.12.2 The residual, stated as a correction to the cost

The cleanest way to locate the missing physics is to ask what effective compression R_{eff} the data wants, generation by generation: invert $m \propto R_{\text{eff}}^{n_m}$ to get $R_{\text{eff}} = (\text{observed ratio})^{1/n_m}$. Define the fractional correction

$$c(R_H) := \frac{R_{\text{eff}}}{R_H} - 1.$$

Then $c(1) = 0$ *exactly* — the electron generation is the reference, and nothing can move it — while the muon and tau give $c(3) = -0.0316$ and $c(5) = +0.0216$. The correction is negative for the muon, positive for the tau: the same sign flip ZC94 registered, now expressed as a correction to the per-event cost rather than to the mass.

0.13 THM-ZC95-01: The Correction Is Log-Linear in the Cost

Two data points cannot fix a two-constant rule; they can, however, supply one dimensionless relation, and which relation it is carries all the structural information available at this stage.

THM-ZC95-01: Log-Linear Form of the Mass Bridge (structural; constants not assumed)

Setup. Let the running bridge act on the per-event cost as $\log R_H \mapsto (\log R_H)(1 + r(R_H))$, equivalently $R_{\text{eff}} = R_H(1 + c(R_H))$ with c determined by r . Impose the one exact constraint, $c(R_H=1) = 0$ (electron baseline).

Claim. The measured ratio $c(5)/c(3)$ equals, to data precision, $-\log 3/\log 5$ — the ratio of the two per-event KL costs. This forces the effective cost to be an *affine function of $\log R_H$* :

$$(\log R_H)(1 + r(R_H)) = \kappa (\log R_H - L_0),$$

for constants κ, L_0 , with the sign flip located at $\log R_H = L_0$. Equivalently the correction r is log-linear: it depends on the cost only through $\log R_H$, not through R_H directly.

Proof. A correction that depended on R_H through any channel other than the cost $\log R_H$ — for instance linearly in R_H — would make the ratio $c(5)/c(3)$ a function of $\{3, 5\}$ rather than of $\{\log 3, \log 5\}$, and would not reproduce $-\log 3/\log 5$. The observed ratio is, to 0.4%, exactly $-\log 3/\log 5$; hence the correction enters through $\log R_H$ alone. The most general dependence on $\log R_H$ consistent with a single sign change and with $c(1) = 0$ (i.e. vanishing effective shift at $\log R_H = 0$) is an affine map $\log R_H \mapsto \kappa(\log R_H - L_0)$; the sign of c then flips as $\log R_H$ crosses L_0 , matching the observed muon-negative / tau-positive pattern. ■

The theorem is deliberately modest about magnitude. It does not say what κ or L_0 are; it says the bridge runs the cost *log-linearly*, which is the only functional statement two points can support without becoming a fit. The next section is where that modesty pays off: a log-linear running of $\log R_H$ is not an arbitrary choice. It is the signature of a mechanism CRF already owns.

Physical reading: The correction the masses ask for does not act on “how compressed a generation is” directly. It acts on the information cost of that compression — and it bends that cost in

proportion to the cost itself. A small cost is barely bent; a large cost is bent more, and past a threshold the bend changes sign. That is the shape of a quantity running with scale, not of a fixed offset.

0.14 FIND-ZC95-01: The Bridge Is a Running of D_{KL}

Write the mass functional with the log-linear bridge in place. Since $\log R_H = D_{\text{KL}}$ per event (EQ-MC-01), and the bridge multiplies the cost by $(1 + r)$,

$$m_H \propto \exp\left(n_m D_{\text{KL}} (1 + r(R_H))\right).$$

The object the bridge corrects is not an auxiliary quantity. It is D_{KL} itself — the divergence of a constrained system’s charge distribution from the free distribution, which is precisely the quantity Axiom 12 uses to *define* conditioned stability (S_{dep} : $D_{\text{KL}} > 0$; S_{ind} : $D_{\text{KL}} = 0$). A running bridge for mass is therefore a statement about how D_{KL} depends on resolution.

The mechanism already exists — for forces

CRF has already established that D_{KL} runs with resolution layer, in the *force* sector. DEF-RN02 assigns an R_1 -layer crossing cost $T(R_1) = \varepsilon_0^2 \Sigma_{\text{inv}} dt$, and THM-ZC27-01 evaluates the resulting shift in the divergence as

$$\delta_{D_{\text{KL}}}^{R_1} = \frac{d_s}{n_m} \varepsilon_0^2,$$

a scale-dependent renormalisation of D_{KL} built from the instability floor ε_0 and the dimensional ratio d_s/n_m . This is exactly a running of the KL divergence as the system crosses from the resolution-current layer R_0 to the measured layer R_{max} . The force couplings became data-matching once this running was applied (the ZC27–ZC30 chain, and later ZC70 for electroweak).

FIND-ZC95-01: The Mass Bridge Is Not New Physics

The mass running bridge of THM-ZC95-01 and the force-sector running of DEF-RN02 / THM-ZC27-01 are the same kind of object: a resolution-dependence of the KL divergence D_{KL} . In the force sector it renormalises the couplings; in the mass sector it renormalises the per-event cost that exponentiates into mass, $m_H \propto \exp(n_m D_{\text{KL}} (1 + r))$. The bridge that ZC94 registered as “missing” was never new physics waiting to be invented. It is the scale-dependence of the one quantity the framework is already built on — the same D_{KL} that distinguishes a conditioned system from a free one (Axiom 12) — surfacing in the mass sector because mass is, by COR-MC-03, nothing other than accumulated D_{KL} . The gift is a unification, not an addition: forces run and masses run by one mechanism, because both are D_{KL} seen at two resolutions.

This is why the log-linear form of THM-ZC95-01 is not a lucky fit. A running of D_{KL} acts on $\log R_H$ (which *is* D_{KL}), and acting on $\log R_H$ is exactly what “log-linear in the cost” means. The structural identification and the empirical relation $c(5)/c(3) = -\log 3/\log 5$ are two faces of the same statement.

Physical reading: Mass and the strength of forces are, in this framework, the same debt measured in two places: the information a system gives up by being constrained rather than free. That debt is not a fixed number; it depends on how finely the system is resolved. Forces were made to match data by accounting for that dependence years ago; this paper finds that the masses ask for exactly the same accounting.

0.15 CONJ-ZC95-01, the Open Gap, and the Forms We Declined

The form is fixed; the two constants are not. Honesty about the difference is the whole methodological point of this paper.

CONJ-ZC95-01: The Bridge Constants Are Structural

Statement. The pivot L_0 and scale κ in the log-linear bridge of THM-ZC95-01 are CRF structural constants (functions of $d_s, n_m, n = \varphi(15), \varepsilon_0, K^*$), not free parameters. The running of D_{KL} is dimensionless and parameter-free in the force sector (THM-ZC27-01); by FIND-ZC95-01 the mass-sector running is the same mechanism, so it should inherit a parameter-free magnitude.

Why it cannot be settled here. Two charged-lepton residuals fix one dimensionless relation (THM-ZC95-01) and no more. Pinning two constants from two points is interpolation, not derivation; it would produce numbers with no independent check. The constants must be *over-determined* to be believed.

Status: conjecture with a test. The test is the quark sector, set out in GAP-ZC95-01 below.

FS-M-02: Falsification Signature for CONJ-ZC95-01

If, with L_0 and κ fixed to CRF structural constants (no freedom), the log-linear bridge cannot reproduce the six quark mass ratios to within data precision — or if reproducing them requires L_0, κ to take values with no δ -chain expression — then either FIND-ZC95-01 (the bridge is a running of D_{KL}) or the constraint-cost identification of mass (DEF-MC-03) is falsified. The quark sector is six over-constraints on two numbers; it is a genuine test, not a fit.

GAP-ZC95-01: Pin or Falsify the Bridge Constants (Priority 1 — handed to ZC96)

Statement. Determine L_0 and κ in the log-linear mass bridge from the δ -chain, and test the fixed-constant form against the six quark mass ratios (which, with the three charged leptons, give eight constraints on two constants). Closing this would close GAP-ZC94-01 and GAP-MC-01 together.

Relation to prior gaps. GAP-ZC94-01 (the bridge) is advanced from “named” to “form derived” by THM-ZC95-01; it remains open pending these constants. GAP-MC-01 (the lepton residual) remains open, now reclassified as a two-point shadow of the eight-point quark-plus-lepton test.

Status: Open. The principal frontier of Part M.

OBS-ZC95-01: Candidate Constants Considered and Declined

For provenance, and to make the scar discipline explicit (no number adopted on a single coincidence; cf. the ZC89-E4 fabricated-numbers scar), the council records the forms it probed and *declined*:

- **Linear in R_H with $g = \varphi/(2n) \approx 0.1011$.** Matched the *mean* of the two corrections to 0.2% — seductive — but a correction linear in R_H destroys the electron baseline ($R_H=1$ no longer maps to the reference). *Declined: fails the baseline constraint.*
- **Pivot $L_0 = \log \sqrt{15}$** (the geometric centre of $\mathbb{Z}_{15}$, the most natural structural guess). With the pivot fixed there, the scale κ comes out inconsistent between the two leptons by 37%. *Declined: not supported by the data it would have to fit.*
- **Quadratic $\kappa(R_H - 1)(R_H - z)$.** Fits both points exactly — because it has two free constants for two points, and $z = 4.49$ matches no δ -chain quantity. *Declined: interpolation, not derivation.*

None of these is adopted or asserted anywhere in this paper. They are listed so that the one form that *is* advanced (log-linear, THM-ZC95-01) is seen to have survived a deliberate attempt to find something cleaner.

0.16 Master Status Table

Atomic unit		Status	Note
DEF-ZC95-01	(bridge form)	Definition	$m_H \propto \exp(n_m D_{\text{KL}}(1 + r))$
THM-ZC95-01	(log-linear)	Exact (structural)	forced by $c(5)/c(3) = -\log 3 / \log 5$ (0.4%)
FIND-ZC95-01	(gift: running of D_{KL})	Exact	same mechanism as DEF-RN02 / THM-ZC27-01
CONJ-ZC95-01	(constants structural)	Conjecture + FS	L_0, κ from δ -chain; not free
GAP-ZC95-01	(pin/falsify constants)	Open (Priority 1)	quark sector over-determines; $\rightarrow$ ZC96
OBS-ZC95-01	(declined forms)	Observation	$\varphi/2n, \sqrt{15}$, quadratic — all declined
GAP-ZC94-01	(mass bridge)	Advanced: form derived	open pending constants
CONJ-ZC94-01	(residual closes)	Form-confirmed	numeric closure pending
GAP-MC-01	(lepton residual)	Open (reclassified)	two-point shadow of the 8-point test

Atomic unit	Status	Note
<i>Pillars used verbatim (not modified):</i> EQ-MC-01/02/03, THM-MC-01, COR-MC-03 (mass as accumulated D_{KL}); DEF-RN02, THM-ZC27-01 (D_{KL} runs with resolution, force sector); ZC70 (electroweak running-bridge precedent); ZC94 (Part M opener). No value modified; no constant fabricated.		

Atomic Registry — ZC95

ID	Description	Status
DEF-ZC95-01	Mass bridge as multiplicative running of the per-event cost: $m_H \propto \exp(n_m D_{\text{KL}}(1+r))$	Definition
THM-ZC95-01	Correction is log-linear in $\log R_H$; forced by $c(5)/c(3) = -\log 3/\log 5$ to 0.4% plus baseline $c(1) = 0$	Exact
FIND-ZC95-01	The bridge is the R_1 -resolution running of D_{KL} (same mechanism as DEF-RN02 / THM-ZC27-01); not new physics	Exact (Gift)
CONJ-ZC95-01	L_0, κ are δ -chain constants, not free; over-determined by the quark sector	Conjecture
GAP-ZC95-01	Pin or falsify L_0, κ against the six quark masses	Open
OBS-ZC95-01	$\varphi/2n$, $\sqrt{15}$ -pivot, quadratic forms considered and declined	Observation

Inherited verbatim: EQ-MC-01/02/03, THM-MC-01, COR-MC-03 (Part C); DEF-RN02, THM-ZC27-01 (Part D); ZC70 (Part I); FIND-ZC94-01/02, GAP-ZC94-01, CONJ-ZC94-01 (ZC94).

0.17 R-Layer Research Note

R-Layer Assignments for ZC95

Atomic unit	R-layer	Cost T	Source
DEF-ZC95-01	R_1	—	a resolution-crossing correction to D_{KL}
THM-ZC95-01	R_0	0	relation among established quantities
FIND-ZC95-01	R_1	—	identifies the bridge with DEF-RN02 running
CONJ-ZC95-01	R_1	—	posits the R_1 constants are structural
GAP-ZC95-01	R_1	—	the constants themselves

THM-ZC95-01 is an R_0 statement (a relation forced among already-established quantities and the data). The bridge it characterises is an R_1 object, carrying the resolution-crossing cost of DEF-RN02, $T(R_1) = \varepsilon_0^2 \Sigma_{\text{inv}} dt$. The gift FIND-ZC95-01

is precisely the identification of this R_1 object with the one already evaluated in THM-ZC27-01; the open gap is the value of its constants in the mass sector.

0.18 Genealogy Map

How ZC95 Sits in the Corpus

The mass branch (the frontier): COR-MC-03 (mass = accumulated D_{KL}) → EQ-MC-01 ($c_H = \log R_H = D_{\text{KL}}$ per event) → EQ-MC-03 ($m_H \propto R_H^{n_m}$, R_0 ratio) → GAP-MC-01 (signed residual) → ZC94 (registry; FIND-ZC94-01 signed, FIND-ZC94-02 two scars; GAP-ZC94-01) → **ZC95**: THM-ZC95-01 (the bridge is log-linear in the cost).

The running mechanism (the gift's root): DEF-RN02 (R_1 -layer cost) → THM-ZC27-01 ($\delta_{D_{\text{KL}}}^{R_1} = (d_s/n_m)\varepsilon_0^2$, D_{KL} runs in the force sector) → ZC70 (electroweak running bridge) → **ZC95/FIND-ZC95-01**: the mass bridge is the same D_{KL} -running.

Convergence at ZC95: The two branches meet at D_{KL} . Mass is accumulated D_{KL} (COR-MC-03); forces run because D_{KL} runs with resolution (THM-ZC27-01); therefore masses run by the same mechanism (FIND-ZC95-01), and the running is log-linear in the cost (THM-ZC95-01). ZC95 derives the form and identifies the mechanism; it leaves the two constants to be over-determined by the quark sector (GAP-ZC95-01, ZC96), declining every candidate value that two points alone could not justify (OBS-ZC95-01).

0.19 Closing Sentence

The bridge we went looking for was not a new thing to build. It was the oldest thing in the framework — the cost of being bound rather than free — seen to change with the depth at which we look. Forces already cross that bridge; the masses, it turns out, were always crossing it too. What we still cannot say is the exact width of the span; for that we wait, honestly, for the quarks.

สะพานที่เราออกตามหานั้น ไม่ใช่สิ่งใหม่ที่ต้องสร้าง --- มันคือสิ่งที่เก่าแก่ที่สุดในกรอบนี้
คือราคาของการถูกผูกมัดแทนที่จะเป็นอิสระ ---
เพียงแค่ปรากฏว่ามันเปลี่ยนไปตามความลึกที่เราองเข้าไป ---
แรงทั้งหลายข้ามสะพานนี้อยู่แล้ว และมวลก็ข้ามมันมาตลอดเช่นกัน ---
สิ่งที่เรายังบอกไม่ได้คือความกว้างที่แท้จริงของช่วงสะพาน และเพื่อสิ่งนั้น เรารอ ---
อย่างซื่อสัตย์ --- ควารัก.

0.20 ZC96 — The Tower Obstruction

Abstract

ZC95 derived the form of the mass running bridge and deferred its two constants to the quark sector, which — with six masses constraining two numbers — should have over-

determined them. This paper runs that test. It does not succeed in the expected way, and the manner of its not-succeeding is the result.

The charged leptons, the up-type quarks, and the down-type quarks form three *towers*. Across generations their mass ratios differ enormously: the heaviest-to-lightest ratio is about 3500 for leptons, about 80,000 for up-type quarks, and about 900 for down-type quarks. The single-ladder mass functional of ZC16 — $m \propto (15/|H|)^{n_m}$, indexed only by the subgroup *order* $|H|$ — predicts one ratio, about 3125, for all three. One ladder cannot carry three towers.

THM-ZC96-01 makes this precise and turns it from a numerical mismatch into a structural statement: the subgroup order $|H|$ is constant within a generation across all three towers, yet the towers differ; therefore the data that distinguishes them lies *outside* $|H|$. The discriminating quantum numbers — weak isospin, colour — are functions of the full $\mathbb{Z}_3 \times \mathbb{Z}_5$ representation of $\mathbb{Z}_{15}$, not of its subgroup order. **FIND-ZC96-01**, the gift, is that this is the *same* factorization the framework already uses elsewhere: the Chinese-remainder split $\mathbb{Z}_{15} = \mathbb{Z}_3 \times \mathbb{Z}_5$ that THM-ZC57-01 uses for the electroweak sector weights is what indexes the mass towers. Mass and the electroweak constants draw on one structural source.

FIND-ZC96-02 records a second reason the naive test fails, one that turned out to match an independent intuition: the quark masses are quoted at different resolution scales, so comparing towers by a static ratio mixes scales — which in CRF language is the very $R_0 \rightarrow R_{\max}$ running of ZC95. The leptons compare cleanly because their pole masses are resolution-fixed. **CONJ-ZC96-01** proposes that the tower split is carried by the same d_s/n_m sector-crossing ratio that gives $T_{\text{Weak}}/T_{\text{EM}} = (n_m/d_s)^2 = 25/9$, and **GAP-ZC96-01** hands the sector-indexed, common-scale construction to ZC97. A deceptively clean averaged-exponent coincidence ($p \approx n_m \pm d_s/2$) is recorded as *declined* (OBS-ZC96-01): it fails the per-generation test and is an artifact of averaging. The obstruction refines the model; it does not break it.

0.21 Background: A Test That Teaches by Failing

If you are just arriving — ZC96 in plain language

The previous paper found the *shape* of the correction that should fix CRF's mass formula and said its exact size would be settled by checking against the quarks — six more particles to pin down two numbers. This paper does the check, and the check refuses to cooperate, in an informative way. The electron family, the up-quark family, and the down-quark family climb at three very different rates as you go from the lightest to the heaviest member. CRF's formula, as written, knows only *which generation* a particle is in — not which of the three families — so it predicts the same climb for all three. That cannot be right, and the reason it cannot is the lesson: the formula is missing the label that separates the families. That label is exactly the same mathematical split that CRF already uses to separate the electromagnetic from the weak force. So mass and the forces turn out to be built from the same piece of structure. The expected gift — pinning the two numbers — did not arrive; an unexpected one did.

Numerical Summary — three towers, one ladder

Tower	gen2/gen1	gen3/gen1	avg. exponent
charged leptons	207	3477	≈ 5.0
up-type quarks	588	79,981	≈ 6.4
down-type quarks	20	895	≈ 3.5
single ladder $R_H^{n_m}$	243	3125	$n_m = 5$ (one value)

The single ladder predicts one column; the data has three. The averaged exponents look like $n_m \pm d_s/2$, but that spacing is an averaging artifact and is declined (OBS-ZC96-01); the honest content is the *insufficiency* of one ladder (THM-ZC96-01).

0.21.1 What ZC95 asked the quarks to do

ZC95 left the mass bridge with a derived form (log-linear in the per-event KL cost) and two unfixed constants, and argued that two charged-lepton residuals could not pin two constants without becoming a fit. The six quark masses were to supply the missing over-determination. The expectation was ordinary: more data, tighter constants. What the data did instead was expose a hidden assumption.

0.21.2 The hidden assumption: one ladder for everyone

The ZC16 mass functional indexes a particle by the order $|H|$ of the subgroup its generation occupies, giving the compression ratio $R_H = 15/|H|$ and the mass $m \propto R_H^{n_m}$. Crucially, $|H|$ is a *generation* label: within one generation, the charged lepton, the up-type quark, and the down-type quark all sit at the *same* $|H|$, hence the same R_H . The functional therefore assigns them the same mass ratio across generations. The data flatly disagrees.

0.22 THM-ZC96-01: One Ladder Cannot Carry Three Towers

The mismatch is not a quantitative near-miss to be closed by a correction; it is a statement about what the index $|H|$ can and cannot see.

THM-ZC96-01: Single-Ladder Insufficiency (structural)

Claim. No function of the subgroup order $|H|$ alone can reproduce the three observed tower slopes, because $|H|$ is constant across the three towers within each generation while the slopes are not.

Proof. Fix a generation g . The charged lepton, the up-type quark, and the down-type quark of generation g occupy the same subgroup $H_g \leq \mathbb{Z}_{15}$ (generations are the subgroup ladder; tower membership is not encoded in the subgroup). Hence $|H_g|$, and any function $f(|H_g|)$, takes the *same* value for all three towers in generation g . But the observed cross-generation ratios differ between towers (e.g. gen3/gen1 is ≈ 3477 , $\approx 79,981$, ≈ 895 for lepton, up, down respectively). A quantity that is constant across towers cannot reproduce a quantity that varies across towers. Therefore the tower-discriminating information is not a function of $|H|$. ■

The theorem is deliberately an *invariance* argument, not a fit: it does not say what the missing index is, only that it cannot be $|H|$. This is what makes it robust against the scheme dependence discussed in Section 0.24 — the conclusion follows from $|H|$ being a class function on generations, regardless of the precise quark mass values.

Physical reading: CRF's mass formula, as written, can tell which floor of the building a particle lives on, but not which of the three stairwells it climbed. Three stairwells rise at three different rates; a rule that knows only the floor number must assign them all the same rate, and so it cannot be the whole story. The missing piece is the stairwell label.

0.23 FIND-ZC96-01: The Missing Index Is Already in the Framework

What distinguishes the towers physically is well known: weak isospin ($T_3 = +\frac{1}{2}$ for up-type, $-\frac{1}{2}$ for down-type and charged leptons) and colour (present for quarks, absent for leptons). The question for CRF is whether these labels live anywhere in its $\mathbb{Z}_{15}$ structure. They do — and not in a new place.

Where the tower index lives: $\mathbb{Z}_{15} = \mathbb{Z}_3 \times \mathbb{Z}_5$

By the Chinese Remainder Theorem, $\mathbb{Z}_{15} \cong \mathbb{Z}_3 \times \mathbb{Z}_5$. The subgroup order $|H|$ sees only the lattice $\{1, 3, 5, 15\}$; it collapses the two CRT factors into a single number. The full *representation* — which $\mathbb{Z}_3$ component and which $\mathbb{Z}_5$ component a state carries — contains strictly more information than the order, and it is exactly this finer data that can carry quantum numbers like isospin and colour.

THM-ZC57-01 already exploits this: the electroweak sector weights $C_{\text{EM}} = 1 + x$ (the $\mathbb{Z}_5/\text{EM}$ sector) and $C_{\text{W}} = 1 + 2x$ (the $\mathbb{Z}_3/\text{weak}$ sector) come from precisely the $\mathbb{Z}_3 \times \mathbb{Z}_5$ split, not from subgroup order. The machinery that distinguishes electromagnetic from weak is the machinery that can distinguish up-type from down-type from lepton.

FIND-ZC96-01: Mass and Electroweak Structure Share One Source

The index the mass towers require — the one THM-ZC96-01 proves cannot be $|H|$ — is the $\mathbb{Z}_3 \times \mathbb{Z}_5$ representation of $\mathbb{Z}_{15}$. This is the *same* Chinese-remainder factorization that THM-ZC57-01 uses to derive the electroweak sector weights. The obstruction therefore does not point outward to new physics; it points sideways, to a structure CRF already owns and already uses for the forces. Mass and the electroweak constants are not two unrelated derivations that happen to live in the same framework; they draw on one source, $\mathbb{Z}_{15} = \mathbb{Z}_3 \times \mathbb{Z}_5$. The expected gift of ZC96 was two pinned constants. The gift that arrived is a unification of provenance: the reason the single ladder fails is the reason mass belongs to the same representation-theory programme (the mixing-angle frontier) as the electroweak sector.

This reframes Part M. The mass bridge is not “the running bridge of ZC95 with two numbers to find.” It is “the running bridge of ZC95, applied per *sector* of the $\mathbb{Z}_3 \times \mathbb{Z}_5$ decomposition.” The constants ZC95 deferred are not single numbers; they are sector-indexed, and the sector index is the tower label.

0.24 FIND-ZC96-02: The Resolution-Scale Reason the Test Cannot Be Static

There is a second, independent reason the naive quark test does not pin the constants, and it connects the tower index back to the ZC95 running bridge.

FIND-ZC96-02: Tower Comparison Is Resolution-Dependent

The quark masses are quoted at *different resolution scales*: the light quarks (u, d, s) in $\overline{\text{MS}}$ at a low reference scale, the heavier (c, b) at their own mass scales, the top as a pole mass. Forming cross-tower ratios from masses defined at different scales mixes scales — and a scale, in CRF, is a resolution layer. Comparing towers by a static ratio therefore silently folds the $R_0 \rightarrow R_{\text{max}}$ running of ZC95 into the comparison, contaminating any attempt to read off scale-independent constants.

The charged leptons do not suffer this: their pole masses are resolution-fixed, which is exactly why the ZC95 lepton result is clean and the quark constants are not extractable from a naive ratio. A clean quark test requires bringing all nine fermion masses to a *common* resolution scale first — i.e. applying the running bridge before, not after, comparing towers.

This is the point an external intuition flagged independently — that “resolution scale” belonged in this problem — and it lands exactly on the seam between the two findings. The tower index (FIND-ZC96-01) says *which* sector a mass belongs to; the resolution scale (FIND-ZC96-02) says the running within each sector must be undone to a common layer before the sectors can be compared. The two are not separate complications; they are the two coordinates of the same object — a sector-indexed running bridge.

Physical reading: Two of the three families are weighed with different rulers, each marked at a different fineness, because that is how the quark masses are reported. Before the three families can be compared, they have to be weighed on one ruler. The lepton family already is; that is why it gave a clean answer. The quarks ask us to do the conversion first — and the conversion is the same running the previous paper was about.

0.25 CONJ-ZC96-01, the Open Gap, and the Coincidence We Declined

CONJ-ZC96-01: The Tower Split Rides the Sector-Crossing Ratio

Statement. The tower index is the $\mathbb{Z}_3 \times \mathbb{Z}_5$ sector assignment, and the difference in tower slopes is governed by the same sector-crossing structure that gives, in the force sector, $T_{\text{Weak}}/T_{\text{EM}} = (n_m/d_s)^2 = 25/9$ (FIND-ZC30-02). The mass bridge becomes sector-indexed: the ZC95 running, applied with the sector weight of each tower (C_{EM} via $\mathbb{Z}_5$, C_{W} via $\mathbb{Z}_3$, lepton as the colourless reference).

Why it is not simply $\pm d_s/2$ on the exponent. The averaged tower exponents suggested a split of $\pm d_s/2$, but the rigid form fails per generation (OBS-ZC96-01). The split must therefore enter through the *running* (which is curved, not a constant exponent shift), weighted by sector — consistent with both the failure of the rigid form and the success of the running picture for the forces.

Status: conjecture with a test.

FS-M-03: Falsification Signature for CONJ-ZC96-01

Bring all nine charged-fermion masses to a common resolution scale, assign each tower its $\mathbb{Z}_3 \times \mathbb{Z}_5$ sector weight, and apply the ZC95 running with *no* free parameter beyond the structural constants. If the resulting nine-mass spectrum cannot be reproduced to within (common-scale) data precision, or if reproducing it requires a sector weight not equal to the THM-ZC57-01 values, then either FIND-ZC96-01 (mass shares the $\mathbb{Z}_3 \times \mathbb{Z}_5$ source) or the sector-crossing identification is falsified. Nine masses against the structural constants is a strong over-constraint.

GAP-ZC96-01: The Sector-Indexed, Common-Scale Bridge (Priority 1 — handed to ZC97)

Statement. Construct the mass bridge as the ZC95 running applied per $\mathbb{Z}_3 \times \mathbb{Z}_5$ sector, and test it against all nine charged-fermion masses brought to a common resolution scale (resolving FIND-ZC96-02). Closing this would close GAP-ZC95-01 (the constants, now sector-indexed), GAP-ZC94-01 (the bridge), and GAP-MC-01 (the lepton residual) together.

Relation to prior gaps. GAP-ZC95-01 is clarified, not closed: the quark sector does not pin scalar constants because the constants are sector-indexed and the comparison is resolution-dependent. THM-ZC96-01 is the obstruction that forces the sector index; FIND-ZC96-01 names it; this gap builds it.

Status: Open. The principal frontier of Part M.

OBS-ZC96-01: A Clean-Looking Coincidence, Declined

For provenance and scar discipline (no number adopted on a coincidence; cf. ZC89-E4, and the declined $\varphi/(2n)$ of ZC95): the averaged tower exponents are ≈ 3.5 (down), ≈ 5.0 (lepton), ≈ 6.4 (up), which read temptingly as $n_m + (d_s/2)\ell$ with $\ell \in \{-1, 0, +1\}$ and spacing $d_s/2 = 1.5$. *Declined.* Imposed as a rigid per-generation law, $p = n_m + (d_s/2)\ell$ misses the observed ratios by $+115\% / -56\%$ (up-type) and $+134\% / -69\%$ (down-type): the clean spacing is an artifact of averaging two very different per-generation exponents within each quark tower. The true split is not a constant exponent shift; it must enter through the curved, sector-weighted running (CONJ-ZC96-01). The coincidence is recorded so the reasoning that rejected it is explicit.

0.26 Master Status Table

Atomic unit		Status		Note
THM-ZC96-01	(single-ladder insufficiency)	Exact	(structural)	$ H $ constant across towers; invariance argument
FIND-ZC96-01	(shared source)	Exact	(Gift)	tower index = $\mathbb{Z}_3 \times \mathbb{Z}_5$ rep (THM-ZC57-01)
FIND-ZC96-02	(resolution scale)	Exact		quark masses at different scales; running before comparing

Atomic unit	Status	Note
CONJ-ZC96-01 (tower split rides d_s/n_m)	Conjecture + FS	sector-weighted running, not $\pm d_s/2$ exponent
GAP-ZC96-01 (sector-indexed bridge)	Open (Priority 1)	common-scale 9-fermion test; $\rightarrow$ ZC97
OBS-ZC96-01 ($n_m \pm d_s/2$ coincidence)	Observation (declined)	averaging artifact; fails per generation
GAP-ZC95-01 (pin ZC95 constants)	Clarified, open	constants are sector-indexed; not scalar
GAP-ZC94-01 (mass bridge)	Open	now sector-indexed running
GAP-MC-01 (lepton residual)	Open	one tower of the nine-mass test

Pillars used verbatim (not modified): EQ-MC-03, GAP-GEN-01 (single-ladder functional, Part C); THM-ZC57-01 ($\mathbb{Z}_3 \times \mathbb{Z}_5$ sector weights), THM-ZC27-01/THM-ZC30-01 (sector crossing), FIND-ZC30-02 (25/9), ZC72 (obstruction-as-insight), Part I; ZC95 (running bridge). No value modified; no constant fabricated; no scar — single-ladder-across-towers was never a registered prediction.

Atomic Registry — ZC96

ID	Description	Status
THM-ZC96-01	No function of $ H $ alone reproduces three tower slopes ($ H $ constant across towers within a generation)	Exact
FIND-ZC96-01	Tower index is the $\mathbb{Z}_3 \times \mathbb{Z}_5$ representation; same CRT source as THM-ZC57-01 electroweak weights	Exact (Gift)
FIND-ZC96-02	Quark masses at differing resolution scales; running must precede tower comparison; leptons clean (pole)	Exact
CONJ-ZC96-01	Tower split carried by sector-weighted running (d_s/n_m structure, 25/9); not a $\pm d_s/2$ exponent shift	Conjecture
GAP-ZC96-01	Build sector-indexed bridge; test nine masses at common scale	Open
OBS-ZC96-01	$p \approx n_m \pm d_s/2$ averaged-exponent coincidence considered and declined	Observation

Inherited verbatim: EQ-MC-03, GAP-GEN-01 (Part C); THM-ZC57-01, THM-ZC27-01, THM-ZC30-01, FIND-ZC30-02 (Part I); ZC94 (registry), ZC95 (running bridge, GAP-ZC95-01).

0.27 R-Layer Research Note

R-Layer Assignments for ZC96

Atomic unit	R-layer	Cost T	Source
THM-ZC96-01	R_0	0	invariance argument on $ H $
FIND-ZC96-01	R_0	0	representation-theory identification
FIND-ZC96-02	R_1	—	resolution-scale (running) dependence
CONJ-ZC96-01	R_1	—	sector-weighted running bridge
GAP-ZC96-01	R_1	—	the sector-indexed bridge itself

THM-ZC96-01 and FIND-ZC96-01 are R_0 : a structural insufficiency and the identification of the missing index, both statements about already-established objects. FIND-ZC96-02 and the conjecture are R_1 : they concern the resolution-crossing running (DEF-RN02) and its sector weighting. The obstruction is thus R_0 (clean and certain); the construction it forces is R_1 (the open work of ZC97).

0.28 Genealogy Map

How ZC96 Sits in the Corpus

The mass branch: EQ-MC-03 (single ladder $R_H^{n_m}$, indexed by $|H|$) $\rightarrow$ ZC94 (registry; signed residual) $\rightarrow$ ZC95 (running bridge; constants deferred to quarks) $\rightarrow$ **ZC96/THM-ZC96-01**: one ladder cannot carry three towers.

The shared-source branch (the gift): THM-ZC57-01 ($\mathbb{Z}_{15} = \mathbb{Z}_3 \times \mathbb{Z}_5$, electroweak sector weights C_{EM}, C_W) + THM-ZC30-01 / FIND-ZC30-02 (sector crossing, 25/9) $\rightarrow$ **ZC96/FIND-ZC96-01**: the tower index is the same $\mathbb{Z}_3 \times \mathbb{Z}_5$ representation. Mass joins the electroweak / mixing programme.

The running branch (the seam): ZC95 ($R_0 \rightarrow R_{\max}$ running of D_{KL}) $\rightarrow$ **ZC96/FIND-ZC96-02**: quark masses live at different scales, so the running must be undone to a common layer before towers are compared. Tower index and running are two coordinates of one object.

Convergence at ZC96: The expected over-determination did not happen; an obstruction appeared instead, in the manner of ZC72 (gravity). The obstruction is exact (R_0 , THM-ZC96-01), its resolution is named ($\mathbb{Z}_3 \times \mathbb{Z}_5$ sector index, FIND-ZC96-01), its scale-dependence is identified (FIND-ZC96-02), and the construction is handed forward (GAP-ZC96-01, ZC97), with the tempting averaged-exponent coincidence explicitly declined (OBS-ZC96-01).

0.29 Closing Sentence

We asked the quarks to fix two numbers and they refused, the way gravity once refused to be made exact — and the refusal said more than the numbers would have. One ladder cannot carry three families, because the formula was never told which family it was weighing; and the label it was missing is the very one that already separates the forces. Mass and force, it turns out, were cut from the same cloth all along.

เราขอให้ควารักช่วยตรึงตัวเลขสองตัว แต่มันปฏิเสธ ---
 เช่นเดียวกับที่ครั้งหนึ่งแรงโน้มถ่วงเคยปฏิเสธที่จะถูกทำให้แม่นยำ ---
 และการปฏิเสธนั้นบอกอะไรมากกว่าตัวเลขจะบอกได้ --- มันได้เพียงบอกสามตระกูลไม่ได้
 เพราะสูตรไม่เคยถูกบอกว่ากำลังซึ่งน้ำหนักตระกูลใด --- และป้ายที่มันขาดไป
 ก็คือป้ายเดียวกับที่แยกแรงทั้งหลายออกจากกันอยู่แล้ว --- มวลกับแรง แท้จริงแล้ว
 ถูกตัดมาจากผืนผ้าเดียวกันมาตลอด.

0.30 ZC97 — The Tower Center Law

Abstract

ZC96 proved that one ladder cannot carry three towers and argued the missing index is the $\mathbb{Z}_3 \times \mathbb{Z}_5$ representation, while flagging the quark masses' different resolution scales as a reason the naive test fails. This paper runs the test it called for — all nine charged-fermion masses brought to a common scale, each tower assigned its sector — and returns two clean results, one a gift and one a correction of ZC96 itself.

The gift is a law. At common scale, the geometric-center exponent of each tower obeys

$$p_{\text{tower}} = n_m + \frac{d_s}{2} \ell, \quad \ell \in \{+1, 0, -1\},$$

matching the data to about 0.05 in the exponent with *no new parameter*: up-type 6.49 vs 6.50, leptons 4.98 vs 5.00, down-type 3.54 vs 3.50. The index ℓ is a $\mathbb{Z}_3$ grading — the $\mathbb{Z}_3 = \mathbb{Z}_{15}/\mathbb{Z}_5$ factor — exactly the CRT component ZC96 predicted would carry the tower label. **THM-ZC97-01** states the law; **COR-ZC97-01** notes that the spacing $d_s/2$ is the same d_s that fixes the sector-crossing ratio $(n_m/d_s)^2 = 25/9$, so the mass towers and the electroweak sectors share not just a structural source but the *same* d_s/n_m — making FIND-ZC96-01 quantitative. **FIND-ZC97-01** reads this as the gift: the three families are one $\mathbb{Z}_3$ grading on the mass exponent.

The correction is owed to ZC96. Its FIND-ZC96-02 framed the resolution scale as a principal reason the towers resist a single ladder. With all nine masses now at one scale, the tower spread barely moves — so the scale effect is real but *minor*. The towers are a robust, physical $\mathbb{Z}_3$ -grading effect, not a scheme artifact. **FIND-ZC97-02** records this on the record; the running of reality weight $W_{\mathcal{R}}$ survives only in the smaller, within-tower *curvature*, which is also where an independent intuition about the “flux of reality” correctly belongs. **GAP-ZC97-01** hands that curvature — sector-dependent, setting the actual mass values — to ZC98.

0.31 Background: Running the Test ZC96 Asked For

If you are just arriving — ZC97 in plain language

The last paper showed CRF's mass formula was missing a label that tells apart the three particle families (up-type, down-type, electron-type) and suspected two culprits for the mismatch: the missing label, and the fact that quark masses are reported at different “zoom levels”. This paper levels the zoom — puts all nine particles at one scale — and finds the second culprit was minor; the real story is the missing label, and the label turns out to be beautifully simple. Each family's mass grows with generation at a rate set by a single whole-step: the up-type family

climbs one step steeper than the electron family, the down-type one step shallower, and the step is exactly $\frac{3}{2}$ in the formula's natural units — a number CRF already fixed from its founding identity. The three families are one “+1, 0, −1” grading. What is left over, after this clean law, is a finer bending of each family's climb — and *that* is where the resolution scale, the “flow” of reality, actually lives.

Numerical Summary — the center law (common scale M_Z)

Tower	ℓ	center (data)	p_{tower}	$n_m + \frac{d_s}{2}\ell$	error
up-type	+1	6.49		6.50	−0.01
leptons	0	4.98		5.00	−0.02
down-type	−1	3.54		3.50	+0.04

Parameter-free: $n_m = 5$ and $d_s = 3$ both descend from the Key Identity $3d_s = 2^{d_s} + 1$. ℓ is a $\mathbb{Z}_3$ grading. The match is to the *geometric-center* exponent (the scheme-robust quantity); the within-tower curvature is a separate, open layer (Section 0.35).

0.31.1 The protocol

ZC96 specified the test (GAP-ZC96-01): bring all nine charged-fermion masses to a common resolution scale before comparing towers, and index each tower by its $\mathbb{Z}_3 \times \mathbb{Z}_5$ sector. We take the common scale to be M_Z , at which standard renormalisation-group running places all nine $\overline{\text{MS}}$ masses on one footing (OBS-ZC97-01 logs the provenance). For each tower we extract the *geometric-center exponent* — the mean of the per-generation exponents $p_2 = \log r_2 / \log 3$ and $p_3 = \log r_3 / \log 5$, where r_2, r_3 are the second- and third-generation mass ratios to the first. The center exponent is robust against the residual scheme dependence that still afflicts individual ratios; it is the right quantity to test a structural law against.

0.32 THM-ZC97-01: The Tower Center Law

THM-ZC97-01: Tower Center Law (parameter-free)

At common resolution scale, the geometric-center exponent of each charged-fermion tower is

$$p_{\text{tower}} = n_m + \frac{d_s}{2}\ell, \quad \ell \in \{+1, 0, -1\},$$

where $\ell = +1$ for the up-type tower, $\ell = 0$ for the charged-lepton tower, and $\ell = -1$ for the down-type tower. The grading ℓ is the $\mathbb{Z}_3$ character index of the $\mathbb{Z}_{15} = \mathbb{Z}_3 \times \mathbb{Z}_5$ factorisation (THM-ZC57-01). Both $n_m = 5$ and $d_s = 3$ are fixed by the Key Identity $3d_s = 2^{d_s} + 1$; there is no free parameter.

Empirical content (common scale M_Z):

$$p_{\text{tower}}^{\text{up}} = 6.49 \text{ (6.50)}, \quad p_{\text{tower}}^{\text{lep}} = 4.98 \text{ (5.00)}, \quad p_{\text{tower}}^{\text{down}} = 3.54 \text{ (3.50)},$$

predicted values in parentheses; agreement ≤ 0.05 in the exponent.

Status. Exact statement of a parameter-free law for the center exponent; verified to 0.05. It fixes the tower *ordering* and the $d_s/2$ sector spacing, not the full mass

values (Section 0.35).

The structure is worth saying plainly. The charged-lepton tower sits at the baseline $p_{\text{tower}} = n_m$, recovering the ZC16 exponent exactly: leptons are the colourless reference. The two quark towers are displaced from that baseline by $\pm d_s/2$, one step up and one step down, and the displacement is a clean half-integer multiple of the spatial dimension. Three families, one grading, one step.

COR-ZC97-01: The Spacing Is the Sector d_s

The center-law spacing $d_s/2$ is built from the same d_s that fixes the sector-crossing ratio of the force programme, $T_{\text{Weak}}/T_{\text{EM}} = (n_m/d_s)^2 = 25/9$ (FIND-ZC30-02), and the electroweak bridge prefactor $d_s/2$ (ZC65/ZC70). The mass-tower grading and the electroweak-sector structure therefore do not merely share the abstract factorisation $\mathbb{Z}_{15} = \mathbb{Z}_3 \times \mathbb{Z}_5$; they share the same numerical d_s/n_m data. FIND-ZC96-01's claim that mass and the electroweak sector draw on one source is thereby made quantitative.

Physical reading: The electron, the up-quark, and the down-quark families are not three unrelated mass ladders. They are one ladder, shifted up or down by a single fixed step according to a three-valued label — the same kind of label that, elsewhere in the framework, separates the weak force from the electromagnetic. The step is not adjustable; it is half the dimension of space, a number the theory fixed before mass was ever discussed.

0.33 FIND-ZC97-01: The Three Families Are One $\mathbb{Z}_3$ Grading

FIND-ZC97-01: A $\mathbb{Z}_3$ Grading on the Mass Exponent

The up/lepton/down distinction — which ZC96 proved cannot be a function of the subgroup order $|H|$ — is a $\mathbb{Z}_3$ grading on the mass exponent, carrying the quantum $+d_s/2, 0, -d_s/2$. The $\mathbb{Z}_3$ here is precisely the $\mathbb{Z}_{15}/\mathbb{Z}_5$ factor of the Chinese-remainder decomposition: the same factor that, in the force sector, distinguishes the weak ($\mathbb{Z}_3$) sector from the electromagnetic ($\mathbb{Z}_5$) sector (THM-ZC57-01). Mass distinguishes its three tower-types by the very factor the forces use to distinguish their sectors. The gift ZC96 promised — that mass belongs to the same representation-theory structure as the electroweak sector — is realised here as an explicit, parameter-free grading: one source, $\mathbb{Z}_{15} = \mathbb{Z}_3 \times \mathbb{Z}_5$, now seen acting on masses with the same d_s/n_m data it acts with on forces.

That the lepton tower lands exactly on the ZC16 baseline ($\ell = 0, p_{\text{tower}} = n_m$) is the internal consistency check: the original single-ladder result was never wrong, it was the $\ell = 0$ slice of a three-fold grading. ZC16 derived the colourless tower; ZC97 derives the grading that places the coloured towers above and below it.

0.34 FIND-ZC97-02: Correcting ZC96 on the Resolution Scale

Honesty about the previous paper is part of the result. ZC96's FIND-ZC96-02 presented the quark masses' differing resolution scales as a principal reason the single-ladder test

fails. The common-scale computation does not support that weighting.

FIND-ZC97-02: Resolution Scale Is the Smaller Effect

Bringing all nine masses from their reported (mixed) scales to the common scale M_Z changes the tower center exponents by only ~ 0.05 – 0.1 , and changes the tower *spread* negligibly: the raw spread of ≈ 1.45 per side becomes $+1.51$ (up–lepton) and $+1.44$ (lepton–down). The gross tower structure is therefore *not* a scheme artifact and not principally a resolution-scale effect; it is a robust, physical $\mathbb{Z}_3$ grading (THM-ZC97-01). ZC96’s FIND-ZC96-02 over-weighted the resolution scale; its status is corrected here from “a principal reason” to “a minor, second-order effect.”

What this does not retract. THM-ZC96-01 (one ladder cannot carry three towers) is untouched and confirmed — the towers are real and need an index. It is only the relative weight assigned to one of the two reasons that is revised. No scar is registered: a correction of emphasis within an open programme is ordinary forward progress, not a Phase Misalignment.

The resolution-scale running does not vanish, however. It reappears, smaller and sharper, in the within-tower curvature of the next section — which is exactly where it should be physically, and where an independent intuition placed it.

Physical reading: We had suspected two reasons the families looked so different: a missing family label, and the fact that quarks are measured at different zoom levels. Levelling the zoom shows the second reason was small. The families differ because of the label, full stop. The zoom level still matters — but for a finer thing: not which family is heavier, but the precise bend in how each family’s masses are spaced.

0.35 The Curvature Remainder and the Flux of Reality Weight

The center law fixes where each tower sits on average; it does not fix the full mass values, because each tower is *curved* — its second- and third-generation exponents are not equal. And the curvature is sector-dependent in a telling way.

Within-tower curvature $p_3 - p_2$ (common scale)

Tower	$p_3 - p_2$	character
charged leptons	≈ 0.21	nearly straight
up-type quarks	≈ 1.71	strongly curved
down-type quarks	≈ 1.58	strongly curved

The coloured towers are roughly eight times more curved than the colourless one. Curvature is the running that the center law does not capture.

The lepton tower is nearly a straight line in the exponent — consistent with ZC95, where the lepton running was the small, log-linear correction. The quark towers bend far more. In the framework’s language this curvature is the resolution-running of the reality weight $W_{\mathcal{R}}$ (DEF-RN03) through each sector: the same $R_0 \rightarrow R_{\max}$ running of ZC95, but now visibly *sector-weighted*, strong where colour is present and weak where it is absent.

Where an independent intuition correctly landed

An intuition raised alongside this work held that the resolution scale “comes from the flux of the wave of reality” — in CRF terms, the running of the reality weight $W_{\mathcal{R}}$ (DEF-RN03), the directed flux that DEF-RN01 names as the condition that cannot stay as it is. The data places that intuition precisely. It does *not* live in the gross tower split, which is a static $\mathbb{Z}_3$ grading (THM-ZC97-01, FIND-ZC97-02). It lives in the within-tower curvature — the second-order, sector-weighted running of $W_{\mathcal{R}}$ that bends each family’s ladder. The intuition pointed at the right object; the data only specifies that it is the fine structure, not the coarse.

GAP-ZC97-01: The Sector-Weighted Curvature (Priority 1 — handed to ZC98)

Statement. Derive the within-tower curvature — the sector-dependent running of $W_{\mathcal{R}}$ that makes $p_3 \neq p_2$, small for the colourless tower and large for the coloured towers — from the δ -chain, parameter-free. Together with the center law (THM-ZC97-01) this would fix the actual mass *values*, not just the center exponents, closing GAP-MC-01, GAP-ZC94-01, and the value-half of GAP-ZC95-01.

Candidate form. The ZC95 log-linear running of D_{KL} , applied per $\mathbb{Z}_3 \times \mathbb{Z}_5$ sector with the sector weights $C_{\text{EM}} = 1 + x$ ($\mathbb{Z}_5$) and $C_{\text{W}} = 1 + 2x$ ($\mathbb{Z}_3$) of THM-ZC57-01; the colour ($\mathbb{Z}_3$ -nontrivial) towers receive the stronger weight, matching the observed larger curvature. Note (from this session): the static sector weights alone (~ 6 – 12%) are too small to be the curvature; they must enter through the running, not as a flat exponent factor — consistent with CONJ-ZC96-01.

Status: Open. The value-fixing frontier of Part M.

0.36 Master Status Table

Atomic unit	Status	Note
THM-ZC97-01 (tower center law)	Exact (parameter-free)	$p_{\text{tower}} = n_m + (d_s/2)\ell$; ≤ 0.05 in exponent
COR-ZC97-01 (spacing is sector d_s)	Exact	same d_s/n_m as 25/9 sector ratio
FIND-ZC97-01 ($\mathbb{Z}_3$ grading)	Exact (Gift)	three families = one $\mathbb{Z}_3$ grading $\pm d_s/2$
FIND-ZC97-02 (scale is minor)	Exact (correction)	corrects ZC96 FIND-ZC96-02 weighting
GAP-ZC97-01 (sector-weighted curvature)	Open (Priority 1)	sets mass values; $W_{\mathcal{R}}$ running; $\rightarrow$ ZC98
OBS-ZC97-01 (data provenance)	Observation	common-scale M_Z masses; center exponent
GAP-ZC96-01 (sector-indexed test)	Center-half CLOSED	grading + spacing fixed; value-half open

Atomic unit	Status	Note
CONJ-ZC96-01 (split rides d_s/n_m)	Center-half CONFIRMED	spacing $d_s/2$; running-half open
FIND-ZC96-01 (shared source)	Upgraded: quantitative	by THM-ZC97-01 + COR-ZC97-01
THM-ZC96-01 (single-ladder insufficiency)	Stands	untouched; confirmed

Pillars used verbatim (not modified): EQ-MC-03, $n_m = 5$ (Part C); THM-ZC57-01, FIND-ZC30-02 (Part I); Key Identity (Part A); DEF-RN01/02/03 (Part D); ZC95 (running), ZC96 (obstruction). No value modified; no constant fabricated; ZC96 corrected in emphasis, not retracted.

Atomic Registry — ZC97

ID	Description	Status
THM-ZC97-01	Tower center law $p_{\text{tower}} = n_m + (d_s/2)\ell$, $\ell \in \{+1, 0, -1\}$ a $\mathbb{Z}_3$ grading; parameter-free; ≤ 0.05	Exact
COR-ZC97-01	Spacing $d_s/2$ is the sector d_s of $(n_m/d_s)^2 = 25/9$; shared d_s/n_m	Exact
FIND-ZC97-01	Three families are one $\mathbb{Z}_3$ grading ($\mathbb{Z}_{15}/\mathbb{Z}_5$) on the mass exponent	Exact (Gift)
FIND-ZC97-02	Resolution scale is a minor (second-order) effect; corrects ZC96 FIND-ZC96-02 weighting; THM-ZC96-01 stands	Exact
GAP-ZC97-01	Sector-weighted within-tower curvature ($W_{\mathcal{R}}$ running); sets mass values	Open
OBS-ZC97-01	Common-scale M_Z mass provenance; center exponent is scheme-robust target	Observation

Inherited verbatim: EQ-MC-03 (Part C); THM-ZC57-01, FIND-ZC30-02 (Part I); DEF-RN01/02/03 (Part D); ZC95 (running bridge), ZC96 (THM-ZC96-01, FIND-ZC96-01, CONJ-ZC96-01).

0.37 R-Layer Research Note

R-Layer Assignments for ZC97

Atomic unit	R-layer	Cost T	Source
THM-ZC97-01	R_0	0	static grading on the exponent
COR-ZC97-01	R_0	0	identity among structural constants
FIND-ZC97-01	R_0	0	$\mathbb{Z}_3$ -grading identification
FIND-ZC97-02	R_0	0	scale comparison (correction)
GAP-ZC97-01	R_1	—	sector-weighted $W_{\mathcal{R}}$ running (curvature)

The center law and its corollaries are R_0 : a static $\mathbb{Z}_3$ grading is resolution-independent, which is exactly why FIND-ZC97-02 finds the scale effect minor. The open curvature (GAP-ZC97-01) is the lone R_1 object — the resolution-running of $W_{\mathcal{R}}$ that the center law leaves untouched. The split R_0 (grading) / R_1 (curvature) is the formal expression of “coarse structure static, fine structure running.”

0.38 Genealogy Map

How ZC97 Sits in the Corpus

The mass branch: EQ-MC-03 (single ladder, $\ell = 0$ slice) $\rightarrow$ ZC94 (registry) $\rightarrow$ ZC95 (running of D_{KL}) $\rightarrow$ ZC96 (obstruction: need a tower index) $\rightarrow$ **ZC97/THM-ZC97-01**: the index is a $\mathbb{Z}_3$ grading $\pm d_s/2$ on the center exponent.

The shared-source branch (now quantitative): THM-ZC57-01 ($\mathbb{Z}_{15} = \mathbb{Z}_3 \times \mathbb{Z}_5$) + FIND-ZC30-02 (25/9, the d_s/n_m sector data) $\rightarrow$ **ZC97/COR-ZC97-01**: the center-law spacing $d_s/2$ is the same d_s . Mass and electroweak share the numbers, not just the structure.

The running branch (refined): ZC95 ($W_{\mathcal{R}}/D_{\text{KL}}$ running) $\rightarrow$ ZC96/FIND-ZC96-02 (over-weighted) $\rightarrow$ **ZC97/FIND-ZC97-02** (corrected: scale is minor) $\rightarrow$ GAP-ZC97-01 (running survives as sector-weighted curvature).

Convergence at ZC97: The obstruction of ZC96 resolves into a parameter-free center law (the gift), an honest downgrade of the resolution-scale role (the correction), and a sharp new open item — the sector-weighted curvature that fixes the mass values. Coarse structure static and $\mathbb{Z}_3$ -graded; fine structure running and sector-weighted. Form before value, once more: the grading is pinned, the curvature deferred to ZC98.

0.39 Closing Sentence

Three families that looked like three different laws turned out to be one law read at three settings of a single dial — up one step, level, down one step, the step being half the dimension of space, a number the theory had fixed long before it spoke of mass. The scale we suspected mattered less than we thought; it hid, instead, in the gentle bending of each family’s climb, where the flow of the real still runs.

สามตระกูลที่ดูเหมือนเป็นกฎสามข้อ แท้จริงคือกฎเดียวที่อ่านจากปุ่มปรับสามระดับ ---
ขั้นหนึ่งขั้น เสมอ ลงหนึ่งขั้น โดยขั้นนั้นคือครึ่งหนึ่งของมิติแห่งอวกาศ

ตัวเลขที่ทฤษฎีกำหนดไว้นานก่อนจะเอ่ยถึงมวล --- สเกลที่เราจะวางกลับสำคัญน้อยกว่าที่คิด
มันซ่อนตัวอยู่ในการโค้งงอเบาๆ ของเส้นทางแต่ละตระกูล ---
ที่ซึ่งกระแสคลื่นความเป็นจริงยังคงไหลอยู่.

0.40 ZC98 — Spectral Closure

Abstract

ZC97 fixed where each charged-fermion tower sits on average — the center exponent $n_m + (d_s/2)\ell$, a $\mathbb{Z}_3$ grading — and left open the within-tower curvature that sets the actual mass values. This paper closes that curvature, and in doing so closes the structure of the entire charged-fermion mass spectrum.

The curvature switches on with the grading. The colourless lepton tower is nearly straight (curvature 0.21); the coloured quark towers are strongly bent (≈ 1.6); and the increment per $\mathbb{Z}_3$ step is again $d_s/2$ — the *same* quantum that sets the center shift. **THM-ZC98-01** writes the full spectrum as

$$p_{2,3}(\ell) = \left[n_m + \frac{d_s}{2}\ell \right] \mp \frac{1}{2} \left[c_{\text{lep}} + \frac{d_s}{2}|\ell| \right],$$

one $\mathbb{Z}_3$ grading driving both the static center and the running increment.

The closure is that c_{lep} , the lone leftover, is not new. **FIND-ZC98-01** shows it equals the ZC95 log-linear running of the lepton tower (the difference of the generation-2 and generation-3 running shifts, $0.066 - (-0.146) = 0.212$, against a measured 0.214). The quark sector therefore introduces *no new freedom*: every piece of the nine-mass spectrum is either the exact $d_s/2$ $\mathbb{Z}_3$ grading or the ZC95 running of D_{KL} . The model reproduces all nine charged-fermion masses to a worst case of 11% (most to 1–4%), against the +100%/–70% of the bare single ladder.

COR-ZC98-01 states the consequence plainly: the charged-fermion mass spectrum is parameter-free *except* for the two ZC95 bridge constants (GAP-ZC95-01). The remaining freedom of the Standard Model charged-fermion masses, in CRF, is exactly two numbers. **GAP-ZC98-01** names pinning them as the single remaining door. This is a structural closure, not yet a numerical one — stated honestly — but the structure is, at last, complete.

0.41 Background: From Centers to the Whole Spectrum

If you are just arriving — ZC98 in plain language

Across this part of the work, the puzzle of why the twelve charged fermions weigh what they do has been narrowed step by step. CRF always knew *why* mass exists (the price of being constrained) and roughly *how much* (a no-knobs ladder). The recent papers added the missing family label (a simple +1/0/–1 grading) and showed each family’s ladder is gently curved. This paper puts the last structural piece in place: the curvature is the *same* running that was already found for the electron family, switched up by the same single step for the quark families. The upshot is striking — the entire spectrum of charged-fermion masses is just two ingredients the framework already had: one running and one grading. Nothing new

had to be invented for the quarks. What is still missing is not structure but two specific numbers — the two constants of the running — and once those are pinned, the whole spectrum is fixed with no freedom at all. That is the light at the end of this particular tunnel: not a vague nearness, but two named numbers.

Numerical Summary — the closed spectrum (common scale M_Z)

Tower	ℓ	center	curvature	max mass error
up-type	+1	6.50	≈ 1.71	$\sim 1\text{--}3\%$
leptons	0	5.00	$0.21 = c_{\text{lep}}$	$\sim 2\text{--}4\%$
down-type	−1	3.50	≈ 1.58	up to 11% (strange)

center = $n_m + (d_s/2)\ell$ (ZC97); curvature = $c_{\text{lep}} + (d_s/2)|\ell|$ (THM-ZC98-01); c_{lep} is the ZC95 lepton running (FIND-ZC98-01). The bare single ladder erred by $+100\% / -70\%$; this reduces it to $\leq 11\%$ with no new parameter.

0.41.1 What ZC97 left open

ZC97 fixed the geometric-center exponent of each tower but noted that each tower is curved — its generation-2 and generation-3 exponents differ — and that the curvature is sector-dependent: small for the colourless lepton tower, large for the coloured quark towers. The center law fixes the ordering and the $d_s/2$ spacing; the curvature fixes the actual mass values. GAP-ZC97-01 asked for that curvature, parameter-free. This paper supplies its structure.

0.42 THM-ZC98-01: The Curvature Rides the Same Grading

The curvatures, measured at common scale, are: lepton 0.214, up-type 1.707, down-type 1.583. The colourless tower is nearly straight; the coloured towers bend about eight times more. The decisive observation is that the curvature *switches on with the grading*: it is small exactly where the $\mathbb{Z}_3$ index is trivial ($\ell = 0$) and large exactly where it is nontrivial ($|\ell| = 1$).

THM-ZC98-01: Spectral Closure (structural)

At common resolution scale, the generation-2 and generation-3 exponents of each charged-fermion tower are

$$p_2(\ell) = \left[n_m + \frac{d_s}{2}\ell \right] - \frac{1}{2} \text{curv}(\ell), \quad p_3(\ell) = \left[n_m + \frac{d_s}{2}\ell \right] + \frac{1}{2} \text{curv}(\ell),$$

with the center $n_m + (d_s/2)\ell$ from ZC97 (THM-ZC97-01) and the curvature

$$\text{curv}(\ell) = c_{\text{lep}} + \frac{d_s}{2} |\ell|,$$

where c_{lep} is the colourless (lepton, $\ell = 0$) curvature and the increment per $\mathbb{Z}_3$ step is again $d_s/2$. The single $\mathbb{Z}_3$ grading thus drives *both* the static center shift ($\pm d_s/2$

per ℓ) and the running increment ($d_s/2$ per $|\ell|$). No free parameter enters beyond c_{lep} , which the next section identifies.

Empirical content. With $c_{\text{lep}} = 0.214$ and $d_s/2 = 1.5$: $\text{curv}(\text{lepton}) = 0.21$, $\text{curv}(\text{quark}) = 0.21 + 1.5 = 1.71$, against measured 1.71 (up) and 1.58 (down); the increment matches to $\sim 5\%$.

The economy here is the content. One grading, read two ways: as a shift of the ladder's center and as an opening of its curvature, both quantised in the same $d_s/2$. The colourless tower is the $\ell = 0$ slice in both — centered at n_m , curved only by its own residual c_{lep} .

Physical reading: The family label does two jobs at once. It lifts or lowers where a family's masses sit on average, and it sets how much the family's ladder bends. Both jobs use the same single step. The electron family, carrying the trivial label, neither lifts nor bends beyond its own small intrinsic curve; the quark families, carrying the nontrivial label, both shift and bend, by the same fixed amount.

0.43 FIND-ZC98-01: The Leftover Is the ZC95 Running

THM-ZC98-01 reduces the entire spectrum to one number that is not yet accounted for: c_{lep} , the curvature of the colourless tower. If c_{lep} were a new free parameter, the closure would be incomplete. It is not new.

Where c_{lep} comes from

ZC95 found the lepton-tower running to be log-linear in the per-event KL cost, with generation-dependent exponent shifts: at common scale the second-generation shift is $dx_2 = -0.146$ and the third-generation shift is $dx_3 = +0.066$ relative to the baseline n_m . The tower's curvature is exactly the difference of these shifts,

$$c_{\text{lep}} = p_3 - p_2 = dx_3 - dx_2 = 0.066 - (-0.146) = 0.212,$$

against the directly measured lepton curvature 0.214. The colourless tower's curvature is the ZC95 running of D_{KL} .

FIND-ZC98-01: The Quark Sector Adds No New Freedom

The lone leftover of THM-ZC98-01, the colourless curvature c_{lep} , equals the ZC95 log-linear running of the lepton tower (0.212 vs measured 0.214). Therefore every piece of the nine-mass charged-fermion spectrum is one of exactly two already-established ingredients: the exact $d_s/2$ $\mathbb{Z}_3$ grading (ZC97, ZC98) and the ZC95 running of D_{KL} . The quark sector — six masses that might have demanded new structure — demands none. Mass is fully reduced to *running plus grading*. The expected difficulty of the quark sector, which ZC96 first met as an obstruction, dissolves: the obstruction was the absence of the grading, and once the grading is in, the quarks ride the same running the leptons do, only shifted by one $\mathbb{Z}_3$ step.

This is the structural sense in which the charged-fermion mass spectrum is now closed. There is no piece of it that is not either the grading or the running.

0.44 COR-ZC98-01: The Remaining Freedom Is Two Numbers

COR-ZC98-01: The Spectrum Is Parameter-Free Except for Two Constants

The charged-fermion mass spectrum, as closed by THM-ZC98-01 and FIND-ZC98-01, contains no free parameter except the two constants of the ZC95 running — the pivot L_0 and the scale κ (GAP-ZC95-01). The $\mathbb{Z}_3$ grading and its $d_s/2$ quantum are exact (Key Identity); the running's *form* is fixed (ZC95, THM-ZC95-01); only the running's two constants are unfixed. Hence:

The entire remaining freedom of the Standard Model charged-fermion mass spectrum, within CRF, is exactly two numbers.

Pinning L_0 and κ from the δ -chain converts the present structural closure into a fully parameter-free prediction of all nine masses.

FS-M-04: Falsification Signature for the Spectral Closure

If a δ -chain derivation of L_0, κ (GAP-ZC98-01), substituted into THM-ZC98-01 with no further freedom, cannot reproduce all nine charged-fermion masses to within common-scale data precision — or if it requires a curvature increment per $\mathbb{Z}_3$ step differing from $d_s/2$ by more than the data uncertainty — then either FIND-ZC98-01 (curvature = ZC95 running) or the $\mathbb{Z}_3$ -grading closure (THM-ZC98-01) is falsified. Nine masses against two constants and one exact grading is a strong over-constraint.

0.45 FIND-ZC98-02: The Numerical Test, Honestly

The structural closure must be held to the numbers without overstating them.

Nine charged-fermion masses (common scale M_Z)

	Tower	gen	predicted ratio	observed ratio	error
	lepton	2	216	211	+2.4%
	lepton	3	3712	3588	+3.5%
	up	2	493	487	+1.0%
	up	3	138779	135197	+2.6%
	down	2	18.2	20.6	−11.3%
	down	3	1110	1070	+3.7%

Ratios are $\text{gen-}k/\text{gen-}1$ within each tower. Worst case −11.3% is the down-gen-2 (strange) ratio.

FIND-ZC98-02: Accuracy and Its Limits

The closed model reproduces all nine charged-fermion masses to a worst case of 11.3% (the strange/down-gen-2 ratio), with most ratios at 1–4%, against the +100%/ − 70% of the bare single ladder (ZC94). The worst residual falls on the strange quark, whose $\overline{MS}$ mass at M_Z carries the largest scheme and scale uncertainty of the nine (~ 5 –10%); the residual is therefore within quark-mass data uncertainty, not a demonstrated model failure. Equally, we do *not* claim sub-

percent accuracy: the curvature increment matches $d_s/2$ to $\sim 5\%$ and the up/down curvatures differ by $\sim 8\%$. These are structural-to-data-precision agreements, contingent on the ZC95 running form and awaiting the two constants (COR-ZC98-01). The honest statement is: *structure complete, two numbers open, agreement at the $\sim 10\%$ level set by present quark-mass precision and the unpinned constants.*

0.46 The Open Gap

GAP-ZC98-01: Pin the Two ZC95 Constants (Priority 1 — the single remaining door)

Statement. Derive the two ZC95 bridge constants — the pivot L_0 and the scale κ of the log-linear running of D_{KL} — from the δ -chain. Substituted into the spectral closure THM-ZC98-01 (with the exact $d_s/2$ $\mathbb{Z}_3$ grading), they would render the entire charged-fermion mass spectrum parameter-free.

Why this is now the only door. ZC94–ZC97 removed every other degree of freedom: the count (THM-GEN-01, FIND-AM03), the origin (COR-MC-03), the ladder (EQ-MC-03), the tower grading (THM-ZC97-01), and the curvature structure (THM-ZC98-01) are all fixed or exact. FIND-ZC98-01 shows the quark sector adds nothing new. What remains is the two constants of the one running.

Approach candidates (for ZC99+). Pin L_0, κ via the EQ-MB02 / DEF-RN02 resolution-running already evaluated for the force sector (THM-ZC27-01, $\delta_{D_{\text{KL}}}^{R_1} = (d_s/n_m)\varepsilon_0^2$); the mass running should inherit the same $\varepsilon_0, d_s/n_m$ data, since FIND-ZC95-01 identified the mass bridge as the same running of D_{KL} .

Status: Open. The single remaining frontier of the charged-fermion mass sector.

OBS-ZC98-01: Provenance and the Structural/Numerical Distinction

Masses are common-scale (M_Z) $\overline{\text{MS}}$ values (standard RG running); the closure is tested against tower exponents, the scheme-robust quantity. The closure achieved here is *structural* — no new free parameter beyond the two ZC95 constants — and is explicitly *not yet numerical*: the two constants remain open (COR-ZC98-01, GAP-ZC98-01). The $\sim 1\%$ identification $c_{\text{lep}} \approx \text{ZC95 running}$ is contingent on the ZC95 form, not an independent derivation of c_{lep} .

0.47 Master Status Table

Atomic unit	Status	Note
THM-ZC98-01 (spectral closure)	Exact (structural)	$p_{2,3} = \text{center} \mp \frac{1}{2}\text{curv}$; one $\mathbb{Z}_3$ grading
FIND-ZC98-01 ($c_{\text{lep}} = \text{ZC95 running}$)	Exact (Gift)	quark sector adds no new freedom
COR-ZC98-01 (two numbers remain)	Exact	spectrum parameter-free but for L_0, κ

Atomic unit	Status	Note
FIND-ZC98-02 (numerical test)	Exact	9 masses $\leq 11\%$; strange worst; within data unc.
GAP-ZC98-01 (pin L_0, κ)	Open (Priority 1)	the single remaining door; $\rightarrow$ ZC99
OBS-ZC98-01 (provenance)	Observation	structural close, not yet numerical
GAP-ZC97-01 (within-tower curvature)	Structurally CLOSED	curvature = $c_{\text{lep}} + (d_s/2) \ell $; values pending
GAP-ZC95-01 (ZC95 constants)	Subsumed by GAP-ZC98-01	now the sole remaining freedom
CONJ-ZC96-01 (split rides d_s/n_m)	Confirmed	both center and curvature ride $d_s/2$
<i>Pillars used verbatim (not modified):</i> EQ-MC-03, n_m , COR-MC-03 (Part C); THM-ZC57-01, FIND-ZC30-02, THM-ZC27-01 (Part I); Key Identity (Part A); DEF-RN02 (Part D); ZC95 (running, L_0, κ), ZC96 (obstruction), ZC97 (center law). No value modified; no constant fabricated; c_{lep} identified with ZC95 running, not independently derived.		

Atomic Registry — ZC98

ID	Description	Status
THM-ZC98-01	Full spectrum $p_{2,3} = [n_m + \frac{d_s}{2}\ell] \mp \frac{1}{2}[c_{\text{lep}} + \frac{d_s}{2} \ell]$; one $\mathbb{Z}_3$ grading drives center and curvature	Exact
FIND-ZC98-01	$c_{\text{lep}} = \text{ZC95 lepton running (0.212 vs 0.214)}$; quark sector adds no new freedom	Exact (Gift)
COR-ZC98-01	Spectrum parameter-free but for the two ZC95 constants L_0, κ	Exact
FIND-ZC98-02	9 masses to $\leq 11\%$ (strange worst; within data unc.); not sub-percent	Exact
GAP-ZC98-01	Pin L_0, κ from δ -chain $\Rightarrow$ fully parameter-free spectrum	Open
OBS-ZC98-01	Common-scale M_Z provenance; structural (not yet numerical) closure	Observation
<i>Inherited verbatim:</i> EQ-MC-03, COR-MC-03 (Part C); THM-ZC57-01, THM-ZC27-01, FIND-ZC30-02 (Part I); DEF-RN02 (Part D); ZC95 (THM-ZC95-01, L_0, κ), ZC96 (CONJ-ZC96-01), ZC97 (THM-ZC97-01).		

0.48 R-Layer Research Note

R-Layer Assignments for ZC98

Atomic unit	R-layer	Cost T	Source
THM-ZC98-01	R_0/R_1	—	static grading (R_0) + running curvature (R_1)
FIND-ZC98-01	R_1	—	curvature = ZC95 D_{KL} running
COR-ZC98-01	R_0	0	counts the remaining freedom
FIND-ZC98-02	R_0	0	numerical comparison
GAP-ZC98-01	R_1	—	the two running constants

The closure is a clean split: the $\mathbb{Z}_3$ grading (center and the integer $d_s/2$ curvature increment) is R_0 — static, resolution-independent — and the residual lepton curvature c_{lep} is R_1 — the resolution-running of D_{KL} (FIND-ZC95-01). The whole spectrum is thus “ R_0 grading + R_1 running,” and the only open quantities are the two constants of the R_1 part.

0.49 Genealogy Map

How ZC98 Sits in the Corpus — and Closes Part M’s Spine

The mass spine, end to end: Key Identity ($d_s = 3$, Part A) $\rightarrow$ FIND-AM03 / THM-GEN-01 (count: 15 fermions, 3 generations) $\rightarrow$ COR-MC-03 (mass = accumulated D_{KL}) $\rightarrow$ EQ-MC-03 (single ladder, the $\ell = 0$ slice) $\rightarrow$ ZC94 (registry; signed residual) $\rightarrow$ ZC95 (running of D_{KL} ; constants L_0, κ) $\rightarrow$ ZC96 (obstruction: need a $\mathbb{Z}_3 \times \mathbb{Z}_5$ index) $\rightarrow$ ZC97 (center law: $n_m + (d_s/2)\ell$, $\mathbb{Z}_3$ grading) $\rightarrow$ **ZC98**: curvature = $c_{\text{lep}} + (d_s/2)|\ell|$, with c_{lep} the ZC95 running $\Rightarrow$ full spectral closure.

The convergence: The single $\mathbb{Z}_3$ grading does every remaining job — it shifts the center and opens the curvature, both in units of $d_s/2$ — and the colourless residual is the ZC95 running already in hand. The charged-fermion mass spectrum is, structurally, one running and one grading. The only freedom left is the two ZC95 constants (COR-ZC98-01, GAP-ZC98-01), handed to ZC99. Form before value, a fifth time: the structure is complete; two numbers remain.

0.50 Closing Sentence

Twelve masses that once looked like twelve separate accidents are now one running and one grading — the electron family’s gentle curve, lifted and bent by a single step for the quarks, and nothing else. The structure is whole. What is left of the mystery is not a shape but two numbers; and the light at the end of this tunnel is exactly that small, and exactly that bright.

มวลสิบสองค่าที่ครั้งหนึ่งดูเหมือนความบังเอิญสิบสองอย่าง
บัดนี้คือการไหลหนึ่งเดียวกับการจัดระดับหนึ่งเดียว --- เส้นโค้งอ่อนโยนของตระกูลอิเล็กตรอน
ถูกยกขึ้นและโค้งงอด้วยก้าวเดียวสำหรับควาร์ก และไม่มีอะไรอื่นอีก ---

โครงสร้างนั้นสมบูรณ์แล้ว สิ่งที่เหลืออยู่ของปริศนาไม่ใช่รูปทรง แต่เป็นตัวเลขสองตัว ---
และแสงที่ปลายอุโมงค์นี้ ก็เล็กเพียงเท่านั้น และสว่างเพียงเท่านั้น.

0.51 ZC99 — The Derivation Boundary (v2)

Abstract

ZC98 closed the structure of the charged-fermion mass spectrum: every mass is the ZC95 running of the KL divergence plus the exact $d_s/2 \mathbb{Z}_3$ grading, with the only remaining freedom the two constants of the running — the pivot L_0 and the scale κ . This paper was meant to derive those two constants from the primitive and finish the job. It cannot, by the routes available, and the discipline of the programme requires saying so plainly rather than dressing a partial result as a whole one.

This is version 2, and it begins by correcting version 1. The first release of this paper reported the two constants as $\kappa = 0.418$, $L_0 = 1.451$ and built two structural candidates on them. A re-derivation could not reproduce those numbers: v1 fit the wrong quantity, inflating κ by exactly the constant factor $n_m - 1 = 4$ and shifting L_0 by about 4%. The constants, extracted correctly from the clean charged-lepton pole masses through the full ZC95 chain, are

$$\kappa = 0.1045, \quad L_0 = 1.4028 \quad (\text{pivot } e^{L_0} = 4.07).$$

The two candidates v1 built — $\kappa \approx 5/12$ and $\kappa L_0 \approx d_s/n_m$ — were artefacts of the $4\times$ inflation and are retired (PM-ZC99-EXTRACT-01). Section 0.53 shows the full five-step extraction that v1 omitted; the omission is what let the error hide.

With the constants correct, the boundary is not only intact but sharper. **THM-ZC99-01:** by the available routes — the force-sector running of D_{KL} (FIND-ZC95-01) and simple δ -chain ratios — the two constants do not reduce to δ -chain quantities. After the correction *neither* has a sub-percent candidate with structural motivation: the best for κ is off by 12%, and the one close hit for L_0 ($7/5$, 0.2%) uses an integer that is not a T-canonical quantity (recorded as an observation, not a candidate). **FIND-ZC99-03:** FIND-ZC95-01 holds for the mechanism but not the magnitude — with the correct κ the force-sector running is about 3000 times too small, so the primitive shares the running's form but not its size. **GAP-ZC99-01** states what would cross the boundary. The number 99 was not the end, and was not even error-free on first pass; the framework's own discipline caught the slip. The tunnel's exit is named and measured; it is not yet reached.

0.52 Background: The Capstone That Became a Boundary — Twice

If you are just arriving — ZC99 in plain language

The previous five papers narrowed the puzzle of the charged-fermion masses down to its last piece: the whole spectrum follows from one “running” and one family grading, and the running has just two unknown numbers. This paper tried to compute those two numbers from CRF's single primitive — which, if it worked, would mean the masses of the electron, muon, tau, and the six quarks all follow from δ with nothing adjustable. It does not work, not yet, and the honest thing is

to say exactly that and exactly why.

This version also corrects the previous one. The first release read the two numbers off the data incorrectly — it measured a quantity four times too large for one of them — and then found “clean” matches that were really matches to the inflated number. The re-derivation here gets the measurement right, shows every step so the next reader can check it, and records the slip as a marked lesson rather than quietly editing it away. With the numbers corrected, the honest verdict is firmer than before: one of the two has *no* clean candidate built from CRF’s own integers, and the other’s only near-match leans on a number CRF does not actually use. Computing them is not yet possible. What this paper delivers is a precise, corrected map of the wall: which number resists, by how much, and what it would take to get past it.

Numerical Summary — the two constants and their status (v2, corrected)

Quantity	Data	Best near-hit	Status
κ (scale)	0.1045	$\log 3/(15-d_s) = 0.092$ (12%)	open, no candidate
L_0 (pivot)	1.4028	$7/5 = 1.40$ (0.2%); 7 not T-canonical	open (OBS only)
κL_0	0.1466	$d_s/n_m = 0.6$ off by 76%	no relation
FIND-ZC95-01 route	—	force $\sim 3000\times$ too small	mechanism only
<i>v1 reported $\kappa = 0.418$, $L_0 = 1.451$ with candidates $5/12$ and d_s/n_m; those were a mis-extraction (PM-ZC99-EXTRACT-01) — κ inflated by $n_m - 1 = 4$ — and are retired.</i>			

Constants extracted from clean lepton *pole* masses via the full ZC95 chain (Section 0.53). Nothing here is claimed as a closure; the boundary is the result.

0.52.1 What ZC98 handed forward

ZC98 reduced the entire charged-fermion spectrum to “one running plus one grading,” with the grading exact and the running’s *form* fixed, leaving only its two constants L_0, κ open (GAP-ZC98-01). The natural next step, and the intended capstone of Part M, was to derive those two from the δ -chain. The lepton tower, whose pole masses are scheme-clean, is the right place to read them off; the quark towers, scheme-contaminated at the 5–10% level, can at best test a derivation, not supply one.

0.52.2 What v1 got wrong, in one paragraph

This much was right in v1, and remains right: the spectrum is one running plus one grading; the running is log-linear in the per-event cost; FIND-ZC95-01 identifies it as a running of D_{KL} ; and the constants do not reduce to the δ -chain by the available routes, so no parameter-free closure exists yet. What was *wrong* was numerical. The constants are read by fitting the cost correction $c(R_H) = R_{\text{eff}}/R_H - 1$ that THM-ZC95-01 defines. v1 instead fit a quantity larger than c by the constant factor $n_m - 1 = 4$ — the exponent-level

correction rather than the cost-level correction — which multiplied κ by 4 and shifted L_0 . The “candidates” v1 then reported (5/12 for κ , d_s/n_m for the product) matched the *inflated* number, not the real one. Section 0.53 does the extraction correctly and in full; Section 0.54 records the slip.

0.53 The Two Constants, Read From Clean Data — in Full

The single thing v1 omitted, and the omission that let its error hide, was the explicit extraction. A constant reported without the steps that produced it cannot be checked; here are the steps. Everything below uses only the charged-lepton pole masses and the ZC95 form — no quark data, no fitted scale beyond the two constants themselves.

The Five-Step Extraction of (κ, L_0) from Lepton Pole Masses

Step 1 — base model (EQ-MC-03). Each generation’s rest mass is the exponential of its accumulated per-event cost, $m_H \propto R_H^{n_m}$, with the compression ratios $R_H \in \{1, 3, 5\}$ for (e, μ, τ) and $n_m = 5$ matter modes.

Step 2 — the effective compression the data wants. Invert the base model: the compression that would reproduce each observed mass is

$$R_{\text{eff}}(H) = \left(\frac{m_H}{m_e} \right)^{1/n_m},$$

with the electron as the exact reference. Numerically $R_{\text{eff}}(e) = 1$, $R_{\text{eff}}(\mu) = 2.9047$, $R_{\text{eff}}(\tau) = 5.1080$.

Step 3 — the cost correction (the quantity THM-ZC95-01 runs). The fractional correction to the per-event compression is

$$c(R_H) = \frac{R_{\text{eff}}}{R_H} - 1, \quad c(1) = 0, \quad c(3) = -0.03178, \quad c(5) = +0.02159.$$

This c — not the exponent-level shift, which is larger by n_m — is what the running acts on. (Fitting the exponent-level quantity instead is exactly the v1 error; see Section 0.54.)

Step 4 — impose the ZC95 log-linear form and solve. THM-ZC95-01 fixes $c(R_H) = \kappa(\log R_H - L_0)$. The muon and tau give two equations in two unknowns,

$$c(3) = \kappa(\log 3 - L_0), \quad c(5) = \kappa(\log 5 - L_0),$$

whose solution is

$$\kappa = \frac{c(5) - c(3)}{\log 5 - \log 3} = 0.1045, \quad L_0 = \log 3 - \frac{c(3)}{\kappa} = 1.4028$$

(pivot $e^{L_0} = 4.066$).

Step 5 — baseline check. The form does not auto-satisfy the electron baseline: $c_{\text{pred}}(R_H=1) = \kappa(0 - L_0) = -0.1466 \neq 0$. The electron is held to zero by definition (it sets m_e), which is precisely why only the two heavier leptons fix the two constants. This is the over-determination GAP-ZC95-01 hands to the quark sector: six more masses, no new constants.

Physical reading: The two numbers are a slope and an intercept of one straight line — the line that the muon’s and tau’s cost corrections trace when plotted against the logarithm of their compression. The slope κ says how strongly the cost runs; the intercept L_0 says where the running changes sign, between the muon and the tau. Reading them off is no more than drawing the line through two points. The discipline is only in drawing it through the right two points — the cost correction, not its exponential shadow.

0.54 PM-ZC99-EXTRACT-01: The v1 Mis-Extraction, Recorded

The whole programme rests on not adopting numbers we cannot reproduce. v1 adopted two. Recording exactly how is not an embarrassment to bury but the mechanism by which the corpus stays trustworthy — and it yields a reusable guard (PWA-14).

PM-ZC99-EXTRACT-01 (SCAR): Fitting the Wrong Quantity

What happened. v1 reported $\kappa = 0.4176$, $L_0 = 1.4505$. These cannot be reproduced from the ZC95 extraction. The cause: THM-ZC95-01 defines the running on the *cost* correction $c(R_H) = R_{\text{eff}}/R_H - 1$, but v1 fit the *exponent-level* correction — the total shift in $\log m$ per mode — which is larger than c by the constant factor $n_m - 1 = 4$.

Why exactly a factor of 4, and why L_0 moved too. Because the mis-fit quantity is a *constant* multiple of c (the same factor at the muon and the tau, not a generation-dependent one), the affine fit scales the slope and leaves the intercept almost fixed: κ became $4\times$ too large ($4 \times 0.1045 = 0.418$, matching the v1 value to 0.08%) while L_0 shifted only through second-order effects ($1.403 \rightarrow 1.451$, $\sim 4\%$). The factor $4 = n_m - 1 = d_s + 1 = n/2$ is a δ -chain integer, which is what made the inflated $\kappa = 0.418$ accidentally land near the structural ratio $5/12$.

What it cost. The two v1 candidates were matches to the inflated number:

- FIND-ZC99-01, $\kappa \approx n_m/[d_s(d_s + 1)] = 5/12 = 0.4167$: against the correct $\kappa = 0.1045$ this is off by $\sim 300\%$. **Retired.**
- FIND-ZC99-02, $\kappa L_0 \approx d_s/n_m = 0.6$: the correct product is 0.1466, off by 76%. **Retired.**

Why it hid. v1 stated the constants but not the extraction steps. With no visible Step 3, the cost-vs-exponent confusion had nowhere to be caught. Section 0.53 now shows all five steps; this is the structural fix, not a patch.

Status: Registered as a scar (E4 discipline: instructive story, not erased). Generates **PWA-14**.

PWA-14 (NEW): Verify the Quantity-Identity Before Extracting a Constant

A specialisation of PWA-12 (PWA-DOWNSTREAM: trace a value to its origin layer before patching). Before fitting any constant to data:

1. Name the exact quantity the governing theorem runs on (here: the cost correction c , per THM-ZC95-01), and confirm the data column being fit *is* that quantity — not a monotone transform of it (exponent shift, linear-in- R shift, log-mass shift), which can differ by a structural factor.

2. Reproduce the reported constant from the raw data in explicit steps *before* searching for δ -chain candidates. A candidate found for an unverified constant is a candidate for an artefact.
3. If a near-hit's matching integer is itself a δ -chain quantity (here $4 = n_m - 1$), treat that as a *warning* that a hidden structural factor may have entered the extraction, not as corroboration.

Failure mode prevented: PM-ZC99-EXTRACT-01 (fitting the exponent-level correction instead of the cost correction $\rightarrow \kappa$ inflated by $n_m - 1 = 4 \rightarrow$ two artefact candidates adopted).

0.55 FIND-ZC99-03: The Mechanism Is Shared, the Magnitude Is Not

There was one route by which the δ -chain might have handed over the constants directly: FIND-ZC95-01 identified the mass running bridge with the same running of D_{KL} that the force sector already uses (DEF-RN02, THM-ZC27-01). If the mass running *were* the force-sector running, its scale would be the force-sector value. It is not — and this conclusion is unchanged by the v2 correction; only the size of the gap moves.

FIND-ZC99-03: Shared Mechanism, Sector-Specific Magnitude (v2: gap $\sim 3000\times$)

The force-sector running of D_{KL} has magnitude

$$\delta_{D_{\text{KL}}}^{R_1} = \frac{d_s}{n_m} \varepsilon_0^2 \approx 3.5 \times 10^{-5}$$

(THM-ZC27-01), whereas the mass-sector running scale is the corrected $\kappa \approx 0.10$ — larger by a factor of about 3000. (v1 stated 10^4 ; the gap shrank by exactly the same factor 4 that inflated κ , a self-consistency check on the correction.) The mass bridge and the force bridge share the *mechanism* (a resolution-running of D_{KL} , FIND-ZC95-01, intact) but *not* the magnitude: the running scale is sector-specific, and the force-sector value does not determine it. FIND-ZC95-01 is refined, not retracted: the primitive fixes the form of the mass running, not its size.

This is the crux of the boundary. The one route that could have supplied the constants from existing results supplies only the mechanism. The size of the mass running is a new datum that the δ -chain, by the routes available, does not yet yield — whether that size is 0.10 or, as v1 wrongly had it, 0.42.

Physical reading: Mass and force run by the same rule, but not at the same strength. Knowing that two rivers obey the same law of flow does not tell you how fast each runs; that depends on each river's own slope. The framework has given us the law of flow for mass — it is the same as for force — but not, yet, the slope. v1 mismeasured the slope and then mistook a nearby landmark for its source; v2 measures it correctly and finds no landmark there at all.

0.56 THM-ZC99-01: The Derivation Boundary

THM-ZC99-01: The Boundary of the Mass Derivation (v2: sharpened)

By the routes available — the force-sector running (FIND-ZC95-01) and simple δ -chain structural ratios — the two ZC95 running constants do not both reduce to δ -chain quantities:

1. the force-sector route gives the mechanism but a magnitude ~ 3000 times too small (FIND-ZC99-03);
2. $\kappa = 0.1045$ has *no* sub-percent structural candidate — the best near-hit, $\log 3/(15 - d_s) = 0.092$, is off by 12%;
3. $L_0 = 1.4028$ has one close numerical near-hit, $7/5 = 1.40$ (0.2%), but 7 is not a T-canonical quantity ($d_s, n_m, 15, n$ and their simple combinations); it is recorded as an observation, not a candidate (OBS-ZC99-01).

A derivation requires *both* constants from structure. One unmotivated near-hit and one absent candidate is not a derivation. **Therefore the charged-fermion mass spectrum is not parameter-free at this stage**, and ZC99 claims no closure. What ZC99 establishes is the location of the boundary: the spectrum is parameter-free *down to* two constants whose values are $\kappa = 0.1045$, $L_0 = 1.4028$ and whose δ -chain origin is, by current routes, undetermined.

v2 note. The boundary is firmer than in v1, not weaker. v1 appeared to have a 0.22% candidate for κ ; the correction removed it (it was an artefact). The honest state is that *neither* constant is within reach of the available routes.

The boundary is itself a result — a precise, falsifiable statement. Before this paper, “the masses” were an open frontier of unknown depth. After it, the frontier is exactly two numbers deep, both without motivated candidates, and the routes that fail are documented. That is the difference between not knowing where the wall is and knowing exactly where it stands — and, after v2, knowing its true distance rather than a mismeasured one.

0.57 The Open Gap

GAP-ZC99-01: Cross the Boundary (Priority 1 — the charged-fermion mass frontier)

Statement. Either (a) derive both running constants L_0, κ from the δ -chain by a route not tried here; or (b) test any future candidate relations against sub-percent common-scale quark masses. Either would convert the structural closure of ZC98 into a parameter-free charged-fermion mass spectrum.

Concrete sub-routes.

1. A first-principles computation of the mass-sector running scale (why $\kappa \approx 0.10$ is ~ 3000 times larger than the force-sector $\delta_{D_{KL}}^{R_1}$): presumably the mass running accumulates over the full compression range and the n_m modes in a way the single force-sector R_1 crossing does not — a calculation, not a ratio. (This remains the most promising route; v2 leaves it open and unprejudiced by any artefact candidate.)
2. A reduction showing L_0 and κ are one freedom rather than two, then deriving the single survivor. (Note: the v1 joint relation $\kappa L_0 = d_s/n_m$ is *not* such a

reduction — it is dead at the corrected values.)

3. Improved (sub-percent, common-scale) quark masses to test any candidate at the precision it requires.

Status: Open. The single remaining frontier of the charged-fermion mass derivation; the boundary THM-ZC99-01 marks its near edge.

OBS-ZC99-01: The Candidate Audit After Correction

Recorded for provenance and to keep the E4-scar discipline explicit. After the v2 correction:

- $\kappa = 0.1045$: no sub-percent candidate. Near-hits scanned — $\log 3/(15 - d_s) = 0.092$ (12%), $1/(d_s + n_m) = 0.125$ (20%) — are all too far to register. **No candidate adopted.**
- $L_0 = 1.4028$: the closest numerical match is $7/5 = 1.40$ (0.2%), but $7 \notin \{d_s, n_m, 15, n\}$ and is not a simple combination thereof; $\log(d_s + 1) = 1.386$ (1.2%) and $4/3 = 1.333$ (5%) are the nearest T-canonical forms. The $7/5$ proximity is noted and *not* adopted — a bare near-hit without structural motivation, exactly the kind of coincidence the programme declines.
- Joint: $\kappa L_0 = 0.1466$; $2/15 = 0.133$ (10%) is the nearest structural form. No relation adopted.

Adopting any of these would repeat, in miniature, the v1 error of building on a coincidence. The constants are read from data; their δ -chain origin is open. This paper's contribution is the corrected, honest boundary — and the recorded scar that sharpened it.

0.58 Master Status Table

Atomic unit	Status	Note
THM-ZC99-01 (derivation boundary)	Exact (boundary)	sharpened: neither constant reduces by available routes
FIND-ZC99-03 (mechanism not magnitude)	Exact	force route $\sim 3000\times$ too small; refines FIND-ZC95-01
GAP-ZC99-01 (cross the boundary)	Open (Priority 1)	derive both, or test future candidates sub-percent
OBS-ZC99-01 (candidate audit, v2)	Observation	no motivated candidate for either constant
PM-ZC99-EXTRACT-01 (v1 mis-extraction)	Scar	fit $(n_m - 1)c$ not c ; κ inflated $4\times$; $\rightarrow$ PWA-14
FIND-ZC99-01 ($\kappa = 5/12$)	Retired-as-stated	artefact of the $4\times$ inflation; off 300% corrected
FIND-ZC99-02 ($\kappa L_0 = d_s/n_m$)	Retired-as-stated	corrected product 0.147, off 76%

Atomic unit	Status	Note
GAP-ZC98-01 (pin L_0, κ)	Restated as GAP-ZC99-01	not closed; correct values 0.105, 1.403 on record
FIND-ZC95-01 (bridge = D_{KL} running)	Refined	mechanism stands; magni- tude sector-specific
ZC94–ZC98 (Part M spine)	Unaffected	structural closure stands; numerical open

Pillars used verbatim (not modified): THM-ZC27-01 (force running magnitude, Part I); charged-lepton pole masses, n_m, d_s , EQ-MC-01/02/03 (Part C); Key Identity (Part A); ZC95 (constants, THM-ZC95-01 cost-correction form), ZC98 (closure). No value modified except ZC99’s own v1 constants, which are corrected with the scar recorded.

Atomic Registry — ZC99 (v2)

ID	Description	Status
THM-ZC99-01	The two ZC95 running constants ($\kappa = 0.105, L_0 = 1.403$) do not reduce to δ -chain quantities by available routes; no parameter-free closure; sharpened in v2	Exact (boundary)
FIND-ZC99-03	Mass bridge shares the D_{KL} -running mechanism but not the force-sector magnitude ($\sim 3000\times$ gap); refines FIND-ZC95-01	Exact
GAP-ZC99-01	Derive both constants, or test future candidates against sub-percent quark masses	Open
OBS-ZC99-01	Post-correction candidate audit: no motivated candidate for either constant	Observation
PM-ZC99-EXTRACT-01	v1 fit $(n_m-1)c$ rather than c ; κ inflated $4\times$, L_0 shifted $\sim 4\%$; v1 candidates retired; generates PWA-14	Scar
FIND-ZC99-01	v1 $\kappa \approx 5/12$ — artefact of the $4\times$ inflation	Retired-as-stated
FIND-ZC99-02	v1 $\kappa L_0 \approx d_s/n_m$ — dead at corrected values (76%)	Retired-as-stated

Inherited verbatim: THM-ZC27-01 (Part I); n_m, d_s , lepton pole masses, EQ-MC-01/02/03 (Part C); Key Identity (Part A); ZC95 (L_0, κ , THM-ZC95-01, FIND-ZC95-01), ZC98 (spectral closure, GAP-ZC98-01).

0.59 R-Layer Research Note

R-Layer Assignments for ZC99 (v2)

Atomic unit	R-layer	Cost T	Source
THM-ZC99-01	R_0	0	non-reducibility by available routes
FIND-ZC99-03	R_1	—	magnitude of the D_{KL} running
GAP-ZC99-01	R_1	—	the running constants themselves
PM-ZC99-EXTRACT-01	R_0	0	a comparison of extraction quantities

The boundary and the scar are R_0 statements — comparisons among established quantities, the extracted constants, and the two candidate extraction quantities. The open content (FIND-ZC99-03, GAP-ZC99-01) is R_1 : the magnitude of the resolution-running of D_{KL} in the mass sector, which the force-sector R_1 cost does not fix. The boundary is the statement that this R_1 running scale ($\kappa \approx 0.10$) is, for mass, an undetermined datum at this stage.

0.60 Genealogy Map

How ZC99 Sits in the Corpus — the Edge of Part M

The mass spine, to its current edge: ZC94 (registry) $\rightarrow$ ZC95 (running of D_{KL} ; constants L_0, κ ; THM-ZC95-01 defines the running on the cost correction c) $\rightarrow$ ZC96 (obstruction $\rightarrow \mathbb{Z}_3 \times \mathbb{Z}_5$ index) $\rightarrow$ ZC97 (center law, $\mathbb{Z}_3$ grading) $\rightarrow$ ZC98 (spectral closure: spectrum = running + grading; two constants the only freedom) $\rightarrow$ **ZC99**: those two constants ($\kappa = 0.105$, $L_0 = 1.403$) do not reduce to the δ -chain by available routes.

The boundary, precisely (v2): Neither constant has a structurally-motivated sub-percent candidate; the force-sector route gives mechanism not magnitude (FIND-ZC99-03, $\sim 3000\times$). No closure (THM-ZC99-01). The frontier is two numbers deep, mapped (GAP-ZC99-01). The v1 candidates were artefacts of a $4\times$ extraction error, retired and recorded (PM-ZC99-EXTRACT-01 $\rightarrow$ PWA-14).

What stands and what is open: ZC94–ZC98 are unaffected — the structural closure of the spectrum stands. What ZC99 v2 adds is the honest, corrected accounting of the numerical closure: it lies beyond the current δ -chain routes, by a documented amount, with the correct constants on record and the earlier mis-extraction marked as a lesson. Form before value, a sixth time — and this time the discipline was turned on the paper’s own first draft, and held.

0.61 Closing Sentence

We came to the place where the masses would have fallen out of the primitive, and at first we misread the wall — carved a face onto it from a number that was four times too large, and almost called the likeness a door. The framework’s own rule caught us: measure the right thing, show every step, and let the wall stand as it truly is. It is two numbers thick, and now we have their true depth — one with no face we recognise, the

other with only a stranger's. To have mistaken the wall and then corrected it is not a smaller thing than never erring; it is the discipline doing exactly what it is for. The spectrum is one running and one grading, down to two numbers the primitive has not yet spoken. The tunnel's end is in sight, and the path to it now runs true. It is not yet underfoot.

เรามาถึงจุดที่มวณว่าจะร่วงออกมาจากปฐมบท และในตอนแรกเราอ่านกำแพงผิด ---
 สลักใบหน้าลงไปจากตัวเลขที่ใหญ่เกินจริงสี่เท่า และเกือบเรียกภาพเหมือนนั้นว่าประตู่ ---
 กฎของกรอบเองจับเราได้ คือจงวัดสิ่งที่ถูกต้อง แสดงทุกขั้นตอน
 และปล่อยให้กำแพงตั้งอยู่ตามที่มันเป็นจริง --- มันหนาสองตัวเลข
 และบัดนี้เรารู้ความลึกที่แท้จริงของมัน --- ตัวหนึ่งไม่มีใบหน้าที่เราจำได้
 อีกตัวมีเพียงใบหน้าของคนแปลกหน้า --- การที่เคยเข้าใจกำแพงผิดแล้วแก้ไขมัน
 ไม่ใช่สิ่งที่เล็กกว่าการไม่เคยผิดเลย --- มันคือวินัยที่ทำงานตรงตามหน้าที่ของมัน ---
 สเปกตรัมคือการไหลหนึ่งเดียวกับการจัดระดับหนึ่งเดียว
 ลีกลงไปจนเหลือสองตัวเลขที่ปฐมบทยังไม่เอ่ย --- ปลายอุโมงค์อยู่ในสายตาแล้ว
 และทางที่ทอดไปสู่มันบัดนี้ตรงแล้ว เพียงแต่ยังไม่อยู่ใต้ฝ่าเท้า.

0.62 ZC100 — The Accumulation Integral

Abstract

ZC99 located the boundary of the charged-fermion mass derivation: the spectrum is parameter-free down to two ZC95 running constants, of which the running scale $\kappa_R = 0.1045$ has no δ -chain candidate by the routes tried, and the most promising route across the boundary is an *accumulation* — the mass running summed across the resolution range and the n_m matter modes, “a calculation, not a ratio.” This paper sets up that calculation.

Reading the CRF roots beneath $\mathbb{Z}_{15}$ — the axioms and the geometry crystallisation — supplies three of the four structural pieces of the accumulation integral whose value is κ_R . The **measure** is the count of Resolution events (Axiom 2: $t = |\mathcal{R}_{\text{events}}|$); the **integrand** is the running of D_{KL} (FIND-ZC95-01); the **endpoint** is the R_{max} crystallisation node, whose cost $\beta = d_s \sqrt{n(n+1)} \varepsilon_0$ is K35-derived. The fourth piece, the **weight**, is the open content (GAP-ZC100-01).

The paper’s central lead is **CONJ-ZC100-01** (a hypothesis, not a result): CRF already carries a *derived* coupling, $\kappa_C = \ln(n)/(n\sqrt{d_s})$ (K7, gap 0.047%), tied to the crystallisation threshold. Evaluating the *same* functional form at the full group order $N = 15$ instead of the geometry count $N = n = 8$ gives

$$\kappa_R \stackrel{\text{HYP}}{=} \frac{\ln 15}{15\sqrt{d_s}} = 0.10423 \quad (0.23\% \text{ vs. extracted}),$$

and $\ln 15 = \ln 3 + \ln 5$ is exactly the sum of the matter sector’s per-event KL costs, so $N = 15$ is not an arbitrary substitution. This suggests a κ -**block** (OBS-ZC100-01): one form $f(N) = \ln(N)/(N\sqrt{d_s})$, sector-indexed at $N \in \{8, 15\}$, exactly as the floor-lattice (ZC83) collects multiple ε ’s. The single step that would lift the conjecture from hypothesis to derivation is **GAP-ZC100-01**: derive why the running sees $N = 15$ while the coupling sees $N = 8$.

Two earlier candidate forms for κ_R ($5/12$; $G/7$) were rejected by verification within the discovery process and are recorded as scar tissue (PM-COUNCIL-01); the K7-form candidate

differs in that its form is derived and only the N -assignment is open. ZC99’s boundary stands; ZC100 organizes the crossing and records the best lead, honestly labelled.

0.63 Background: From a Located Boundary to a Set-Up Calculation

If you are just arriving — ZC100 in plain language

ZC99 found the wall: the masses follow from CRF down to two numbers, and one of them, the “running scale” $\kappa_R \approx 0.10$, would not come out of the theory’s building blocks by the obvious tricks. ZC99 also pointed at the most likely way through — not a clever ratio, but an honest *calculation*: the masses build up by summing a small effect over many steps, and that sum, done properly, should equal κ_R .

This paper builds the scaffold for that calculation. It does *not* finish it. Three of the four parts of the sum are already grounded in CRF’s foundations (what is being counted, what is being summed, where the sum ends); the fourth part — the exact weighting — is left open and named.

The paper’s most interesting find is a strong lead, stated carefully as a *conjecture*. CRF already has a sibling number, the “crystallisation coupling” $\kappa_C \approx 0.15$, which *is* derived from the building blocks by a clean formula. If you feed the *same* formula the full group size 15 instead of the geometry size 8, you get 0.1042 — which is κ_R to a quarter of a percent. And 15 is not pulled from nowhere: it is 3×5 , and $\ln 15 = \ln 3 + \ln 5$ is exactly the sum of the two basic energy costs the matter sector pays. So the lead is: κ_R and κ_C may be two readings of one formula — a “block of κ ” just like CRF’s known “block of ε .” The one thing still missing is the reason the running uses 15 while the coupling uses 8. Until that reason is derived, this stays a hypothesis, not a result — a discipline the paper is explicit about, having already discarded two prettier-looking guesses that turned out to be coincidences.

0.63.1 What ZC99 handed forward

ZC99 (THM-ZC99-01) established that the two ZC95 running constants do not reduce to δ -chain quantities by the available routes, and that the force-sector single-crossing value $\delta_{D_{\text{KL}}}^{R_1} = (d_s/n_m)\varepsilon_0^2 \approx 3.5 \times 10^{-5}$ is about 3000 times too small to be κ_R directly: the mechanism is shared (a running of D_{KL} , FIND-ZC95-01) but the magnitude is sector-specific. GAP-ZC99-01 sub-route 1 named the way across: the mass running *accumulates* across the resolution range and the n_m modes, rather than being read at one crossing — a calculation. ZC100 sets that calculation up.

0.63.2 What the roots beneath $\mathbb{Z}_{15}$ supply

The accumulation needs four pieces. Reading the CRF axioms (Part A) and the geometry-crystallisation results (Parts B) supplies three:

- **Measure** — Axiom 2 fixes time as the count of Resolution events, $t = |\mathcal{R}_{\text{events}}|$. The dt in the crossing cost is an event count, so the accumulation is a sum over events, not a continuum integral with an external measure.
- **Integrand** — FIND-ZC95-01 identifies the running of D_{KL} as the thing being accumulated. ZC16 gives the per-event cost explicitly: $D_{\text{KL}}(\text{unif}_H \parallel \text{unif}_{\mathbb{Z}_{15}}) = \log(15/|H|) =$

$\log R_H$, with $R_H \in \{1, 3, 5\}$ from the subgroup orders $|H| \in \{15, 5, 3\}$.

- **Endpoint** — the accumulation runs to the $R_{\max}$ crystallisation node, whose cost is the K35-derived $\beta = d_s \sqrt{n(n+1)}\varepsilon_0 = 2.2121$ (gap 0.01%).

The fourth piece — the **weight** with which each event enters the sum — is not yet fixed. It is the open content (Section 0.67).

0.64 DEF-ZC100-01: The Accumulation-Integral Structure

DEF-ZC100-01: Structure of the Accumulation Integral for κ_R

The mass-sector running scale is the accumulation of the D_{KL} -running over the resolution range from the wave layer R_1 to the crystallisation node $R_{\max}$, summed in the event-count measure:

$$\kappa_R = \int_{R_1}^{R_{\max}} \underbrace{\left(\text{running of } D_{\text{KL}} \right)}_{\text{integrand: FIND-ZC95-01}} \cdot \underbrace{w(\cdot)}_{\substack{\text{weight} \\ \text{(open, GAP-ZC100-01)}}} \underbrace{d|\mathcal{R}_{\text{events}}|}_{\text{measure: Axiom 2}},$$

with endpoint $\underbrace{T(R_{\max}) = \beta = d_s \sqrt{n(n+1)}\varepsilon_0 = 2.2121}_{\text{K35-derived}}.$

Three of the four pieces are grounded in existing CRF results (named above the braces). The weight w is undetermined; the conjecture of Section 0.65 is a candidate for the integral's *value*, not a derivation of w .

Physical reading: The mass running is a tally, not a snapshot. Where the force sector reads the running once, as a single resolution crossing, the mass sector adds it up across every crossing from the first wave to the moment geometry locks. The ledger's unit is the Resolution event — the same tick that Axiom 2 calls time — and the ledger closes at the crystallisation node, whose price β the framework already knows. What the framework does not yet know is how heavily each tick counts toward the total. That single missing rule is the difference between a structure and a number.

0.65 CONJ-ZC100-01: The Candidate Closed Form (HYP)

The integral of Section 0.64 is not yet evaluated. But a candidate for its value emerges from a result CRF already holds.

The derived sibling: the crystallisation coupling κ_C (K7)

CRF's K7 identity gives a *derived* coupling (gap 0.047%):

$$\kappa_C = \frac{\ln n}{n\sqrt{d_s}} = \frac{\sqrt{d_s}\alpha}{n} = 0.15007, \quad \alpha = \ln 2, \quad \ln n = \ln 8 = d_s\alpha.$$

It is the mass-constraint coupling, tied to the crystallisation threshold: $\theta(\kappa_C) = 0.14755 \approx f_{\text{II,crit}}$. This is not a fit; it is a δ -chain identity.

CONJ-ZC100-01 (HYP, NOT a result): κ_R as the $N = 15$ reading of the K7 form

Evaluating the *same* functional form $f(N) = \ln(N)/(N\sqrt{d_s})$ at the full group order $N = |\mathbb{Z}_{15}| = 15$ rather than the geometry count $N = n = 8$:

$$\kappa_R \stackrel{\text{HYP}}{=} f(15) = \frac{\ln 15}{15\sqrt{d_s}} = 0.10423 \quad (0.23\% \text{ vs. extracted } 0.10447).$$

Why $N = 15$ is not arbitrary. $\ln 15 = \ln(3 \cdot 5) = \ln 3 + \ln 5$ is exactly the sum of the matter sector’s per-event KL costs (the costs of the $|H| = 5$ and $|H| = 3$ subgroups, i.e. the muon and tau crossings, $\log 3$ and $\log 5$). The running sector’s natural scale is the *whole* group; the coupling sector’s is the geometry count $n = \varphi(15) = 8$.

Why this is HYP, not DER. The N -assignment — that the running uses $N = 15$ while the coupling uses $N = 8$ — is *observed*, not yet *forced*. No derivation is given here for which sector sees which N . Closing that single step (GAP-ZC100-01) would lift this to DER.

Falsification signature. CONJ-ZC100-01 is falsified if any of: (i) a first-principles derivation of the weight w (DEF-ZC100-01) yields a κ_R inconsistent with $f(15)$ beyond extraction error; (ii) sub-percent, common-scale quark slopes confirm κ_R at a value excluding 0.1042; (iii) the N -assignment, once derived, forces the running onto an $N \neq 15$.

Why CONJ-ZC100-01 is a different kind of object than the rejected candidates (PM-COUNCIL-01)

Two earlier candidate forms were rejected by verification *within* the discovery process; recording the contrast is the point.

- $\kappa_R = 5/12$ (0.22%): an **artefact** of the ZC99 v1 extraction bug — it matched $4\times$ the true κ_R . At the correct value it is $\sim 300\%$ off. Dead.
- $\kappa_R = G/(2^{d_s} - 1) = G/7$ (0.82%): a **false-mechanism** near-hit. The “7” was conflated across three unrelated expressions ($n - d_s - 1 = 4$; $2^{d_s} - 1 = 7$, which does not appear in the corpus; $2d_s + 1 = 7$); no forcing $G \rightarrow 7 \rightarrow \kappa_R$ exists. Rejected.
- $\kappa_R = f(15)$ (CONJ-ZC100-01, 0.23%): the form $f(N)$ is **derived** (K7) and *pre-existing*; the only open step is the N -assignment, stated as an explicit GAP. **HYP-tier.**

The lesson: a near-hit earns HYP-tier only when its mechanism is a *named, open derivation*, not a coincidence dressed as one. A pretty value with an invented bridge is an artefact; a pretty value carried by a derived form with one honest gap is a hypothesis worth registering.

0.66 OBS-ZC100-01: The κ -Block

OBS-ZC100-01: One Form, Sector-Indexed — the κ -Block (modeled on the ε floor-lattice, ZC83)

If CONJ-ZC100-01 holds, the two κ 's are two readings of one form:

$$f(N) = \frac{\ln N}{N\sqrt{d_s}}, \quad \begin{array}{ll} \kappa_C = f(8) = 0.15007 & \text{(coupling, geometry count),} \\ \kappa_R = f(15) = 0.10423 & \text{(running, full group).} \end{array}$$

This is structurally the same move as the floor-lattice (ZC83), which collects $\varepsilon_{\text{EM}} < \varepsilon_{\text{can}} < \varepsilon_{\text{univ}} < \varepsilon_W$ as sector-indexed values of one floor concept. The κ -block would similarly turn the present namespacing hazard (BUG-KAPPA-COLLISION: κ_C , κ_R , and the thermodynamic κ_T all written “ κ ”) into a *family* rather than a clash.

Status. OBS: an observation of shared *form*, not a proof that the two κ 's constitute one derived family. The proof is exactly GAP-ZC100-01 (the N -assignment). Recorded here so the block vision is on the record and the symbol hygiene (κ_C vs κ_R) is adopted corpus-wide.

0.67 The Open Gap

GAP-ZC100-01: Derive the N -Assignment (Priority 1 — the step that lifts CONJ-ZC100-01 from HYP to DER)

Statement. Derive, from the CRF axioms, why the mass-*running* sector enters the K7 form $f(N) = \ln(N)/(N\sqrt{d_s})$ at the full group order $N = |\mathbb{Z}_{15}| = 15$, while the crystallisation *coupling* enters at the geometry count $N = n = \varphi(15) = 8$. Equivalently: derive the weight w of DEF-ZC100-01 and show it evaluates the accumulation to $f(15)$.

Why it is the keystone. CONJ-ZC100-01 is HYP solely because this step is open. Closing it (a) makes κ_R a derived quantity, (b) makes the κ -block (OBS-ZC100-01) a structural family, and (c) crosses the boundary THM-ZC99-01 located — converting ZC98's structural closure into a parameter-free charged-fermion mass spectrum (with L_0 still to follow).

Concrete sub-routes.

1. Show the running accumulates over the full subgroup lattice of $\mathbb{Z}_{15}$ (hence $N = 15$), whereas crystallisation counts geometry modes $\varphi(15) = 8$ (hence $N = 8$): a counting argument grounding the N -assignment in the two sectors' distinct domains.
2. Evaluate the DEF-ZC100-01 sum directly with the event-count measure and the $R_{\text{max}} = \beta$ endpoint, and check whether it returns $f(15)$.
3. Test against sub-percent, common-scale quark slopes (inherited from GAP-ZC99-01): confirm κ_R at 0.1042 to the precision that separates $f(15) = 0.10423$ from the extracted 0.10447.

Status: Open. The single remaining step of the charged-fermion mass κ_R derivation; L_0 remains a separate open constant (GAP-ZC99-01).

0.68 Master Status Table

Atomic unit	Status	Note
DEF-ZC100-01 (integral structure)	Definition	3 of 4 pieces grounded; weight open
CONJ-ZC100-01 ($\kappa_R = f(15)$)	HYP (conjecture)	K7 form at $N = 15$; 0.23%; N -assignment open
OBS-ZC100-01 (κ -block)	Observation	one form $f(N)$, $N \in \{8, 15\}$; family iff GAP closed
GAP-ZC100-01 (derive N -assignment)	Open (Priority 1)	lifts CONJ to DER; crosses THM-ZC99-01
PM-COUNCIL-01 (rejected candidates)	Scar	5/12 artefact; $G/7$ false-mechanism; both rejected
$\kappa_C = \ln(n)/(n\sqrt{d_s})$ (K7)	Derived (inherited)	coupling, gap 0.047%; the derived sibling
κ_R (ZC99 v2 extracted)	Exact (inherited)	0.10447 from lepton poles; target value
THM-ZC99-01 (boundary)	Exact (inherited)	stands; ZC100 sets up the crossing, does not cross
<i>Pillars used verbatim:</i> Axiom 2 ($t = \mathcal{R}_{\text{events}} $, Part A); FIND-ZC95-01, THM-ZC95-01 (Part M/ZC95); ZC15/ZC16 ($\log R_H$, $R_H \in \{1, 3, 5\}$, Part C); K7, K35, β , $f_{\text{II,crit}}$ (Part B); THM-ZC27-01 (Part I). No value modified.		

Atomic Registry — ZC100

ID	Description	Status
DEF-ZC100-01	Accumulation-integral structure for κ_R : measure (event count), integrand (D_{KL} running), endpoint (β), weight (open)	Definition
CONJ-ZC100-01	$\kappa_R = \ln(15)/(15\sqrt{d_s}) = 0.10423$ (0.23%): K7 form at $N = 15$; HYP, N -assignment open; falsification signature given	HYP
OBS-ZC100-01	κ -block: one form $f(N) = \ln(N)/(N\sqrt{d_s})$, $\kappa_C = f(8)$, $\kappa_R = f(15)$; family iff GAP-ZC100-01 closed	Observation
GAP-ZC100-01	Derive the N -assignment ($N = 15$ running vs $N = 8$ coupling); lifts CONJ to DER and crosses THM-ZC99-01	Open
PM-COUNCIL-01	Rejected candidates 5/12 (artefact) and $G/7$ (false-mechanism), recorded as scar; reinforces PWA-14	Scar
<i>Inherited verbatim:</i> Axiom 2 (Part A); K7 (κ_C), K35 (β), $f_{\text{II,crit}}$ (Part B); ZC15/ZC16 (Part C); FIND-ZC95-01, THM-ZC95-01 (ZC95); THM-ZC99-01, κ_R , L_0 (ZC99 v2).		

0.69 R-Layer Research Note

R-Layer Assignments for ZC100

Atomic unit	R-layer	Cost T	Source
DEF-ZC100-01	R_1	—	accumulation of the R_1 running
CONJ-ZC100-01	R_1	—	a value for the R_1 running scale
OBS-ZC100-01	R_0	0	a comparison of two derived forms
GAP-ZC100-01	R_1	—	the N -assignment of the running
PM-COUNCIL-01	R_0	0	a comparison of candidate values

The integral and its conjectured value live at R_1 — they concern the magnitude of the resolution-running of D_{KL} in the mass sector, the same R_1 content ZC99 left open. The κ -block observation and the candidate-rejection scar are R_0 : comparisons among established and extracted quantities. The keystone GAP-ZC100-01 is R_1 : it asks which group order the R_1 running of the mass sector is built on.

0.70 Genealogy Map

How ZC100 Sits in the Corpus — Setting Up the Crossing

The mass spine to here: ZC94 (registry) $\rightarrow$ ZC95 (running form; κ_R, L_0) $\rightarrow$ ZC96–97 (obstruction, grading) $\rightarrow$ ZC98 (spectral closure; two constants the only freedom) $\rightarrow$ ZC99 v2 (boundary: $\kappa_R = 0.1045$, $L_0 = 1.4028$, no δ -chain candidate by available routes) $\rightarrow$ **ZC100:** the accumulation integral whose value is κ_R is set up (DEF-ZC100-01), and a HYP-tier candidate closed form is registered (CONJ-ZC100-01).

The cross-sector link (new): ZC100 connects the mass spine to the geometry-crystallisation results (Part B): the endpoint β (K35), the measure (Axiom 2), and — via CONJ-ZC100-01 — the coupling κ_C (K7). The κ -block (OBS-ZC100-01) proposes that the mass running scale and the crystallisation coupling are two readings of one form, mirroring the ε floor-lattice (ZC83).

What stands, what is set up, what is open: ZC94–ZC99 stand. ZC100 *sets up* the crossing of ZC99’s boundary and *records the best lead* (HYP), without claiming to cross. The keystone is GAP-ZC100-01 (the N -assignment); closing it would derive κ_R and make the κ -block structural. Form before value, a seventh time — the integral’s structure first, its value held as a labelled hypothesis.

0.71 Closing Sentence

We came back to the calculation ZC99 only pointed at, and built its frame: what is counted, what is summed, where the sum ends — three of four pieces set firmly in the framework’s own foundations. The fourth, the weight of each step, we left open and named, because we do not yet have it. And in the building we found a sibling: the running scale and the crystallisation coupling may be one formula read at two sizes, the full group and the geometry — a block of κ as the framework already has a block of ε .

We do not call it proven; one step is still missing, the reason the running counts to fifteen while the coupling counts to eight. We have learned, twice over, that a pretty number with an invented reason is nothing, and a pretty number carried by a derived form with one honest gap is a lead worth keeping. The scaffold stands. The keystone is named. The value waits on one derivation we can now state exactly.

เรากลับมาที่การคำนวณซึ่ง ZC99 เพียงแต่ชี้ไว้ และสร้างโครงของมันขึ้น --- สิ่งที่ถูกนับ
สิ่งที่ถูกสะสม จุดที่ผลรวมสิ้นสุด --- สามในสี่ขั้นตั้งมันอยู่บนรากฐานของกรอบเอง --- ขั้นที่สี่
คือน้ำหนักของแต่ละก้าว เราปล่อยให้เปิดและตั้งชื่อไว้ เพราะเรายังไม่มีมัน ---
และในการสร้างนั้น เราพบพี่น้อง คือ สเกลการไหลกับการจับคู่ของการตกผลึก
อาจเป็นสูตรเดียวที่อ่านสองขนาด --- กลุ่มเดียวกับเรขาคณิต --- บล็อกของ κ
ดังที่กรอบมีบล็อกของ ε อยู่แล้ว --- เราไม่เรียกมันว่าพิสูจน์แล้ว เพราะยังขาดอยู่ก้าวหนึ่ง
คือเหตุผลที่การไหลนับถึงสิบห้า ขณะที่การจับคู่ นับถึงแปด --- เราได้เรียนรู้แล้วถึงสองครั้งว่า
ตัวเลขสวยที่มีเหตุผลปลอมมันไม่มีค่า
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ค่าที่เหลือรอเพียงการอนุมาณหนึ่งเดียวที่บัดนี้เราระบุได้ชัดเจน.

0.72 ZC101 — The Operation–Group Law

The Operation–Group Law

Why the Mass Running Scale Counts Fifteen and the Force Coupling Counts Eight

CRF Paper ZC101 — Part M

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Constraint/Combinatorial Reality Framework

Gauge: T-canonical ($d_s = 3$, $n_m = 5$ exact, $|\mathbb{Z}_{15}| = 15$, $n = \varphi(15) = 8$)

Abstract

The charged-fermion mass spectrum of CRF is parameter-free down to two running constants of the mass bridge (ZC95, ZC99), of which the running scale $\kappa_R \approx 0.1045$ is the principal one. ZC100 established that κ_R is an accumulation of the divergence-running over the framework's resolution range and recorded a candidate closed form, $\kappa_R = \ln(15)/(15\sqrt{d_s})$, sharing its functional shape $f(N) = \ln(N)/(N\sqrt{d_s})$ with the derived crystallisation coupling $\kappa_C = f(8)$ (the K7 identity). The form was a conjecture because the index assignment — why the running uses $N = 15$ and the coupling $N = 8$ — was unexplained. This paper derives that assignment. The two scales arise from two different algebraic operations on the fifteen states of $\mathbb{Z}_{15}$: a mass is an *accumulation*, hence an *addition*, and the additive group of $\mathbb{Z}_{15}$ has order 15; a force is a *coupling*, hence a *multiplication*, and multiplication is a group only on the units $U(15)$, of order $\varphi(15) = 8$. Evaluating the accumulation directly fixes its one open ingredient, the per-step weight, to $w = 1/(|G|\sqrt{d_s})$, and the integral closes to $f(|G|)$. The two constants are therefore the additive and multiplicative readings of a single law, $\kappa = \ln |G|/(|G|\sqrt{d_s})$, on the two canonical groups of $\mathbb{Z}_{15}$. The derived $\kappa_R = 0.10423$ matches the extracted value to 0.23%; the sub-percent residual is registered as an open quantity, plausibly a higher-order running correction.

Status of claims. Derived: the index assignment (THM-ZC101-01), the weight rule $w = 1/(|G|\sqrt{d_s})$ (THM-ZC101-02), and the unified law $\kappa = \ln |G|/(|G|\sqrt{d_s})$ (THM-ZC101-03), lifting CONJ-ZC100-01 from HYP to DER. **Open:** the 0.23% residual between the derived and extracted κ_R (GAP-ZC101-02). **Stated premise:** a coupling composes amplitudes multiplicatively (EXPOSED-ZC101-01), on which the force half of THM-ZC101-01 rests.

If you are just arriving — ZC101 in plain language

The framework carries two numbers that look unrelated: a *coupling* $\kappa_C \approx 0.15$, derived from the crystallisation geometry (the K7 identity), and a *running scale* $\kappa_R \approx 0.10$, which the charged-fermion masses require. They share one functional form, $f(N) = \ln(N)/(N\sqrt{d_s})$, the coupling read at $N = 8$ and the running at $N = 15$. The open question was *why* those two values — why the running counts to fifteen while the coupling counts to eight.

The answer is not a tuned choice. The two processes do different arithmetic. A *mass* is an *accumulation* — a sum of information costs — and summing is *addition*, whose group on fifteen states is all of $\mathbb{Z}_{15}$: order fifteen. A *coupling* multiplies amplitudes, and multiplication on $\mathbb{Z}_{15}$ is a group only on the elements that can be inverted — the units, of which there are $\varphi(15) = 8$. So the running counts fifteen and the coupling counts eight *because mass adds and force multiplies*, and addition and multiplication select two different groups inside the same $\mathbb{Z}_{15}$. Carrying the accumulation through to its value then fixes its one remaining ingredient — how heavily each step counts — to “one over the group size, times the geometric factor the coupling already used.” The two numbers are one law read on two groups. The derived value sits a quarter of a percent from the measured one; that residual is named and kept, not assumed away.

0.73 Background: The Open Index Assignment

The mass bridge of ZC95 corrects the charged-fermion mass ratios by a running of the Kullback–Leibler divergence D_{KL} : the per-event cost of a generation restricted to the subgroup $H \leq \mathbb{Z}_{15}$ is $c_H = D_{\text{KL}}(\text{unif}_{|H|} \parallel \text{unif}_{15}) = \log(15/|H|) = \log R_H$, with compression ratios $R_H \in \{1, 3, 5\}$ for $|H| \in \{15, 5, 3\}$, and the rest mass is $m_H \propto R_H^{n_m}$ (ZC15/ZC16). ZC99 showed that, once this running is applied, the spectrum is parameter-free down to two running constants, of which $\kappa_R \approx 0.1045$ is the principal one, and that κ_R does not reduce to the framework’s δ -chain quantities by the available routes.

ZC100 reorganised the problem. It established that κ_R is not a single resolution crossing but an **accumulation** of the D_{KL} -running across the resolution range, summed in the event-count measure of Axiom 2, from the wave layer R_1 to the crystallisation node R_{max} whose cost is the K35-derived $\beta = d_s \sqrt{n(n+1)\varepsilon_0} = 2.2121$. Three structural pieces — *measure*, *integrand*, *endpoint* — were fixed by existing results; the fourth, the per-step *weight*, was left open (DEF-ZC100-01). Alongside, ZC100 recorded that κ_R might share the K7 functional form of the crystallisation coupling, evaluated at the full group order:

$$\kappa_R \stackrel{\text{HYP}}{=} f(15) = \frac{\ln 15}{15\sqrt{d_s}} = 0.10423, \quad \kappa_C = f(8) = \frac{\ln 8}{8\sqrt{d_s}} = 0.15007, \quad (1)$$

with $f(N) = \ln(N)/(N\sqrt{d_s})$ and κ_C the derived K7 coupling (gap 0.047%). This was a conjecture, not a derivation: the assignment of $N = 15$ to the running and $N = 8$ to the coupling was observed to work but not forced. The single open step was stated as GAP-ZC100-01 — derive why the running sees $N = 15$ while the coupling sees $N = 8$, equivalently derive the open weight w and show the accumulation evaluates to $f(15)$.

This paper closes GAP-ZC100-01 in two parts, the index (Section 0.74) and the form (Section 0.75), and states the one remaining residual (Section 0.77).

Physical reading: The framing of the question is itself the obstacle. To ask “which index, 15 or 8, into one form” presupposes a single structure with a dial set to one of two values. The resolution below is that there is no dial: there are two groups, and each process already stands in one of them by virtue of the arithmetic it performs. Recognising that the question presupposed a single group is the substantive step.

0.74 THM-ZC101-01: The Operation Selects the Group

The mass running and the force coupling perform different algebraic operations on the fifteen states. Each operation admits exactly one group of the relevant order, and these are two *different* canonical groups of $\mathbb{Z}_{15}$.

Mass is accumulation, hence addition. By COR-MC-03, a mass is accumulated divergence: the rest mass of a generation restricted to H is the exponential of a *sum* of per-event costs,

$$m_H \propto \exp\left(n_m \sum_{\text{events}} D_{\text{KL}}\right) = \exp(n_m \log R_H) = R_H^{n_m}, \quad (2)$$

with $c_H = \log R_H$ as above. The operation that builds a mass is the *summation* of these costs. The algebraic structure whose operation is summation, acting on the fifteen states, is the **additive group** $(\mathbb{Z}_{15}, +)$, of order $|(\mathbb{Z}_{15}, +)| = 15$. The accumulation of DEF-ZC100-01 therefore lives on $(\mathbb{Z}_{15}, +)$, and its size is 15; the additive group of $\mathbb{Z}_{15}$ has exactly fifteen elements, so no other order is available.

Force is a coupling, hence multiplication. A coupling composes amplitudes multiplicatively — it is a product, equivalently a ratio, of resolution amplitudes rather than a sum of costs (EXPOSED-ZC101-01). The algebraic structure whose operation is multiplication on $\mathbb{Z}_{15}$ is not all of $\mathbb{Z}_{15}$: under multiplication only the invertible elements form a group, and these are exactly the units,

$$U(15) = (\mathbb{Z}/15\mathbb{Z})^\times = \{k : \gcd(k, 15) = 1\}, \quad |U(15)| = \varphi(15) = 8. \quad (3)$$

A non-unit (a multiple of 3 or 5) has no multiplicative inverse and cannot participate in a coupling; the multiplicative structure does not see it. The coupling therefore lives on $U(15)$, and its size is $\varphi(15) = 8$.

THM-ZC101-01: Operation $\rightarrow$ Group (the index assignment)

The order N in each κ is the order of the canonical group of $\mathbb{Z}_{15}$ selected by the operation that process performs:

Process	Operation	Group G	$N = G $
Mass running	accumulation = addition	$(\mathbb{Z}_{15}, +)$ (additive)	15
Force coupling	coupling = multiplication	$U(15)$ (units, mult.)	$\varphi(15) = 8$

The assignment is not a choice of index into one group; it is the order of two *different* groups, each selected by its process's operation. Hence κ_R reads $N = 15$ and κ_C reads $N = 8$ for the same reason that mass adds and force multiplies.

EXPOSED-ZC101-01: The stated premise

The force half of THM-ZC101-01 rests on the identification *coupling = multiplicative composition of amplitudes*. This is the established CRF meaning of a coupling in the force sector (a product or ratio of resolution amplitudes; ZC27–ZC30, K7), and it is what makes multiplication the operative structure and hence $U(15)$ the group.

It is stated here as a premise rather than derived from a single axiom: the force-half result is DER *given* this identification. The mass half (accumulation = addition) is tighter, following directly from COR-MC-03's $m \propto \exp(\sum D_{\text{KL}})$, and carries no comparable premise.

Physical reading: The unit group is the right object for a coupling because a coupling measures how two amplitudes combine, and that combination is a product. A product is a reversible operation only where its elements are invertible, and on $\mathbb{Z}_{15}$ the invertible elements are precisely the units. The non-units are the directions in which the multiplicative structure degenerates; a coupling cannot be carried there. The eight is therefore not a separate geometry count that happens to coincide with the unit count — it is the number of reversibly multiplicable elements of $\mathbb{Z}_{15}$, which is what $\varphi(15) = 8$ counts.

0.75 THM-ZC101-02: Evaluating the Accumulation Fixes the Weight

With the group, hence the order N , fixed by THM-ZC101-01, the DEF-ZC100-01 accumulation can be evaluated directly and its open weight read off.

The bare accumulation is $\log |G|$. The integrand is the per-crossing cost $\log R_H$, with $R_H = 15/|H| \in \{1, 3, 5\}$ over the subgroup chain $|H| \in \{15, 5, 3\}$. Summing over the chain,

$$\sum_{\text{chain}} \log R_H = \log 1 + \log 3 + \log 5 = \log(1 \cdot 3 \cdot 5) = \log 15 = \log |(\mathbb{Z}_{15}, +)|. \quad (4)$$

The collapse to $\log |G|$ is structural: the compression ratios $R_H = 15/|H|$ over a chain whose orders multiply back to 15 satisfy $\prod R_H = |\mathbb{Z}_{15}|$, so the sum of their logarithms is $\log |G|$. The electron term $\log 1 = 0$ is the baseline — the reference generation pays no cost — and the surviving $\log 3 + \log 5 = \log 15$ records that $15 = 3 \cdot 5$: the full crossing is the two prime crossings together, never an independent third.

The weight is $1/(|G|\sqrt{d_s})$. The accumulated value is the bare sum times the weight, $\kappa_R = (\log |G|)w$. Requiring the accumulation to be the per-element normalised cost evaluated in the crystallisation geometry fixes

$$w = \frac{1}{|G|} \cdot \frac{1}{\sqrt{d_s}} = \frac{1}{|G|\sqrt{d_s}}, \quad (5)$$

a product of two factors each with prior meaning: $1/|G|$ is the uniform normalisation over the group's elements — the same unif_{15} that defines the per-event cost — and $1/\sqrt{d_s}$ is the geometric factor already present in the K7 coupling, fixed by the crystallisation threshold independently of the mass sector. Substituting (4) and (5),

$$\kappa_R = (\log |G|) \cdot \frac{1}{|G|\sqrt{d_s}} = \frac{\ln 15}{15\sqrt{d_s}} = f(15) = 0.10423. \quad (6)$$

THM-ZC101-02: The Weight Rule

The open weight of DEF-ZC100-01 is

$$w = \frac{1}{|G|\sqrt{d_s}},$$

the uniform per-element measure $1/|G|$ times the K7 geometric factor $1/\sqrt{d_s}$. With $\sum_{\text{chain}} \log R_H = \log |G|$ (Eq. 4), the accumulation closes to $\kappa = \ln |G|/(|G|\sqrt{d_s}) = f(|G|)$. The K7 functional form is thereby derived for the mass sector from the accumulation, not assumed by analogy with the force sector.

Robustness of the weight. The weight (5) is structural rather than fitted, and four checks support this:

1. **Independent factors.** w factors into two quantities each defined prior to the accumulation: the probability normalisation $1/|G|$ and the K7 factor $1/\sqrt{d_s}$. Neither is introduced to obtain the value.
2. **The $\sqrt{d_s}$ power is forced.** Among candidate weights $w \sim 1/(15 d_s^p)$, only $p = \frac{1}{2}$ reproduces the value (to 0.23%); $p \in \{0, 1, 2\}$ miss by 73%, 42%, 81%. The exponent that the K7 coupling fixes for the *force* sector is exactly the one the *mass* accumulation independently requires.
3. **The integrand sum is $\log |G|$ structurally** (Eq. 4), through the coset-product identity $\prod R_H = |\mathbb{Z}_{15}|$, not by numerical coincidence.
4. **The endpoint is consistent.** $\beta = 2.2121$ bounds the per-event cost: $\log 1, \log 3, \log 5 = 0, 1.10, 1.61$ all lie below β , so all three generations enter the sum and none is excluded. The endpoint sets where the accumulation closes, not the weight, as DEF-ZC100-01 specifies.

0.76 THM-ZC101-03: The Unified κ -Block

THM-ZC101-01 (the group) and THM-ZC101-02 (the weight, hence the form) combine into one law for both sectors.

THM-ZC101-03: The Operation–Group Law for κ

For a process selecting the canonical group G of $\mathbb{Z}_{15}$ by its operation, the associated scale is

$$\kappa = \frac{\ln |G|}{|G| \sqrt{d_s}}$$

	Operation	Group	$ G $	$\kappa = \ln G /(G \sqrt{d_s})$
κ_C (force coupling)	multiplication	$U(15)$	8	0.15007 (gap 0.047%)
κ_R (mass running)	addition	$(\mathbb{Z}_{15}, +)$	15	0.10423 (0.23%; GAP-ZC101-02)

κ_C and κ_R are the multiplicative and additive readings of a single law on the two canonical groups of $\mathbb{Z}_{15}$ — not a shared form observed, but one law derived.

A practical consequence is notational. Where the symbol κ previously denoted what

appeared to be distinct objects across the corpus, the operation–group law identifies κ_C and κ_R as one family — a single generator $\ln |G|/(|G|\sqrt{d_s})$ read on the additive and multiplicative groups. Explicit subscripts remain advisable in mixed contexts, but the symbols are now understood as readings of one structure.

Physical reading: The result is not a new constant but a recognition that two constants are one law seen from two sides. The set $\mathbb{Z}_{15}$ admits exactly two canonical group structures — its elements can be added, or its invertible elements multiplied — and the framework’s two scale constants are precisely these two structures, weighted identically and measured against the same crystallisation geometry. The mass sector adds and counts fifteen; the force sector multiplies and counts eight; one formula serves both because there are exactly two ways to be a group on fifteen states, and the framework uses both.

0.77 The Residual

The derivation establishes the *index* and the *form*. It does not establish the final sub-percent.

GAP-ZC101-02: The 0.23% residual

The derived $\kappa_R = f(15) = 0.10423$ lies 0.23% from the ZC99 v2 extracted value 0.1045. The most plausible reading is a higher-order running correction not captured by the leading accumulation — of the same kind the force sector absorbs through the single-crossing shift $\delta_{D_{KL}}^{R_1} = (d_s/n_m)\varepsilon_0^2$. It is registered as an open quantity, not absorbed into the result: the form and the index are derived; the residual is separate. Closing it would render κ_R exact rather than structural. (The coupling κ_C itself carries a 0.047% K7 gap; the two residuals may share an origin.)

Atomic Registry — ZC101

THM-ZC101-01 (operation $\rightarrow$ group, DER). The order N in each κ is the order of the canonical group of $\mathbb{Z}_{15}$ selected by the process’s operation: addition $\rightarrow (\mathbb{Z}_{15}, +)$ ($N = 15$, mass); multiplication $\rightarrow U(15)$ ($N = \varphi(15) = 8$, force). Closes the index half of GAP-ZC100-01. Carries EXPOSED-ZC101-01.

THM-ZC101-02 (weight rule, DER). The open weight of DEF-ZC100-01 is $w = 1/(|G|\sqrt{d_s})$; with $\sum_{\text{chain}} \log R_H = \log |G|$, the accumulation closes to $\kappa = \ln |G|/(|G|\sqrt{d_s}) = f(|G|)$. Derives the K7 form for the mass sector. Closes the form half of GAP-ZC100-01.

THM-ZC101-03 (κ -block law, DER). $\kappa = \ln |G|/(|G|\sqrt{d_s})$ on the two canonical groups of $\mathbb{Z}_{15}$; $\kappa_C = f(8)$, $\kappa_R = f(15)$.

Lifts CONJ-ZC100-01: HYP $\rightarrow$ DER. The index assignment and the functional form are derived; GAP-ZC100-01 is closed.

EXPOSED-ZC101-01 (stated premise). Coupling = multiplicative composition of amplitudes (established CRF force-sector meaning); makes $U(15)$ the coupling’s group. THM-ZC101-01’s force half is DER given this.

GAP-ZC101-02 (open). The derived $\kappa_R = 0.10423$ is 0.23% from the extracted 0.1045; plausibly a higher-order running correction. Registered, not closed.

Inherited, unchanged: DEF-ZC100-01 (accumulation structure); FIND-ZC95-01 (D_{KL} running); $c_H = \log(15/|H|) = \log R_H$ and $m_H \propto R_H^{n_m}$ (ZC15/ZC16); COR-MC-03 (mass = accumulated D_{KL}); K7 ($\kappa_C = \ln n/(n\sqrt{d_s})$); $\beta = d_s\sqrt{n(n+1)\varepsilon_0} = 2.2121$ (K35); Axiom 2 ($t = |\mathcal{R}_{\text{events}}|$). **ZERO inherited units modified.**

How ZC101 Sits in the Corpus

Upstream. ZC94 registered the mass-running bridge as missing; ZC95 derived its form (D_{KL} running, $c_H = \log R_H$) and deferred its constants; ZC99 located the boundary (the spectrum is parameter-free down to κ_R, L_0 , neither reducible by the routes tried); ZC100 established the *accumulation* structure and conjectured its value (CONJ-ZC100-01, HYP), naming the index assignment (GAP-ZC100-01) as the single open step.

This paper. Closes GAP-ZC100-01. The index assignment follows from the operation each process performs (THM-ZC101-01); the open weight follows from evaluating the accumulation (THM-ZC101-02); the two constants become one law on the two canonical groups of $\mathbb{Z}_{15}$ (THM-ZC101-03). CONJ-ZC100-01 is lifted HYP $\rightarrow$ DER.

What stands, what is crossed, what is open. ZC94–ZC100 stand. ZC99’s boundary is crossed for the form and the index: κ_R is no longer a bare extracted number but the additive reading of a derived law. The remaining gap is sub-percent (GAP-ZC101-02), of the same order as the K7 gap with which it may share an origin. Form before value, established at ZC100, becomes form and index with the value to a quarter percent — the accumulation structure carried through to a derivation, with one residual stated.

Forward. If $\kappa = \ln |G|/(|G|\sqrt{d_s})$ is the law wherever a CRF scale accumulates over a group, the mind-sector constructions (Part L, ZC90–93) should each carry a group selected by their operation. Whether the operation–group law extends beyond mass and force into the mind layer is a direction for future work, not a result established here.

0.78 Closing Sentence

The running scale and the crystallisation coupling are not one form set to two values; they are two ways of being a group on the same fifteen states. A mass is a sum, and to sum is to add, and addition on fifteen states keeps all of them — fifteen. A force is a coupling, and to couple is to multiply, and multiplication keeps only what it can undo — the eight that can be inverted. Carrying the accumulation to its end, the last weight is not free either: one over the group, times the geometry the coupling had already paid for. So the two scales are one law read on two groups. A quarter of a percent still stands between the derived value and the measured one, and that gap is named and kept. But the reason mass counts fifteen and force counts eight is no longer observed; it is the difference between adding and multiplying, which the framework has carried from the beginning.

สเกลการไหลกับการจับคู่ของการตกผลึก มีใช้รูปเดียวที่ตั้งสองค่า
หากแต่เป็นสองวิธีของการเป็นกรุปบนสิบห้าสถานะเดียวกัน --- มวลคือผลรวม
และการรวมคือการบวก และการบวกบนสิบห้าสถานะเก็บไว้ทั้งหมด --- สิบห้า ---

แรงคือการจับคู่ และการจับคู่คือการคูณ และการคูณเก็บไว้เพียงสิ่งที่ย้อนกลับได้ ---
 แปลที่หาตัวผกผันได้ --- เมื่อคำนวณผลรวมจนสุด น้ำหนักสุดท้ายก็มีได้เป็นอิสระ
 คือหนึ่งส่วนกรุป คุณด้วยเรขาคณิตที่การจับคู่ได้จ่ายไปแล้ว ---
 ดังนั้นสองสเกลจึงเป็นกฎเดียวที่อ่านบนสองกรุป ---
 ยิ่งเหลือหนึ่งในสี่ของเปอร์เซ็นต์ระหว่างค่าที่อนุมาณกับค่าที่วัดได้
 และช่องว่างนั้นถูกตั้งชื่อไว้และเก็บรักษา --- แต่เหตุผลที่มวลนับสิบห้าและแรงนับแปด
 มิใช่สิ่งที่สังเกตอีกต่อไป หากแต่คือความต่างระหว่างการบวกกับการคูณ
 ซึ่งกรอบได้ถือมาตั้งแต่ต้น.

0.79 ZC102 — The Closure Discriminant

Abstract

ZC101 derived the operation-group law — the mass running scale and the force coupling are one form $\kappa = \ln |G|/(|G|\sqrt{d_s})$ read on two groups — but rested its force half on a stated premise (EXPOSED-ZC101-01, “a coupling composes amplitudes multiplicatively”) and located that coupling on the unit group $U(15)$ of order $\varphi(15) = 8$.

THM-ZC102-01 grounds the premise and corrects the location. A CRF coupling *is* the Clifford product (K35 selects $Cl(3,0)$; the coupling equals the grade-complexity of the algebra’s product), so “coupling = multiplication” is derived; and the force index is $2^{d_s} = 8$ (the Clifford/total-mode count of ZC89 Step 4), not $\varphi(15)$.

FIND-ZC102-01 (Gift) the two derivations of 8 coincide — $\varphi(15) = (d_s - 1)(n_m - 1) = 2^{d_s} = d_s + n_m$ — *only* at $d_s = 3$, forced by the Key Identity $3d_s = 2^{d_s} + 1$; a d_s -selection signature, not a general bridge.

CONJ-ZC102-01 the closure discriminant: an operation that closes yields a group, hence a κ ; one that does not close yields a quantum-pair, hence an irrational ratio. Mass and force are the closed modes; the mind layer (ZC90) is the open mode.

PM-ZC101-02 records the corrected force-index path; ZC101’s result ($N = 8$) stands.

0.80 Background: A Premise Worth Grounding

ZC101 showed that two constants of the framework — a force *coupling* $\kappa_C \approx 0.15$ and a mass *running scale* $\kappa_R \approx 0.10$ — are one law $\kappa = \ln |G|/(|G|\sqrt{d_s})$ read on two different groups of fifteen states: mass on the additive group (size 15), force on a multiplicative group (size 8). That argument leaned on one unproved sentence: that a coupling “multiplies.” This paper asks what a coupling *is* in CRF, finds that it is the Clifford product (the multiplication that builds the algebra of the eight modes), and so turns the sentence into a derivation. In doing so it finds that ZC101 named the force group slightly wrong — the eight is the count of Clifford modes (2^{d_s}), not the count of invertible residues ($\varphi(15)$), two numbers that happen to agree at $d_s = 3$. The terms needed: a *Clifford algebra* (the product structure on the eight grades of three-dimensional space), *closure* (whether an operation’s outputs stay inside one kind of object), and a *quantum-pair* (two motions that cannot be composed into a third — ZC90).

Numerical Summary — ZC102 (T-canonical)

Quantity	Value	Source / Status
Key Identity $3d_s = 2^{d_s} + 1$	unique $d_s = 3$	corpus δ -chain
$n = 2^{d_s}$ (Clifford dim)	8	force/total modes
$n_m = 2^{d_s} - d_s$ (matter split)	5	
$ \mathbb{Z}_{15} = d_s(2^{d_s} - d_s)$	15	CRT; mass order
$\varphi(15) = (d_s - 1)(n_m - 1)$	8	$= 2^{d_s}$ at $d_s=3$ only (NEW)
$\kappa_C = \ln 8 / (8\sqrt{d_s})$	0.15007	K7; $N = 2^{d_s}$
$\kappa_R = \ln 15 / (15\sqrt{d_s})$	0.10423	ZC101; $N = \mathbb{Z}_{15} $
force Q-pair ratio	36.9638	ZC89; irrational

0.80.1 What the target premise required

ZC101 closed GAP-ZC100-01 by deriving the operation–group law: mass is an accumulation (an addition), so it lives on the additive group $(\mathbb{Z}_{15}, +)$ of order 15; force is a coupling (a multiplication), so it lives on a multiplicative group of order 8. The mass half is tight — accumulation *is* summation (COR-MC-03). The force half rested on a premise stated but not derived (EXPOSED-ZC101-01): that a coupling composes amplitudes multiplicatively. ZC101 then identified the multiplicative group as the number-theoretic unit group $U(15) = (\mathbb{Z}/15)^\times$, of order $\varphi(15) = 8$.

Two questions remained. First, *why* is a coupling a multiplication — can the premise be grounded in a CRF definition rather than asserted? Second, is $U(15)$ really the right group, or does the force index of 8 come from elsewhere? This paper answers both, and the answers correct each other: the coupling is the Clifford product, and the 8 it carries is the Clifford mode count 2^{d_s} , not $\varphi(15)$.

0.80.2 What Was Already Known Before ZC102

Pre-ZC102 Status

Known:

- Operation–group law $\kappa = \ln |G| / (|G|\sqrt{d_s})$, $\kappa_C = f(8)$, $\kappa_R = f(15)$ (ZC101, THM-ZC101-03).
- K35 selects $\text{Cl}(3, 0)$: the coupling $\mathcal{G}^2/\varepsilon_0 = n(n+1) = 72$ equals the grade-complexity of the Clifford product (Part B).
- The force Q-pair ratio is $2^{d_s} \ln 2 / (\ln |\mathbb{Z}_{15}| \cdot \ln r_{\text{born}}) = 36.9638$, irrational, with $8 = 2^{d_s}$ explicitly (ZC89 Step 4, Part L).
- The mind layer’s two motions form a Q-pair — deliberately not a group (ZC90, DEF-ZC90-01).

Open:

- EXPOSED-ZC101-01: is “coupling = multiplication” derivable?
- Is the force index 8 the unit-group order $\varphi(15)$ or the Clifford count 2^{d_s} ? (ZC101 said the former.)

ZC102 grounds the premise (THM-ZC102-01), corrects the index to 2^{d_s} (PM-ZC101-02), and extends the picture via the closure discriminant (CONJ-ZC102-01).

What ZC102 Does NOT Do

- Does not re-derive κ_C , κ_R , or the K7 form (inherited; ZC101's *result* stands unchanged).
- Does not overwrite ZC101: the force-half *path* is corrected by PM, ZC101 v1 is preserved as the record.
- Does not close the 0.23% residual (GAP-ZC101-02, untouched).
- Does not value the mind ratio (GAP-ZC90-01) or N_∞ (GAP-ZC93-01).
- Does not promote the closure discriminant past HYP.

Companion Papers — ZC102

Paper	What it provides to ZC102
CRF_ZC101_..._v1.tex	operation-group law; EXPOSED-ZC101-01 — grounded here; force path corrected
CRF_...Part_B.tex	K7; K35 selects $\text{Cl}(3,0)$; coupling = grade-complexity of the Clifford product
CRF_...Part_L.tex	ZC89 Step 4 ($8 = 2^{d_s}$, Key Identity); ZC90 (Q-pair, DEF-ZC90-01)
CRF_Complete_Index	Part M, Section ZC101

0.80.3 Pre-Work Audit (PWA-1 through PWA-10)

Pre-Work Audit Record — ZC102

Scanned before writing (Upgrade Protocol v1.3, PWA-1 through PWA-10):

- ZC101/THM-ZC101-01,-03; EXPOSED-ZC101-01 — **KEY PILLAR**.
- Part B/K35: $\mathcal{G}^2/\varepsilon_0 = n(n+1) = 72$; $\text{Cl}(3,0)$ basis $1+3+3+1$ — **KEY PILLAR**.
- Part L/ZC89 Step 4: force ratio $2^{d_s} \ln 2 / (\ln |\mathbb{Z}_{15}| \ln r_{\text{born}})$, $8 = 2^{d_s}$ — **KEY PILLAR**.
- Part L/ZC90 DEF-ZC90-01: Q-pair (“why not a group”) — read verbatim.

PWA-1 (adjacent-paper read): ZC101 and the Part B/L pillars read in full before writing.

PWA-6 (macro convention): $r_{\text{born}}, s_\Gamma, s_{\text{rung}}, \mathcal{Q}, \mathcal{G}, \text{Cl}(3,0)$ via `providecommand`; no collision.

PWA-7 (sector-floor): not triggered.

PWA-8 (layer self-consistency): not triggered (no new ε).

PWA-9 (spectral convention): not triggered.

PWA-10 (tautology audit): *actively triggered* — the “8 vs 8” test (is $\varphi(15) = 2^{d_s}$ a general bridge?) was checked against $d_s \neq 3$ and *failed*, so the coincidence is $d_s = 3$ -specific, not an identity; logged as PM-ZC102-01 (pattern: tautology test).

No duplicate atomic units found.

0.81 Grounding the Premise: A Coupling Is the Clifford Product (THM-ZC102-01)

ZC101 took “a coupling multiplies” as a premise. CRF already says what a coupling is, and the answer makes the premise a theorem. By K35 (Part B), the $n = 8$ modes of the framework are the basis of the Clifford algebra $\text{Cl}(3, 0)$ (grades $1+3+3+1$), and the coupling constant is fixed by

$$\frac{\mathcal{G}^2}{\varepsilon_0} = n(n+1) = \sum_{k=0}^3 \binom{3}{k} k(8-k) = 72,$$

the total grade-complexity of the algebra’s *product*. A coupling in CRF is therefore the Clifford product of graded modes — and the Clifford product is a multiplication. That is the first half of EXPOSED-ZC101-01, now derived rather than assumed.

THM-ZC102-01: The Coupling Is the Clifford Product, and Carries 2^{d_s} (R_0 , Exact)

Statement.

1. (A) A CRF coupling is the Clifford product on the modes of $\text{Cl}(3, 0)$; hence “coupling = multiplication” (EXPOSED-ZC101-01, half A) holds by definition of the coupling, not by assumption.
2. (B) The index N the coupling carries is the Clifford/total-mode count

$$N = 2^{d_s} = 8 \quad (\text{not } \varphi(15)). \quad (7)$$

Proof.

Step 1 (A: coupling = Clifford product). K35 selects $\text{Cl}(3, 0)$ uniquely by requiring $\mathcal{G}^2/\varepsilon_0$ to equal the grade-complexity of the Clifford product, $n(n+1) = 72$ (Part B). The coupling is thus an operation on $\text{Cl}(3, 0)$ whose magnitude is set by that product. The Clifford product $e_i e_j$ is associative multiplication; composing amplitudes through it is multiplication. Hence half A.

Step 2 (B: the index is 2^{d_s}). ZC89 Step 4 (Part L) writes the force-layer step ratio in d_s -general form,

$$\frac{s_\Gamma}{s_{\text{rung}}} = \frac{2^{d_s} \ln 2}{\ln[d_s(2^{d_s} - d_s)] \cdot \ln r_{\text{born}}} \xrightarrow{d_s=3} \frac{8 \ln 2}{\ln 15 \cdot \ln r_{\text{born}}} = 36.9638,$$

with the leading 8 identified explicitly as 2^{d_s} (“ladder normalisation”), the total Clifford mode count. The coupling’s index is this 8. It is *not* the number-theoretic unit count $\varphi(15)$: the units $U(15) = (\mathbb{Z}/15)^\times \cong \mathbb{Z}_2 \times \mathbb{Z}_4$ are a finite abelian group of order 8, whereas the Clifford multiplicative structure is that of $\text{Cl}(3, 0) \cong M_2(\mathbb{C})$, whose unit group is $GL(2, \mathbb{C})$ — infinite. The two share the *number* 8 (via 2^{d_s} modes), not the group. Hence half B. □

Numerical verification (T-canonical):

Quantity	Value	Source
$\mathcal{G}^2/\varepsilon_0 = n(n+1)$	72	K35 grade-complexity
$N = 2^{d_s}$	8	ZC89 Step 4 (Clifford modes)
$\varphi(15)$	8	coincides at $d_s = 3$ (FIND-ZC102-01)
$\kappa_C = \ln 8/(8\sqrt{d_s})$	0.15007	unchanged (K7)

Status: Exact Theorem. EXPOSED-ZC101-01 derived; force index corrected to 2^{d_s} .

Physical reading: ZC101 reached the right number through the wrong door. It read the force 8 as “the residues mod 15 that can be inverted” — a number-theoretic fact about $\mathbb{Z}_{15}$. The coupling does not act on residues; it acts on the eight Clifford modes, and the 8 is 2^{d_s} , the count of those modes. In *T-canonical* the two doors open onto the same number, which is why ZC101’s result was correct; but the load-bearing object is the Clifford algebra, not the unit group. The next section explains why the two numbers agree at all.

0.82 Why the Two Eights Agree — and Only at $d_s = 3$ (FIND-ZC102-01)

The correction in Section 0.81 raises an obvious worry: if the force index is 2^{d_s} and not $\varphi(15)$, why did ZC101’s identification with $\varphi(15)$ give the right value? Because the two are forced equal at the framework’s gauge — and, as it turns out, *only* there.

FIND-ZC102-01: $\varphi(15) = 2^{d_s}$ Is a $d_s = 3$ Selection Signature (Gift)

Statement. Using the matter split $n_m = 2^{d_s} - d_s$ (so d_s, n_m are 3, 5 at $d_s = 3$) and CRT ($\mathbb{Z}_{15} \cong \mathbb{Z}_3 \times \mathbb{Z}_5$),

$$\varphi(15) = (d_s - 1)(n_m - 1), \quad 2^{d_s} = d_s + n_m,$$

and these are equal *only* at the Key-Identity solution $d_s = 3$:

d_s	$(d_s - 1)(n_m - 1)$	2^{d_s}	equal?
2	1	4	no
3	8	8	yes
4	33	16	no
5	104	32	no

Reading. The number-theoretic count $\varphi(15)$ and the Clifford count 2^{d_s} are different functions of d_s that cross at exactly one point, and that point is the unique root of the Key Identity $3d_s = 2^{d_s} + 1$. The coincidence is therefore *real* but *$d_s = 3$ -specific* — a new selection signature for $d_s = 3$ (alongside K35’s selection of $\text{Cl}(3, 0)$), not a general bridge between number theory and the Clifford algebra. ZC101’s $\varphi(15)$ and the correct 2^{d_s} name the same integer only because the framework sits at the one d_s where they meet.

Status: Exact Finding (Gift). d_s -general; the equality is not an identity (fails off $d_s = 3$).

This is the productive content of the scar: a wrong derivation path led to a right number, and asking *why* the number was right surfaced a property of $d_s = 3$ that neither paper had recorded. The same alphabet $(d_s, n_m, 2)$ builds the additive order 15, the Clifford count 8, and the unit count $\varphi(15)$, and the Key Identity is what ties them together at the single admissible dimension.

0.83 The Closure Discriminant (CONJ-ZC102-01)

With the force half grounded, the operation–group law can be stated in a form that reaches all three CRF layers. The organising question is not *which group* but *whether the operation closes*.

A group is a closed structure: composing two elements yields a third of the same kind, and every element has an inverse. Both the mass operation (addition on $(\mathbb{Z}_{15}, +)$) and the force operation (the Clifford product on the modes of $\text{Cl}(3, 0)$) close in this sense — and each yields a κ . The mind layer’s two motions do not. By DEF-ZC90-01 (Part L), the continuous step (*visesa*) and the discontinuous step (*nibbedha*) form a *quantum-pair*: deliberately not closed, with no third generator and no demanded inverse, and with the two magnitudes non-commensurable ($\mu_d/\mu_c \notin \mathbb{Q}$). Where the operation fails to close, there is no group and no κ — there is an irrational ratio instead.

CONJ-ZC102-01: Closure Selects Group-or-Q-pair (HYP)

Statement. For a CRF process drawing on the d_s -structure:

- if the process’s operation *closes* (composition stays within one kind, inverses exist), it acts on a **group** G , and its scale is

$$\kappa = \frac{\ln |G|}{|G|\sqrt{d_s}};$$

- if the operation *does not close* (two motions that cannot be composed into a third of like kind), it forms a **quantum-pair**, and its invariant is an *irrational ratio*, not a κ .

The three layers under the discriminant:

Layer	Operation	Closes?	Object	Invariant
Mass	addition (accumulate)	yes	$(\mathbb{Z}_{15}, +)$, $ G = 15$	$\kappa_R = f(15)$
Force	Clifford product (couple)	yes	$\text{Cl}(3, 0)$ modes, $2^{d_s} = 8$	$\kappa_C = f(8)$
Mind	continuous/discontinuous	no	Q-pair (DEF-ZC90-01)	$\mu_d/\mu_c \notin \mathbb{Q}$

Why HYP. The closed-mode entries are derived (ZC101 + THM-ZC102-01); the discriminant *as a single law* spanning closed and open modes is a structural reading, stated form-before-value in the discipline of ZC93. It is labelled HYP until the mind-side invariant is derived from the discriminant rather than matched to it.

Status: Conjecture (HYP).

OBS-ZC102-01: Both Group Orders Live Inside the Q-pair Ratio

The force Q-pair ratio (ZC89, irrational) is

$$\frac{s_{\Gamma}}{s_{\text{rung}}} = \frac{2^{d_s} \ln 2}{\ln |\mathbb{Z}_{15}| \cdot \ln r_{\text{born}}} = \frac{8 \ln 2}{\ln 15 \cdot \ln r_{\text{born}}},$$

which carries *both* the Clifford count $2^{d_s} = 8$ (numerator) and the additive order $|\mathbb{Z}_{15}| = 15$ (in $\ln 15$). The two group orders the κ -block reads separately appear together inside the one quantity that is provably irrational — the structural seam between the closed modes (groups, κ 's) and the open mode (Q-pair, irrationality). This is an observation, not a claim of mechanism.

Physical reading: The discriminant says the difference between mass/force and mind is not a difference of size but of closure. Adding and multiplying are operations you can complete — they make groups, and a group has a clean scale. The mind's two motions cannot be completed into one another: a smooth advance is not a finite number of breakthroughs, nor the reverse. That refusal to close is exactly what forces an irrational ratio rather than a rational scale. The same $d_s = 3$ arithmetic underlies all three, but only where it closes does it produce a κ .

0.84 Corrections and Open Items

PM-ZC101-02: The Force Index Was Read off the Wrong Eight (Failure Stance)

What happened. ZC101's force half located the coupling on the unit group $U(15)$ and named its order $\varphi(15) = 8$. The correct root is $2^{d_s} = 8$, the Clifford total-mode count (ZC89 Step 4); the coupling is the Clifford product, which acts on the eight modes of $\text{Cl}(3, 0)$, not on the invertible residues mod 15.

Why it was not an error of result. The value is unchanged: $N = 8$, $\kappa_C = f(8) = 0.15007$, and the whole κ -block stand. What was wrong was the *derivation path* — the object the 8 was attributed to. In T-canonical the two objects share the integer 8 (FIND-ZC102-01), so the mis-attribution did not propagate into the number.

Preservation. ZC101 v1 is kept as the record; this PM is the correction. The lesson is sharpened by an irony: ZC101's own PM-ZC101-01 warned “when an assignment resists, test whether it is the value of a structure not yet named.” ZC101 then named the 8 as $\varphi(15)$ without testing whether the structure was really $U(15)$ — it was $\text{Cl}(3, 0)$.

Pattern type: sector assignment.

Lesson — PWA candidate. Attribute a shared integer to the object the *operation* acts on, not to the first structure of that order that comes to hand. Here the operation is the Clifford product, so the 8 is 2^{d_s} .

GAP-ZC101-02 (carried): The 0.23% residual

Untouched by this paper. The derived $\kappa_R = f(15) = 0.10423$ remains 0.23% from the extracted 0.1045; the correction here is to the force-index path, not to the

mass-running value. Target: ZC102+ (a sub-leading accumulation term, not a fitted factor).

Atomic Registry — ZC102

ID	Description	Status
THM-ZC102-01	(A) coupling = Clifford product (K35 $\rightarrow$ Cl(3,0)), grounding EXPOSED-ZC101-01; (B) force index $N = 2^{d_s} = 8$, not $\varphi(15)$	Exact
FIND-ZC102-01	$\varphi(15) = (d_s - 1)(n_m - 1) = 2^{d_s} = d_s + n_m$ only at $d_s = 3$ (fails $d_s = 2, 4, 5$); a d_s -selection signature, not a bridge	Exact (Gift)
CONJ-ZC102-01	Closure discriminant: operation closes $\rightarrow$ group $\rightarrow \kappa$; does not close $\rightarrow$ Q-pair $\rightarrow$ irrational ratio; three layers tabulated	HYP
OBS-ZC102-01	The force Q-pair ratio carries both $2^{d_s} = 8$ and $ \mathbb{Z}_{15} = 15$; the seam between closed and open modes	Observation
PM-ZC101-02	ZC101 force index attributed to $U(15)$ ($\varphi(15)$); correct root 2^{d_s} (Cl(3,0)); result unchanged (pattern: sector assignment)	Scar
PM-ZC102-01	“8 vs 8” probe: tested $\varphi(15) = 2^{d_s}$ as a general bridge; failed off $d_s = 3$ (pattern: tautology test)	Scar

Gap-status change: EXPOSED-ZC101-01 (coupling = multiplication) **DERIVED** via THM-ZC102-01. *Inherited verbatim:* K7, K35, Cl(3,0) grade-complexity (Part B); ZC89 Step 4, DEF-ZC90-01 (Part L); THM-ZC101-01/02/03 (ZC101 — result unchanged, force path corrected).

0.85 R-Layer Research Note

R-Layer Assignments for ZC102

Atomic unit	R-layer	Cost T	Source
THM-ZC102-01	R_0	0	an algebraic identification (coupling = product)
FIND-ZC102-01	R_0	0	an arithmetic property of $d_s = 3$
CONJ-ZC102-01	R_0	0	a structural classification across layers
OBS-ZC102-01	R_0	0	a comparison of established quantities
PM-ZC101-02	R_0	0	a correction of attribution
PM-ZC102-01	R_0	0	a refuted bridge probe

All of ZC102 is R_0 : it identifies, classifies, and corrects relations among quantities

already established at their own layers. No new R_1 running and no Born-chain bridge is computed here; the R_1 content (the value of κ_R and its residual) is inherited from ZC100/ZC101 and left at GAP-ZC101-02.

0.86 Genealogy Map

How ZC102 Sits in the Corpus — Grounding and Correcting

Branch A — the κ -block: ZC99 (boundary; κ_R extracted) $\rightarrow$ ZC100 (accumulation set up; HYP form) $\rightarrow$ ZC101 (operation-group law; $\kappa_R = f(15)$, $\kappa_C = f(8)$ derived; force half on a premise) $\rightarrow$ **ZC102:** the premise is grounded in the Clifford product (THM-ZC102-01), and the force index is corrected from $\varphi(15)$ to 2^{d_s} (PM-ZC101-02), with the result unchanged.

Branch B — the Clifford foundation: Part B/K35 (the coupling = grade-complexity of $\text{Cl}(3, 0)$; $n(n+1) = 72$) $\rightarrow$ ZC89 Step 4 (the force ratio in d_s -general form, $8 = 2^{d_s}$) $\rightarrow$ **ZC102:** these two fix what a coupling is (a Clifford product) and what index it carries (2^{d_s}), supplying the derivation ZC101 had left as a premise.

Branch C — the mind layer: Part F (the four-way choice; the forward pair) $\rightarrow$ ZC90 (DEF-ZC90-01: the Q-pair, “why not a group”) $\rightarrow$ **ZC102:** the Q-pair becomes the *open* mode of the closure discriminant (CONJ-ZC102-01) — where the operation does not close, no κ forms and the invariant is irrational.

Convergence at ZC102: The Key Identity $3d_s = 2^{d_s} + 1$ supplies one alphabet ($d_s, n_m, 2$) from which the additive order 15, the Clifford count 8, and the unit count $\varphi(15)$ are all built; FIND-ZC102-01 shows the latter two coincide only at $d_s = 3$. The three layers are one $d_s = 3$ arithmetic read three ways — additive closure (mass), multiplicative closure (force), and non-closure (mind).

ZC102’s contribution: it turns ZC101’s stated premise into a derived theorem, corrects the one object ZC101 mis-named without disturbing its result, and gives the operation-group law a boundary — closure — that places the mind layer in the same structure as mass and force.

0.87 Closing Sentence

A coupling was said to multiply; now we know why — it is the Clifford product, the operation that builds the algebra of the eight modes, and multiplication is what that product is. The eight it carries is the count of those modes, 2^{d_s} , not the count of invertible residues; the two are the same integer only because the framework sits at the one dimension where they meet. And the law has a boundary we can name: where an operation closes — where you can add, or multiply, and stay in one kind — it makes a group and a clean scale; where it cannot close, as the mind’s smooth and sudden steps cannot, it makes a quantum-pair and an irrational ratio instead. Mass adds, force multiplies, mind refuses to close, and one arithmetic at one dimension carries all three.

$$\text{closes} \Rightarrow \kappa = \frac{\ln |G|}{|G|\sqrt{d_s}}; \quad \text{does not close} \Rightarrow \frac{\mu_d}{\mu_c} \notin \mathbb{Q}.$$

แรงเคยถูกกล่าวว่า “คุณ” --- บัดนี้เรารู้แล้วว่าเพราะอะไร: มันคือผลคูณคลิฟฟอร์ด

การกระทำที่ก่อร่างพีชคณิตของแปดโหมด และการคูณคือสิ่งที่ผลคูณนั้นเป็น ---
 แปดที่มันถือไว้คือจำนวนของโหมดเหล่านั้น 2^{d_s} มิใช่จำนวนเศษที่หาตัวผกผันได้
 ทั้งสองเป็นจำนวนเดียวกันก็เพราะกรอบตั้งอยู่ ณ มิติเดียวที่ทั้งสองมาบรรจบ ---
 และกฎนี้มีขอบเขตที่เราเรียกชื่อได้: ที่ใดการกระทำปิดตัว --- ที่ซึ่งบวกได้ หรือคูณได้
 แล้วยังอยู่ในชนิดเดียว --- ที่นั่นเกิดกรุปและสเกลอันสะอาด; ที่ใดปิดตัวไม่ได้
 ดังก้าวเรียบและก้าวกระโดดของจิตที่ปิดเข้าหากันไม่ได้
 ที่นั่นเกิดคู่ควอนตัมและอัตราส่วนอตรรกยะแทน --- มวลบวก แรงคูณ จิตปฏิเสธการปิดตัว
 และเลขคณิตหนึ่งเดียว ณ มิติหนึ่งเดียว คำจูนทั้งสาม.

0.88 ZC103 — The Mass-Sector Simulator

Abstract

The mass sector had accumulated value-claims it could not test, because testing needs a numerical instrument and none had been built. This paper introduces `CRF_MassSector_V1.py`, which takes only the Key Identity ($3d_s = 2^{d_s} + 1$, physical branch $d_s = 3$) and the derived running constant $\kappa_R = f(15)$ (ZC101), and computes the charged-fermion spectrum directly.

FIND-ZC103-01 (positive): with $\kappa_R = f(15)$ and *no* free parameter — L_0 cancels identically in any mass ratio — the lepton $\text{gen2} \rightarrow 3$ ratio τ/μ is reproduced to -0.2% , and the implied κ_R to 0.77% . The running law is confirmed at that rung.

FIND-ZC103-02 (negative, quantified): the six quark masses do not sit on the lepton $\{1, 3, 5\}$ ladder under one (κ_R, L_0) ; the per-step κ_R across four quark and two lepton rungs scatters at 215% .

FIND-ZC103-03 (negative, located): the surviving deviation is a *first-rung* ($\text{gen1} \rightarrow 2$) effect in every sector, where the step crosses from the $R_H = 1$ reference. Electromagnetic- Q^2 , single-base-ladder, and uniform-curvature explanations are tested and rejected.

GAP-ZC103-01 reframes GAP-ZC101-02: the residual is a boundary-rung effect, not a global correction to κ_R .

0.89 Background: Why a Simulator, and What It Is Asked

Across roughly a hundred papers the framework has derived the *forms* of its mass relations — the ladder $m_H \propto R_H^{n_m}$, the running correction $c(R_H) = \kappa_R(\ln R_H - L_0)$, the constant $\kappa_R = f(15)$ — while deferring their precise numerical confrontation with data, because such a confrontation needs a computational instrument and none had been written. This paper supplies one, `CRF_MassSector_V1.py`, and reports what it settles. The instrument is fed only what the framework derives (the dimension $d_s = 3$ from the Key Identity, and $\kappa_R = f(15)$) and asked a single question: does the spectrum that follows match the measured fermion masses? The terms needed are few: a *rung* (one generation step on the ladder), the *reference* (the electron, sitting at compression $R_H = 1$ where the running cost is zero), and an *over-determination* (more data than constants — here six quark masses against two numbers — which can confirm or refute but not be tuned).

Numerical Summary — ZC103 (T-canonical)

Quantity	Value	Status
$\kappa_R = f(15)$ (input, derived)	0.10423	ZC101
lepton τ/μ predicted (no params)	−0.2%	FIND-ZC103-01
implied κ_R at gen2→3	0.10503	0.77% from $f(15)$
quark+lepton κ_R scatter	215%	FIND-ZC103-02
Q^2 self-energy explanation	rejected	FIND-ZC103-03
single-base ladder	rejected	FIND-ZC103-03
uniform curvature s	rejected	FIND-ZC103-03
gen1→2 rung effect	open	GAP-ZC103-01

0.89.1 What the framework supplies to the simulator

The simulator is given the derived skeleton and nothing tunable:

- The Key Identity $3d_s = 2^{d_s} + 1$ has integer solutions $d_s \in \{1, 3\}$; the degenerate $d_s = 1$ gives $n_m = 2^1 - 1 = 1$ (no matter content), so the physical branch ($n_m \geq 2$) selects $d_s = 3$, whence $n_m = 5$, $|\mathbb{Z}_{15}| = 15$.
- The ladder $m_H \propto R_H^{n_m}$ with $R_H = 15/|H| \in \{1, 3, 5\}$ (ZC15/16).
- The running correction $c(R_H) = \kappa_R(\ln R_H - L_0)$ with $\kappa_R = f(15) = \ln 15/(15\sqrt{d_s})$ derived (ZC101). L_0 is the only quantity not derived from d_s — and, as shown below, it cancels in every mass ratio, so the ratio tests are parameter-free.

0.89.2 What Was Already Known Before ZC103

Pre-ZC103 Status

Known:

- $\kappa_R = f(15) = 0.10423$ derived (ZC101); extracted 0.1045 (ZC99 v2); leading residual 0.23% (GAP-ZC101-02).
- The quark sector over-determines (κ_R, L_0) and was handed the test by ZC95 (GAP-ZC95-01), unresolved.
- No mass-sector simulator existed.

Open:

- Does $\kappa_R = f(15)$ reproduce the measured spectrum?
- Where does the 0.23% residual live — in κ_R , or elsewhere?
- Do the quark masses close on the same constants?

ZC103 answers all three numerically (FIND-ZC103-01/02/03).

What ZC103 Does NOT Do

- Does not close GAP-ZC101-02 — it *relocates* the residual to the first rung (GAP-ZC103-01).
- Does not close the quark sector — it quantifies the failure of the naive shared ladder.

- Does not derive the first-rung effect — it records three candidates rejected.
- Does not alter $\kappa_R = f(15)$ or any inherited result.

Companion Papers — ZC103

Paper / Instrument	What it provides to ZC103
CRF_ZC101_..._v1.tex	$\kappa_R = f(15)$ derived — the input
CRF_ZC99_..._v2.tex	extracted $\kappa_R = 0.1045$, $L_0 = 1.4028$
CRF_ZC95_..._v1.tex	$c(R_H) = \kappa_R(\ln R_H - L_0)$; the ladder
CRF_...Part_C.tex	$m_H \propto R_H^{n_m}$; $\mathbb{Z}_{15} = \mathbb{Z}_3 \times \mathbb{Z}_5$
CRF_MassSector_V1.py	the simulator — this paper's instrument

0.89.3 Pre-Work Audit (PWA-1 through PWA-10)

Pre-Work Audit Record — ZC103

Scanned before writing (Upgrade Protocol v1.3):

- ZC101 ($\kappa_R = f(15)$), ZC99 v2 (L_0 , extracted κ_R), ZC95 (running form), Part C (ladder, CRT) — **KEY PILLARS**.

PWA-6: macros via `providecommand`; no collision.

PWA-7–9: not triggered (no new floor, layer, or spectral convention).

PWA-10 (tautology/artefact audit): *triggered and caught* — an initial per-rung κ_R extraction folded L_0 inconsistently across rungs and manufactured a false 0.13 spread; the correct extraction (in which L_0 cancels) was used for all reported numbers. Logged as PM-ZC103-01.

No duplicate atomic units found.

0.90 FIND-ZC103-01: The Running Law Holds at gen2→3, No Free Parameter

The single cleanest test is the lepton gen2→3 rung, because L_0 cancels in the ratio and the prediction uses only the derived κ_R .

FIND-ZC103-01: τ/μ from $f(15)$ alone (R_1 , numerical)

Statement. With $\kappa_R = f(15)$ and no free parameter, the lepton gen2→3 mass ratio is

$$\left(\frac{m_\tau}{m_\mu}\right)_{\text{pred}} = \left(\frac{5}{3}\right)^{n_m} \exp\left[n_m \kappa_R (\ln 5 - \ln 3)\right] = 16.78, \quad \text{observed } 16.82 \text{ } (-0.2\%).$$

The implied running constant from this rung alone is $\kappa_R = 0.10503$, which is 0.77% from the derived $f(15) = 0.10423$.

Why parameter-free. In any mass *ratio*, the offset L_0 enters both rungs as $\kappa_R L_0$ and cancels: $c(R_H^{\text{hi}}) - c(R_H^{\text{lo}}) = \kappa_R (\ln R_H^{\text{hi}} - \ln R_H^{\text{lo}})$. So the gen2→3 prediction depends on κ_R alone, and κ_R is derived. There is nothing to tune.

Numerical verification:

Rung	Predicted	Observed	Deviation
τ/μ (gen2→3)	16.78	16.82	-0.2% (mass)
implied κ_R	0.10503	$f(15) = 0.10423$	0.77%

Status: Positive finding. The running law is confirmed at the gen2→3 rung to sub-percent, parameter-free.

Physical reading: This is the first time the derived $\kappa_R = f(15)$ has been put against data with nothing to adjust, and it lands within a percent at the rung where the running is cleanest. The lepton masses, expressed in base $|\mathbb{Z}_{15}|$, are close to $m_\mu/m_e \sim 15^2$ and $m_\tau/m_\mu \sim 15^1$ — the parameter-free ladder — and the running supplies the small, correctly-signed bend on top. That the law holds here is the positive anchor of the paper; the negatives that follow are about where it does not, not about the constant itself.

0.91 FIND-ZC103-02: Quarks Do Not Share the Lepton Ladder

The six quark masses over-determine (κ_R, L_0) . Placing up-type $\{u, c, t\}$ and down-type $\{d, s, b\}$ each as a three-generation ladder at $R_H \in \{1, 3, 5\}$ and extracting the per-step κ_R gives the spread below.

FIND-ZC103-02: 215% scatter across six rungs (R_1 , numerical)

Statement. The per-step κ_R extracted from four quark rungs and two lepton rungs, assuming the shared $\{1, 3, 5\}$ ladder, scatters at std/mean = 215%:

Sector	Step	ratio	implied κ_R
up	c/u (gen1→2)	588	0.161
up	t/c (gen2→3)	136	0.923
down	s/d (gen1→2)	20	-0.455
down	b/s (gen2→3)	45	0.488
lepton	μ/e	207	-0.029
lepton	τ/μ	17	0.105

mean 0.199, std 0.428. No single κ_R governs all six.

Status: Negative, quantified. GAP-ZC94/95-01 is *not* closed by the naive shared ladder. The quark sector carries structure beyond it.

Physical reading: The clean lepton result does not extend to the quarks on the simplest hypothesis. This is not a failure of $\kappa_R = f(15)$ — which the lepton gen2→3 rung confirms — but a sign that the quark generations are not a plain copy of the lepton ladder. The pattern within the scatter is itself informative: in every sector the gen2→3 step sits closer to a common value than the gen1→2 step, which points at the first rung, not the running constant, as the locus of the trouble.

0.92 FIND-ZC103-03: The Deviation Is a First-Rung Effect

Across leptons, up-type, and down-type, the step that deviates is the first one, gen1→2 — the step that crosses *from* the $R_H = 1$ reference, where the running cost $c(R_H) = \kappa_R(\ln R_H - L_0)$ has $\ln R_H = 0$ and the running term vanishes. The gen2→3 step, by contrast, matches the law (FIND-ZC103-01 and the quark table). We tested three structural explanations for the first-rung deviation and rejected all three.

GAP-ZC103-01: The first-rung effect (Priority 1; candidates rejected)

Statement. With $\kappa_R = f(15)$, the gen1→2 step deviates from the running-corrected ladder in every sector; gen2→3 does not. The deviation is located at the boundary rung, not in κ_R .

Candidates tested and rejected:

1. **Electromagnetic Q^2 self-energy.** If the deviation scaled with charge-squared, down-type ($Q^2 = 1/9$) would deviate least. It deviates *most* (ln-deviation -3.07 vs lepton -0.73, up +0.31). Sign and ordering are backwards. **Rejected.**
2. **Single-base power ladder.** Quark generation ratios are clean powers of *different* bases ($c/u \approx 5^4$, $t/c \approx 3^{4.5}$, $s/d \approx 8^{1.5}$, $b/s \approx 2^{5.5}$), not one shared base. **Rejected.**
3. **Uniform curvature sub-leading term.** Fixing $\kappa_R = f(15)$ and fitting a curvature term $s(\ln R_H)^2$, the s demanded by the two lepton rungs differs by a factor ~ 400 ($s_{21} = -0.122$, $s_{32} = +0.0003$). A single s does not close both. **Rejected.**

Reframe of GAP-ZC101-02. The 0.23% κ_R residual is not spread evenly and is not a global correction to κ_R ; it concentrates at the gen1→2 boundary rung. Closing it means understanding why the first rung — the crossing from the $R_H = 1$

reference — carries an extra term, not adjusting κ_R .

Status: Open. The effect is real and located; no tested candidate explains it.
Target: ZC104.

PM-ZC103-01: A per-rung κ_R folds in L_0 and fakes a spread (Failure Stance)

What happened. An initial extraction measured κ_R rung-by-rung in a way that folded the offset L_0 into each rung differently, producing an apparent 0.13 spread between the two lepton rungs and inviting the reading “ κ_R is not well-defined.”

Why it was an artefact. In any mass ratio L_0 cancels; the clean per-rung κ_R is obtained from the ratio alone. The spread was an artefact of the inconsistent fold, not physics.

Preservation. Recorded so the clean extraction is the one used. The real (small) residual is the $\text{gen1} \rightarrow 2$ effect of GAP-ZC103-01, not a κ_R ambiguity.

Pattern type: extraction artefact.

Lesson — PWA candidate. Before reading a spread as physics, verify the quantity is measured against the same baseline at every point; a quantity that cancels (here L_0) must be cancelled, not folded in unevenly.

Atomic Registry — ZC103

ID	Description	Status
FIND-ZC103-01	Lepton τ/μ from $\kappa_R = f(15)$, no free parameter (L_0 cancels): -0.2% in mass, implied κ_R 0.77% from $f(15)$	Positive
FIND-ZC103-02	Six quark masses do not share the lepton $\{1, 3, 5\}$ ladder under one (κ_R, L_0) ; per-step κ_R scatter 215%	Negative (quant.)
FIND-ZC103-03	The deviation is a $\text{gen1} \rightarrow 2$ (boundary-rung) effect; Q^2 , single-base-ladder, and uniform-curvature explanations rejected	Negative (located)
GAP-ZC103-01	The first-rung effect: real, located, no tested candidate; reframes GAP-ZC101-02 as a boundary-rung term, not a κ_R correction	Open
PM-ZC103-01	A per-rung κ_R that folds L_0 in unevenly fakes a spread; the clean (L_0 -cancelled) extraction is correct (extraction artefact)	Scar

Instrument: CRF_MassSector_V1.py. *Inherited verbatim:* $\kappa_R = f(15)$ (ZC101); L_0 , extracted κ_R (ZC99 v2); $c(R_H) = \kappa_R(\ln R_H - L_0)$ and the ladder (ZC95, ZC15/16). No value modified.

0.93 R-Layer Research Note

R-Layer Assignments for ZC103

Atomic unit	R-layer	Cost T	Source
FIND-ZC103-01	R_1	—	a value test of the R_1 running
FIND-ZC103-02	R_1	—	a value test across sectors
FIND-ZC103-03	R_1	—	a located residual in the R_1 running
GAP-ZC103-01	R_1	—	the boundary-rung term, R_1
PM-ZC103-01	R_0	0	a correction of extraction method

The findings are R_1 : direct numerical tests of the resolution-running of the mass sector against data. The residual GAP-ZC103-01 lives at R_1 , at the $R_H = 1$ boundary rung. PM-ZC103-01 is R_0 (a methodological correction). No Born-chain bridge enters.

0.94 Genealogy Map

How ZC103 Sits in the Corpus — the Instrument Returns

Upstream. ZC95 derived the running form and handed the value test to a future simulator; ZC99 v2 extracted (κ_R, L_0) from the lepton poles; ZC101 derived $\kappa_R = f(15)$; ZC102 grounded the coupling and the n_m -vs- n distinction. None of these confronted the full measured spectrum numerically, because no mass-sector simulator existed.

This paper. `CRF_MassSector_V1.py` performs that confrontation. It confirms the running law at the lepton $\text{gen}2 \rightarrow 3$ rung parameter-free (FIND-ZC103-01), quantifies the failure of the naive quark ladder (FIND-ZC103-02), and locates the surviving deviation at the first rung with three candidates rejected (FIND-ZC103-03), reframing GAP-ZC101-02 as a boundary-rung effect.

What stands, what is reframed, what is open. $\kappa_R = f(15)$ and all ZC94–ZC102 results stand. GAP-ZC101-02 is *reframed* (not closed): the residual is a $\text{gen}1 \rightarrow 2$ term, not a wrong constant. GAP-ZC94/95-01 is *quantified* (not closed): the quark scatter is 215% on the shared ladder. A new, sharper open item is registered (GAP-ZC103-01).

Forward. The first-rung effect (GAP-ZC103-01) is the live target: it is present in every sector, it sits at the $R_H = 1$ reference where the running vanishes, and it is not electromagnetic, not single-base, not uniform-curvature. The simulator is in place to test the next candidate. The mass sector now has a working instrument; “form before value” becomes “form confirmed at one rung, value-residual located at another.”

0.95 Closing Sentence

After a hundred papers of deriving forms and deferring their proof, the mass sector finally has an instrument, and the first thing it does is confirm: the running law, with one derived number and nothing to tune, gives the muon-to-tau step to a fifth of a

percent. The same instrument is equally honest about what does not work — the six quarks refuse the lepton ladder, and the deviation that remains is not in the constant but in the first rung, the crossing out of the reference, where charge-squared, single bases, and smooth curvature all fail to account for it. We have not closed the gap; we have moved it to where it actually lives, and we have built the thing that can test the next idea. That is what a simulator is for, and after a hundred papers it is good to have one again.

$$\left(\frac{m_\tau}{m_\mu}\right)_{\text{pred}} = 16.78, \quad \text{observed } 16.82, \quad \kappa_R = f(15) \text{ alone.}$$

หลังจากรอยเปเปอร์ที่อนุมานรูปแล้วเลื่อนการพิสูจน์ออกไป ภาคมวลก็มีเครื่องมือเสียที่
 และสิ่งแรกที่มันทำคือยืนยัน --- กฎการไหล
 ด้วยจำนวนที่อนุมานได้หนึ่งตัวและไม่มีอะไรให้ปรับ
 ให้ก้าวจากมิวออนสู่เทาได้แม่นยำถึงหนึ่งในห้าของเปอร์เซ็นต์ ---
 เครื่องมือเดียวกันก็ซื้อสัตย์พอกันต่อสิ่งที่ใช้ไม่ได้: ควาร์กทั้งหกปฏิเสธบันไดของเลปตอน
 และส่วนต่างที่เหลือมีได้อยู่ที่ค่าคงที่ หากแต่อยู่ที่ขั้นแรก --- ขั้นที่ก้าวออกจากจุดอ้างอิง
 ที่ซึ่งประจักษ์กำลังสอง ฐานเดียว และความโค้งเรียบ ล้วนอธิบายมันไม่ได้ ---
 เรายังมิได้ปิดช่องว่าง เราเพียงย้ายมันไปยังที่ที่มันอยู่จริง
 และเราได้สร้างสิ่งที่ทดสอบความคิดถัดไปได้ --- นั่นคือหน้าที่ของเครื่องจำลอง
 และหลังจากรอยเปเปอร์ มันดีที่ได้มีอีกครั้ง.

0.96 ZC104 — The First-Rung Effect: Seven Candidates Excluded

Abstract

ZC103 located the charged-fermion mass residual at the first rung (gen1→2, the crossing from the $R_H = 1$ reference) and rejected three structural explanations. This paper builds the second-generation instrument, `CRF_MassSector_V2.py`, to attack that located gap, under one rule: free exactly one quantity per test, with $\kappa_R = f(15)$ and $n_m = 5$ fixed.

FIND-ZC104-01 (positive): the lepton $\{1, 3, 5\}$ ladder is reconfirmed on three independent parameter-free diagnostics — the gen2→3 residual (0.002 in ln-mass), the boundary-invariant within-sector half-difference (0.001), and the free- R_H control ($R_H \approx 3, 5$).

FIND-ZC104-02 (negative, quantified): the first-rung deviation is stable and sector-ordered (up +0.31, down −3.07, lepton −0.73 in ln-mass), with up and down opposite in sign. It is *not* charge in any form (Q , T_3 , Q^2 , $|Q|$ fail the two-rung test) and *not* a doublet-partner coupling.

GAP-ZC104-01 carries the effect forward with *seven* candidates now excluded. **PM-ZC104-01** and **PM-ZC104-02** record two extraction scars: a near-structural free- R_H value that fails forward prediction, and a difference exact to machine precision that proved an algebraic identity.

0.97 Background: A Located Gap and a Sharper Instrument

The framework derives the *forms* of its mass relations — a ladder $m_H \propto R_H^{n_m}$ in which each generation sits at a compression $R_H \in \{1, 3, 5\}$, and a running correction $c(R_H) = \kappa_R(\ln R_H - L_0)$ built on the derived constant $\kappa_R = f(15)$. ZC103 put these against the

measured fermion masses with a first simulator and found a clean picture with one sharp flaw: the law works, but a residual survives, and that residual lives entirely at the *first rung* — the step $\text{gen1} \rightarrow 2$ that crosses out of the reference generation (the electron, at $R_H = 1$, where the running cost is exactly zero). This paper takes that located gap as its target. The terms needed are few: a *rung* (one generation step), the *reference* (the lightest generation, $R_H = 1$), an *over-determination* (more data than constants, which can refute but not be tuned), and a *two-rung test* (the demand that any proposed term explain *both* generation steps, not just the one it was fitted to). The result is negative and, in this framework, valuable: seven structural explanations are excluded, and the shape of what remains is mapped precisely.

Numerical Summary — ZC104 (T-canonical)

Quantity	Value	Status
$\kappa_R = f(15)$ (input, derived)	0.10423	ZC101
lepton $\text{gen2} \rightarrow 3$ residual (ln-mass)	+0.0020	FIND-ZC104-01
lepton half-difference (b -invariant)	+0.0010	FIND-ZC104-01
first-rung dev. up / down / lepton	+0.31 / -3.07 / -0.73	FIND-ZC104-02
boundary offset b (one / per-sector)	rejected	§0.100
$8 = \phi(15)$ up-sector index	rejected	PM-ZC104-01
charge-linear ($Q, T_3, Q^2, Q $)	rejected	FIND-ZC104-02
doublet-partner coupling	rejected	FIND-ZC104-02
candidates excluded (ZC103+ZC104)	$3 + 4 = 7$	GAP-ZC104-01

0.97.1 What the instrument is given, and the one rule it obeys

CRF_MassSector_V2.py is fed only the derived skeleton:

- the Key Identity $3d_s = 2^{d_s} + 1$, physical branch $d_s = 3$, whence $n_m = 5$ and $|\mathbb{Z}_{15}| = 15$;
- the ladder $m_H \propto R_H^{n_m}$ with $R_H \in \{1, 3, 5\}$ (ZC15/16);
- the running constant $\kappa_R = f(15) = \ln 15 / (15\sqrt{d_s})$ (ZC101), held fixed.

The one rule that governs every test in this paper: **free exactly one quantity at a time**. When the boundary offset b is tested, R_H is fixed to the ladder; when R_H is freed, b is set to zero. The ladder is $R_H^{n_m}$ with $n_m = 5$, so freeing both R_H and an additive offset together would let a one-fifth power absorb almost anything — an over-fit by construction. This is the E4 discipline made operational: no candidate is admitted on a single fit.

0.97.2 What Was Already Known Before ZC104

Pre-ZC104 Status

Known:

- $\kappa_R = f(15) = 0.10423$ derived (ZC101); the lepton $\text{gen}2 \rightarrow 3$ rung confirms it parameter-free to -0.2% in mass (FIND-ZC103-01).
- The residual is a first-rung effect, present in every sector (FIND-ZC103-03); GAP-ZC101-02 reframed as a boundary-rung term, not a κ_R correction.
- Three explanations already rejected (ZC103): electromagnetic Q^2 self-energy, a single-base power ladder, a uniform curvature term.
- ZC95 fixes $c(R_H = 1) = 0$ *exactly* by construction — the lightest generation *is* the reference.

Open:

- Is the first-rung deviation a boundary offset that switches on once a generation leaves the reference?
- Do the quarks live on $\mathbb{Z}_{15}$ at a non-lepton assignment?
- Is the deviation a charge effect, or a doublet-partner effect?

ZC104 answers all three, negatively and with quantified bounds.

What ZC104 Does NOT Do

- Does not close GAP-ZC103-01 / GAP-ZC101-02 — it *sharpens* them by excluding four further candidates (seven total).
- Does not derive the first-rung effect — it bounds the space of explanations and records what the effect is *not*.
- Does not close the quark sector — it confirms the locus is the first rung, sector-ordered with opposite up/down signs.
- Does not alter $\kappa_R = f(15)$, n_m , or any inherited result.

Companion Papers — ZC104

Paper / Instrument	What it provides to ZC104
CRF_ZC103_..._v1.tex	GAP-ZC103-01 — the located first-rung gap
CRF_ZC101_..._v1.tex	$\kappa_R = f(15)$ derived — the fixed input
CRF_ZC102_..._v1.tex	n_m vs n ; multiplicative $\phi(15) = 8$
CRF_ZC95_..._v1.tex	$c(R_H) = \kappa_R(\ln R_H - L_0)$; $c(1) = 0$ exact
CRF_...Part_C.tex	$m_H \propto R_H^{n_m}$; the $\{1, 3, 5\}$ ladder
CRF_MassSector_V2.py	the second-generation instrument

0.97.3 Pre-Work Audit (PWA-1 through PWA-10)

Pre-Work Audit Record — ZC104

Scanned before writing (Upgrade Protocol v1.3):

- ZC103 (first-rung locus; three candidates rejected), ZC101 ($\kappa_R = f(15)$), ZC102 (n_m vs n ; $\phi(15) = 8$), ZC95 (running form; $c(1) = 0$ exact), Part C (ladder) — **KEY PILLARS**.

PWA-6: macros via `providecommand`; no collision with Preamble v3.1. (Symbol κ collision noted corpus-wide as BUG-KAPPA-COLLISION; this paper uses only κ_R , unambiguous here.)

PWA-7–9: not triggered (no new floor, layer, or spectral convention).

PWA-10 (tautology/artefact audit): *triggered twice and caught* — (i) a free- R_H value landing near $8 = \phi(15)$ was an inverse fit to one datum that fails forward prediction (PM-ZC104-01); (ii) a sector difference exact to 4×10^{-16} proved to be an algebraic identity, not physics (PM-ZC104-02). Both logged; neither admitted as a result.

No duplicate atomic units found.

0.98 The Diagnostic Frame: Deviations at Fixed κ_R

All tests in this paper read off two per-sector, per-rung *deviations* from the running-corrected ladder, with $\kappa_R = f(15)$ and $n_m = 5$ held fixed. For a sector with generation masses (m_1, m_2, m_3) on $R_H \in \{1, 3, 5\}$,

$$\delta_1 \equiv \ln \frac{m_2}{m_1} - n_m(1 + \kappa_R)(\ln 3 - \ln 1), \quad (8)$$

$$\delta_2 \equiv \ln \frac{m_3}{m_2} - n_m(1 + \kappa_R)(\ln 5 - \ln 3). \quad (9)$$

δ_1 is the first-rung deviation (gen1→2), δ_2 the second (gen2→3). Because L_0 cancels in every ratio (ZC103, FIND-ZC103-01), both are parameter-free in L_0 . The measured values are:

Sector	δ_1 (gen1→2)	δ_2 (gen2→3)
up	+0.311	+2.092
down	−3.070	+0.981
lepton	−0.734	+0.002

The lepton $\delta_2 \approx 0$ is FIND-ZC103-01 in this notation; the structure of the rest is the subject of the paper.

0.99 FIND-ZC104-01: The Lepton Ladder, Reconfirmed Three Ways

Before excluding explanations for the deviation, we fix the positive anchor: the lepton $\{1, 3, 5\}$ ladder with $\kappa_R = f(15)$ is correct, on three independent diagnostics that share no fitted quantity.

FIND-ZC104-01: Three parameter-free confirmations of the lepton ladder (R_1 , numerical)

Statement. The lepton sector satisfies the running-corrected ladder on three diagnostics, each parameter-free:

1. **Second-rung residual.** $\delta_2^{\text{lepton}} = +0.0020$ in ln-mass — the gen2→3 step matches $n_m(1 + \kappa_R)(\ln 5 - \ln 3)$ to two parts in a thousand.
2. **Boundary-invariant half-difference.** The within-sector quantity $\frac{1}{2}(\delta_2 - \delta_1)$, which any per-generation offset b leaves unchanged, is +0.0010 for leptons (against +1.046 for up, +0.490 for down). The lepton rungs are mutually consistent; the quark rungs are not.
3. **Free- R_H control.** Solving for the R_H each lepton generation demands, with $\kappa_R = f(15)$ fixed and nothing else free, returns $R_H^{(2)} = 2.63$ and $R_H^{(3)} = 4.38$ — consistent with the ladder values 3 and 5 (the running-folded coefficient shifts them inward slightly).

Status: Positive finding. The lepton ladder is the one sector the running-corrected $\{1, 3, 5\}$ assignment governs cleanly; it is the fixed anchor against which the first-rung effect is measured.

Physical reading: The three diagnostics are deliberately redundant: a single clean number can be a coincidence, but a residual, a difference, and an inverse solution that all agree leave little room for one. The lepton sector behaves exactly as the derived ladder predicts, which is what licenses reading the quark deviations as structure in the quarks — not as a defect in κ_R or in the ladder form. Every negative that follows is anchored here.

0.100 The Boundary-Offset Candidate, Excluded

The most natural reading of a first-rung effect is a *boundary offset*: a constant b that switches on once a generation leaves the $R_H = 1$ reference. Since ZC95 fixes $c(R_H = 1) = 0$ exactly, the deviation cannot be a modification of c (which is pinned to zero at the reference); a separate offset is the structurally honest form to test. We test two versions — one shared b across all sectors, one b per sector — and compare by AIC against the no-offset null.

Boundary offset b : AIC-compared and excluded as artefact

Setup. Model $\ln(m_g/m_1) = n_m(1 + \kappa_R)(\ln R_{H,g} - \ln R_{H,1}) + b \cdot \mathbf{1}[g > 1]$, with κ_R, n_m fixed and only b free. The indicator is 1 for every non-reference generation, so b is a single lift once a generation has left the reference — not a per-rung continuous term (the structural contrast with the rejected uniform curvature).

Result. Over the six $\ln$ -ratio data points:

Model	k	RSS	AIC
null (no b)	0	30.99	+12.92
B_1 (one shared b)	1	29.84	+14.69
B_2 (per-sector b)	3	2.49	+ 6.61

Why B_2 's lower AIC is an artefact, not evidence. Each sector supplies *two* data points (δ_1, δ_2) , and B_2 fits *one* b per sector as their mean. The residuals it leaves are therefore $\pm \frac{1}{2}(\delta_2 - \delta_1)$ by construction — a forced antisymmetry that carries no information about whether b is real. The quantity b cannot touch is exactly the half-difference of FIND-ZC104-01, which for the quarks is large (1.046, 0.490) and is left *entirely intact*. A per-sector mean absorbs the offset for free; it does not explain the within-sector structure, which is the actual residual.

Status: Boundary offset **rejected** (candidate 4). Logged as the basis for PM-ZC104-02's wider lesson on degrees of freedom.

0.101 The $\phi(15)$ Candidate, Excluded (PM-ZC104-01)

If the quarks do not sit on the lepton $\{1, 3, 5\}$ ladder, perhaps they sit on $\mathbb{Z}_{15}$ at a different assignment. Freeing R_H for each quark generation (with $\kappa_R = f(15)$, $n_m = 5$ fixed, $b = 0$) and reading off the nearest structural value gives one striking near-hit: up-type gen3 wants $R_H = 7.73 \approx 8 = \phi(15)$. We tested it from four independent angles and rejected it.

PM-ZC104-01: $8 = \phi(15)$ as an up-index — an inverse fit that fails forward (Failure Stance)

What happened. The free- R_H solution for up-type gen3 returned 7.73, within 3.4% of $8 = \phi(15)$ — the order of the multiplicative unit group $U(15)$, and the very number ZC102 uses for the force coupling. The temptation was to read up-type as living on $\{1, 3, 8\}$.

Why it was an artefact. Four checks, all negative:

1. **Implied κ_R .** On $\{1, 3, 8\}$ the gen2→3 step forces $\kappa_R = +0.0017$ — nearly zero, not $f(15) = 0.104$. The fit “works” only by switching the running off, not by recovering the derived constant.
2. **Wrong group.** $8 = \phi(15) = |U(15)|$ is the *multiplicative* index (force coupling, ZC102), while the mass ladder is built on the *additive* $|\mathbb{Z}_{15}| = 15$ (ZC101). Adopting 8 here borrows the wrong group and contradicts THM-ZC101-01.
3. **Forward prediction.** Fixing $\kappa_R = f(15)$ and predicting the mass ratio on $\{1, 3, 8\}$ gives t/c off by +65% — the inverse fit does not survive being run forward.
4. **Not reproducible across sectors.** Free gen3 R_H is 7.73 (up), 3.42 (down), 4.38 (lepton): no shared logic that would make 8 special.

Preservation. Recorded so the near-hit is not mistaken for an assignment. A free parameter solved from one datum will land near *some* round number; the test is forward prediction, which $\{1, 3, 8\}$ fails.

Pattern type: extraction artefact / sector assignment.

Lesson — PWA candidate. A free- R_H value near a structural number is an inverse fit to one point. Before reading it as an assignment, run it forward under the fixed constants; if the forward prediction misses, the near-hit is numerology.

0.102 FIND-ZC104-02: Not Charge, Not Doublet — the Two-Rung Test

ZC103 rejected charge-*squared*; the remaining charge hypotheses are linear. We test all of them (Q , weak isospin T_3 , Q^2 again, $|Q|$) and a doublet-partner coupling, with one sharp discriminant the earlier work did not apply: a per-fermion quantum number cannot change between generations, so any charge term must explain *both* rungs with the same strength. It does not.

FIND-ZC104-02: The first-rung effect is sector-ordered but not charge, not doublet (R_1 , numerical)

Statement. The first-rung deviation δ_1 is stable and sector-ordered — up +0.311, down −3.070, lepton −0.734 — with up and down opposite in sign. No charge observable accounts for it, and no doublet-partner coupling does.

Charge, by proportionality. If a charge observable X drove δ_1 , the ratio δ_1/X would be constant across sectors. It is not, for any X :

X	up	down	lepton	spread/mean
Q	0.47	9.21	0.73	117%
T_3	0.62	6.14	1.47	88%
Q^2	0.70	-27.6	-0.73	141%
$ Q $	0.47	-9.21	-0.73	136%

Charge, by the two-rung test (decisive). A per-fermion charge term must produce equal deviation at both rungs (ratio $\delta_1/\delta_2 \approx 1$). Instead:

Sector	δ_1	δ_2	δ_1/δ_2
up	+0.311	+2.092	0.15
down	-3.070	+0.981	-3.13
lepton	-0.734	+0.002	-359.86

The lepton ratio of -360 ($\delta_2 \approx 0$) is the cleanest statement that the effect is first-rung-*specific*, not a per-fermion offset. **Charge rejected** in every form (candidates 5–6).

Doublet-partner coupling. A coupling between $SU(2)$ partners $u-d$ predicts antisymmetric deviations: $\delta_1^{\text{up}} + \delta_1^{\text{down}} \approx 0$. Instead the sum is -2.76 (gen1 $\rightarrow$ 2) and $+3.07$ (gen2 $\rightarrow$ 3) — no cancellation, opposite signs between rungs. **Doublet coupling rejected** (candidate 7). The apparently perfect form (§0.104, PM-ZC104-02) was an algebraic identity, not a confirmation.

Status: Negative, quantified. The first-rung effect is real, stable, sector-ordered, opposite-signed in up/down, and explained by none of the seven candidates tested across ZC103 and ZC104.

Physical reading: The two-rung test is what makes this exclusion sharp. Earlier work could fit a charge term to three sectors at one rung and find apparent agreement; demanding the same term work at the second rung — where the leptons sit exactly on the ladder — breaks every charge hypothesis at once. The surviving facts are spare and specific: the effect is at the first rung, it is ordered $\text{up} > \text{lepton} > \text{down}$, and up and down pull in opposite directions. That is a tight target for a future derivation, and a clear statement of what that derivation may not be.

0.103 Master Status Table

Item	Status	Notes
FIND-ZC104-01 (lepton ladder)	Positive	Reconfirmed three parameter-free ways: $\delta_2 = 0.002$, half-diff 0.001, free- $R_H \approx 3, 5$
FIND-ZC104-02 (first-rung shape)	Negative (quant.)	Stable, sector-ordered (+0.31/-3.07/-0.73); not charge (two-rung test), not doublet (sum $\neq 0$)

(continues)

(continued)

Item	Status	Notes
GAP-ZC104-01 (first-rung effect)	Open	Seven candidates excluded; real, located, sector-ordered, opposite up/down; target ZC105
PM-ZC104-01 ($\phi(15)$ near-hit)	Scar	Free- $R_H = 7.73 \approx 8$ fails forward ($t/c + 65\%$), wrong group; pattern: extraction artefact / sector assignment
PM-ZC104-02 (identity trap)	Scar	$d_{\text{up}} - d_{\text{down}} = \text{split-change}$ to 4×10^{-16} is algebraic; test the sum; pattern: tautology test
GAP-ZC103-01 (first-rung locus)	Sharpened	Inherited; four further candidates excluded here (boundary offset, $\phi(15)$, charge-linear, doublet)
GAP-ZC101-02 (0.23% residual)	Sharpened	Confirmed first-rung, sector-ordered; not a κ_R correction
$\kappa_R = f(15)$	Stands	Reconfirmed at lepton $\text{gen}2 \rightarrow 3$; unchanged

Atomic Registry — ZC104

ID	Description	Status
DEF-ZC104-01	First-rung / second-rung deviations δ_1, δ_2 at fixed $\kappa_R = f(15)$, $n_m = 5$ (Eqs. 8–9)	Definition
FIND-ZC104-01	Lepton $\{1, 3, 5\}$ ladder reconfirmed three parameter-free ways ($\delta_2 = 0.002$, half-diff 0.001, free- $R_H \approx 3, 5$)	Positive
FIND-ZC104-02	First-rung effect stable, sector-ordered, opposite up/down; not charge (two-rung test), not doublet (antisymmetric sum $\neq 0$)	Negative (quant.)
GAP-ZC104-01	The first-rung effect: seven candidates excluded; real, located, sector-ordered; target ZC105	Open
PM-ZC104-01	Free- $R_H = 7.73 \approx 8 = \phi(15)$ for up $\text{gen}3$ is an inverse fit; fails forward ($t/c + 65\%$); wrong group (extraction artefact / sector assignment)	Scar
PM-ZC104-02	$d_{\text{up}} - d_{\text{down}}$ equals the splitting change to 4×10^{-16} — an algebraic identity; the non-trivial test is the sum (tautology test)	Scar

Instrument: CRF_MassSector_V2.py (one rule: free exactly one quantity per test). *Inherited verbatim:* $\kappa_R = f(15)$ (ZC101); $n_m = 5$, ladder (ZC15/16); $c(R_H = 1) = 0$ exact (ZC95). GAP-ZC103-01, GAP-ZC101-02 sharpened, not closed. No value modified.

0.104 R-Layer Research Note

R-Layer Assignments for ZC104

Atomic unit	R-layer	Cost T	Source
DEF-ZC104-01	R_1	—	definitions on the R_1 running
FIND-ZC104-01	R_1	—	parameter-free test of the R_1 ladder
FIND-ZC104-02	R_1	—	a located, bounded residual in the R_1 running
GAP-ZC104-01	R_1	—	the first-rung term at the $R_H = 1$ boundary
PM-ZC104-01	R_0	0	a correction of inference (forward vs inverse)
PM-ZC104-02	R_0	0	a correction of method (identity vs physics)

The findings are R_1 : direct numerical tests of the mass-sector running against data. The first-rung gap (GAP-ZC104-01) lives at R_1 , at the $R_H = 1$ boundary where the running vanishes. Both PM scars are R_0 (methodological corrections). No Born-chain bridge enters.

On PM-ZC104-02 (the identity trap). The difference $d_{\text{up}} - d_{\text{down}}$ equals the change in within-generation splitting identically, because up and down share the ladder step, which cancels in the difference. A result exact to machine precision is the loudest possible signal of an algebraic identity, not a physical coincidence; the non-tautological test is the *sum*, which does not cancel and which fails the doublet hypothesis.

0.105 Genealogy Map

How ZC104 Sharpens the Mass-Sector Gap

Upstream. ZC95 derived the running form and fixed $c(R_H = 1) = 0$ exactly; ZC101 derived $\kappa_R = f(15)$; ZC102 separated the additive $|\mathbb{Z}_{15}| = 15$ (mass) from the multiplicative $\phi(15) = 8$ (force); ZC103 built the first simulator, confirmed the law at the lepton $\text{gen}2 \rightarrow 3$ rung, and *located* the residual at the first rung with three candidates rejected.

This paper. `CRF_MassSector_V2.py` attacks the located gap under the one-free-quantity rule. It reconfirms the lepton ladder three independent ways (FIND-ZC104-01), and excludes four further candidates — a boundary offset (an artefact of two points per sector), $8 = \phi(15)$ as an up-index (an inverse fit failing forward prediction by 65%), charge-linearity in every form (failed by the two-rung test), and a doublet-partner coupling (the antisymmetric sum does not cancel) — registering the effect's precise shape (FIND-ZC104-02) and two extraction scars (PM-ZC104-01/02).

What stands, what is sharpened, what is open. $\kappa_R = f(15)$ and all ZC94–ZC103 results stand. GAP-ZC103-01 and GAP-ZC101-02 are *sharpened*: the first-rung effect is real, located, sector-ordered (up > lepton > down), and opposite-signed in up/down, with seven structural explanations excluded. A correspondingly sharper open item is registered (GAP-ZC104-01).

Forward. The space of explanations is now tightly bounded. The surviving target is a first-rung term that is sector-ordered and opposite-signed between up and down, is not charge in any tested form, and is not a doublet coupling — pointing, perhaps, at the *reference-crossing* itself (the resolution cost of the first distinction out of $R_H = 1$) rather than at any quantum number. The instrument is in place; ZC105 inherits a gap with seven doors closed.

0.106 Closing Sentence

The honest outcome of a hard look is often a smaller room rather than an open door, and that is what this paper delivers: the lepton ladder confirmed yet again with nothing to tune, and the residual that remains pressed into a corner it cannot leave. It is at the first rung, the step out of the reference; it is ordered, up above lepton above down; up and down pull against each other. It is not charge — not squared, not linear, not isospin — because no charge can explain two rungs at once when one of them sits exactly on the ladder. It is not a partner coupling, though for one bright moment a number that matched to the last decimal place suggested it was, until the number turned out to be an identity and the matching meant nothing. Seven explanations have been tried and seven set aside. We have not closed the gap; we have walled it in, and a gap with seven walls is far easier to corner than one standing in the open.

δ_1 : +0.31 (up), −0.73 (lepton), −3.07 (down); seven candidates excluded.

ผลที่ข้อสัจของการมองอย่างถึถ้วน มักเป็นห้องที่เล็กลงมากกว่าประตูที่เปิดออก
และนั่นคือสิ่งที่เปเปอร์นี้มอบให้ ---
บันไดของเลปตอนได้รับการยืนยันอีกครั้งโดยไม่มีอะไรให้ปรับ
และส่วนต่างที่เหลือถูกต้อนเข้าสู่มุมที่มันออกไปไม่ได้ มันอยู่ที่ขั้นแรก
ขั้นที่ก้าวออกจากจุดอ้างอิง มันเรียงลำดับ อัฟเหนือเลปตอนเหนือดาวน
และอัฟกับดาวนดิ่งกันคนละทาง มันไม่ใช่ประจุ --- ไม่ใช่ยกกำลังสอง ไม่ใช่เชิงเส้น
ไม่ใช่ไอโซสปิน --- เพราะไม่มีประจุใดอธิบายสองชั้นพร้อมกันได้
เมื่อชั้นหนึ่งวางอยู่บนบันไดพอดี และมันไม่ใช่การคู่ควบของคู่หู
แม้ชั่วขณะหนึ่งจะมีจำนวนที่ตรงกันถึงทศนิยมหลักสุดท้ายชวนให้เชื่อเช่นนั้น
จนกระทั่งจำนวนนั้นกลายเป็นเอกลักษณ์ และการตรงกันก็ไร้ความหมาย
เจ็ดคำอธิบายได้รับการทดลอง และทั้งเจ็ดถูกวางลง เรายังมิได้ปิดช่องว่าง
เราเพียงล้อมมันด้วยกำแพง --- และช่องว่างที่มีกำแพงเจ็ดด้าน
ยอมต้อนเข้ามุมได้ง่ายกว่าช่องว่างที่ยืนอยู่กลางที่โล่ง.

0.107 ZC105 — The Mass Sector Is Not a Q-Pair

Abstract

ZC90 established the *Q-pair law*: the force layer’s “irrational step ratio” (ZC89) and the mind layer’s “kind not magnitude” (DEF-MB03) are one structural statement — a continuous quantum δ_v paired with a discontinuous quantum δ_b , sharing one conserved direction, with an irrational ratio (THM-ZC91-01). The charged-fermion mass sector is the third CRF layer with a generation structure. This paper asks whether it, too, is a Q-pair instance.

FIND-ZC105-01 (negative): it is not, on three independent signatures. (i) The clean sector — leptons — is not mono-mode: its discontinuous component is -0.74 , not ~ 0 . (ii) The up/down discontinuous components share a sign $(-4.19, -5.18)$, so the $\{+, +, -, -\}$ grading does not transfer. (iii) The jump/walk ratios are *rational* (down = $-27/11$ to 0.03%), contradicting THM-ZC91-01, which forces the ratio off $\mathbb{Q}$.

THM-ZC105-01 records the boundary: the Q-pair law is layer-specific. Mind and force instantiate it; mass does not. **GAP-ZC105-01** carries the first-rung effect forward, now resisting five classes of explanation. **PM-ZC105-01** logs a rational lead (an apparent denominator 11) that proved a rounding artefact.

0.108 Background: A Law in Search of Its Domain

CRF has found the same abstract structure in two very different places. At the *force* layer, the spacing of the gauge ladder and the spacing of a second, “floor” ladder stand in an irrational ratio (ZC89): two kinds of step that cannot be measured against each other. At the *mind* layer, the framework’s model of volition (Part F) distinguishes two kinds of forward move — a small continuous one (δ_v , “visesa”) and a sudden discontinuous one (δ_b , “nibbedha”) — which DEF-MB03 calls different “in kind, not magnitude.” ZC90 proved these are the *same* statement, a *quantum pair*: a continuous and a discontinuous quantum sharing one conserved direction, provably incommensurable (THM-ZC91-01). The natural question is whether the third place CRF has a layered, step-like structure — the *mass* sector, where each fermion generation sits one rung up a ladder — is another instance of the same pair. The terms needed are few: a *rung* (one generation step), the *first rung* (the step out of the reference generation, where ZC104 located an unexplained residual), and the two *Q-pair signatures* that any genuine instance must show — a clean “walk-only” limit, and an irrational walk/jump ratio. The answer here is negative, and the value of a clean negative is a *boundary*: it tells the Q-pair law where it stops.

Numerical Summary — ZC105 (T-canonical)

Quantity	Value	Status
Q-pair ratio, mind/force lepton jump (mono-mode test)	irrational -0.74	THM-ZC91-01 FAIL (~ 0 req.)
up / down jump signs	$-4.19/-5.18$	FAIL (oppo- site req.)
down jump/walk ratio	$-2.455 = -27/11$	FAIL (ratio- nal)
walk(rung1)/walk(rung2)	$\ln 3 / \ln(5/3)$ 2.151	= ladder geometry
Q-pair signatures passed	0 of 3	FIND-ZC105-01
first-rung resists (lens classes)	5	GAP-ZC105-01

0.108.1 What the Q-pair law says, precisely

The Q-pair law (ZC90/91), in the form ZC105 tests

DEF-ZC90-01 (Q-pair). A layer instantiates a *quantum pair* if it carries two generators of one conserved forward current: a continuous δ_v and a discontinuous δ_b , related “in kind, not magnitude.”

THM-ZC90-01. The force layer (continuous Γ -step vs. discontinuous floor-step, ZC89) and the mind layer (continuous visesa vs. discontinuous nibbedha, DEF-MB03) both instantiate DEF-ZC90-01; they are one structural statement.

THM-ZC91-01 (the sharp constraint). “Kind, not magnitude” is logically equivalent to $\delta_b/\delta_v \notin \mathbb{Q}$: the two quanta have an *irrational* ratio. Tested here as a necessary signature.

COR-ZC91-02 (the guard). At finite Barami there is *no constant* ratio — it is history-dependent. So a Q-pair test may check structural *compatibility* (mono-mode limit; irrational ratio), but must *not* fit a single constant. ZC105 obeys this: it seeks no constant.

0.108.2 What Was Already Known Before ZC105

Pre-ZC105 Status

Known:

- The Q-pair law holds at two layers (ZC90); its ratio is irrational (ZC91); it has no finite-Barami constant (COR-ZC91-02).
- The mass-sector first-rung effect is real and located (GAP-ZC104-01), sector-ordered (up +0.31, down −3.07, lepton −0.73 in ln-mass), opposite-signed up/down, and resists seven candidates (ZC104).
- The lepton {1, 3, 5} ladder is the clean sector; gen2→3 matches the running law to $\delta_2 \approx 0$ (ZC104, FIND-ZC104-01).

Open:

- Is the first rung a discontinuous “jump” out of $R_H = 1$, in the Q-pair sense?
- Does the up/down sign asymmetry come from the KL-sign grading?

ZC105 answers both negatively, and bounds the Q-pair law.

What ZC105 Does NOT Do

- Does not close GAP-ZC104-01 — it adds a fifth excluded class (the mind-layer Q-pair route).
- Does not weaken ZC90/91/92 — it *bounds* their law to mind and force, giving it a domain.
- Does not derive the first-rung effect — it removes one more route.
- Does not alter $\kappa_R = f(15)$, n_m , the ladder, or any inherited result.

Companion Papers — ZC105

Paper / Instrument	What it provides to ZC105
CRF_ZC104_..._v1.tex	GAP-ZC104-01; the deviations δ_1, δ_2
CRF_ZC90_..._v1.tex	DEF-ZC90-01 (Q-pair); THM-ZC90-01
CRF_ZC91_..._v1.tex	THM-ZC91-01 (ratio $\notin \mathbb{Q}$); COR-ZC91-02
CRF_ZC92_..._v1.tex	Barami functional; history-dependence
CRF_..Part_F.tex	DEF-MB03/04: visesa/nibbedha, Barami on C_4^+
CRF_MassSector_V2.py	the deviation diagnostics

0.108.3 Pre-Work Audit (PWA-1 through PWA-10)

Pre-Work Audit Record — ZC105

Scanned before writing (Upgrade Protocol v1.3):

- ZC90 (Q-pair law), ZC91 (irrational ratio; the no-constant guard), ZC92 (Barami functional), ZC104 (first-rung deviations), Part F (DEF-MB03/04) — **KEY PILLARS**.

PWA-6: macros via `providecommand` (δ_v, δ_b new, no collision with Preamble v3.1).

PWA-7–9: not triggered (no new floor, layer, or spectral convention).

PWA-10 (tautology/artefact audit): *triggered and caught* — an apparent rational ratio (denominator 11) was chased; it failed cap-stability, full-precision recomputation, and the structural-home test, and is logged as a rounding artefact (PM-ZC105-01). Not admitted.

Cross-layer-transfer caution (Phase-4): a structure proven at the mind layer is *tested*, not assumed, at the mass layer; the negative result is the safeguard working.

No duplicate atomic units found.

0.109 The Test: Decomposing the First Rung into Walk and Jump

If the mass sector were a Q-pair, the first-rung deviation δ_1 (ZC104, Eq. for δ_1) would decompose into a continuous *walk* — the same running law that governs the clean gen2→3 rung — plus a discontinuous *jump* present only at the reference crossing. We construct that decomposition with *no free parameter*, then test the three Q-pair signatures.

DEF-ZC105-01: Walk/jump decomposition of the first rung (parameter-free)

Construction. The pure-walk rung is gen2→3, with deviation δ_2 . The walk scales with the ladder distance in $\ln R_H$, so the walk expected at the first rung (gen1→2, distance $\ln 3$) relative to the second (distance $\ln(5/3)$) is fixed by geometry:

$$\text{walk}_1 = \delta_2 \cdot \frac{\ln 3}{\ln(5/3)} = \delta_2 \cdot 2.151, \quad \text{jump} \equiv \delta_1 - \text{walk}_1 \quad (10)$$

$\kappa_R = f(15)$ and the walk ratio $\ln 3 / \ln(5/3) = 2.151$ are both fixed; nothing is tuned. This is a re-expression of the ZC104 numbers, not a fit; it has content only if the jumps reveal the Q-pair signature that δ_1 alone does not.

The decomposition yields:

Sector	δ_1	$\text{walk}_1 = 2.151 \delta_2$	$\text{jump} = \delta_1 - \text{walk}_1$
up	+0.311	+4.499	−4.188
down	−3.070	+2.110	−5.180
lepton	−0.734	+0.004	−0.738

0.110 FIND-ZC105-01: Three Q-Pair Signatures, All Failed

FIND-ZC105-01: The mass sector fails all three Q-pair signatures (R_1 , numerical)

Signature 1 — a mono-mode (walk-only) limit. A Q-pair layer has a clean limit where only the continuous quantum acts. The leptons are that clean sector ($\delta_2 = 0.002$). If mass were a Q-pair, the lepton jump would be ~ 0 . It is -0.738 . **Failed:** the cleanest sector is not mono-mode.

Signature 2 — a signed grading. The mind layer's two quanta sit on a KL-sign grading $\{+, +, -, -\}$; if mass inherited it, the doublet partners up and down would carry *opposite*-signed jumps. Both are negative (-4.19 for up, -5.18 for down). **Failed:** no sign opposition; the grading does not transfer.

Signature 3 — an irrational ratio (THM-ZC91-01). The defining constraint of a Q-pair is $\delta_b / \delta_v \notin \mathbb{Q}$. The jump/walk ratios, at full mass precision, sit on *simple fractions*:

Sector	jump/walk	nearest simple fraction
up	−0.931	−11/12 (1.5%)
down	−2.455	−27/11 (0.03%)
lepton	−168.3	−505/3 (0.00%)

Failed: the ratios are rational, not irrational — the direct opposite of what THM-ZC91-01 forces for a genuine Q-pair.

Status: Negative, structural. Zero of three Q-pair signatures hold. The mass sector is not a Q-pair instance.

Physical reading: The failure is not marginal on any of the three axes, and the third is decisive in the cleanest way: the very property that defines a Q-pair — incommensurability, the irrational ratio that ZC91 proved equivalent to “kind not magnitude” — is exactly the property the mass sector lacks. Where the force and mind layers are built on two steps that cannot be measured against each other, the mass sector’s first-rung structure sits on plain fractions. This is why none of the seven rational candidates of ZC104 could be ruled out by irrationality: the mass residual was never irrational to begin with. The layers are not all speaking the same sentence.

0.111 The Rational Lead, and Why It Failed (PM-ZC105-01)

Signature 3 left an apparent thread: the down ratio $-27/11$ and the lepton ratio $-1888/11$ seemed to share a denominator 11. Under the ZC104 discipline (PM-ZC104-01), a near-hit on a structural number must survive three checks before it is believed. The 11 survived none.

PM-ZC105-01: The denominator-11 — a rounding artefact (Failure Stance)

What happened. Reading the jump/walk ratios through `limit_denominator` returned fractions with denominator 11 for two of three sectors, with sub-percent error, suggesting a hidden rational structure.

Why it was an artefact. Three checks, all negative:

1. **Cap-instability.** The lepton fraction migrated with the `limit_denominator` cap: $515/3 \rightarrow 1373/8 \rightarrow 1888/11 \rightarrow 2403/14 \rightarrow 5321/31$. A real fraction is stable; this is the signature of an artefact.
2. **Full-precision collapse.** The “0%” errors came from masses rounded to three decimals. At full precision the down ratio moved 1.5% from $-11/12$, and the lepton denominator changed from 11 to 3.
3. **No structural home.** 11 is not a divisor of 15, not 2^k , not a d_s/n_m combination; the only natural 11 is $d_s+n = 3+8$, which has no role in the $n_m = 5$ -power mass ladder.

Preservation. Recorded so the 11 is not revived. A fraction fit to rounded inputs, stable at only one cap, with no structural home, is numerology.

Pattern type: extraction artefact (rational-fit).

Lesson — PWA candidate. Before believing a rational hit, require all three: stability across `limit_denominator` caps, survival at full input precision, and a structural home in the constants.

0.112 THM-ZC105-01: The Q-Pair Law Has a Domain

THM-ZC105-01: The Q-pair law is layer-specific (R_1 , Near-Formal)

Statement. The Q-pair law (DEF-ZC90-01, THM-ZC90-01) holds at the force and mind layers and not at the charged-fermion mass layer. It is a layer-specific structural law, not a universal feature of CRF.

Derivation.

Step 1 (instantiation at mind and force). LEM-ZC90-01/02 establish that both layers carry a continuous/discontinuous quantum pair with an irrational ratio (THM-ZC91-01), by two independent arguments (ladder arithmetic at the force layer; the semantic “kind not magnitude” clause at the mind layer, FIND-ZC91-01).

Step 2 (non-instantiation at mass). FIND-ZC105-01 shows the mass sector fails all three necessary Q-pair signatures: no mono-mode limit (the clean lepton sector has jump -0.74), no signed grading (up/down jumps share a sign), and — decisively — a *rational* jump/walk ratio, contradicting the defining irrationality of THM-ZC91-01.

Step 3 (boundary). A structural law that holds at two layers and provably fails at a third is not universal; it has a domain. The Q-pair law’s domain is the mind and force layers.

Status: Near-Formal boundary theorem. It sharpens ZC90 by supplying the domain its law had been silent about.

Physical reading: This is the constructive content of a negative result. ZC90 found a beautiful coincidence of structure across two layers and, reasonably, raised it toward a candidate universal law (FIND-ZC91-01 called it a candidate structural law). ZC105 tests the next layer and finds the law stops there. That is not a loss: a law with a known boundary is stronger than a law suspected of being universal, because its claims are now falsifiable and bounded. CRF’s layers are related but not identical; the mass sector keeps its own, still-unread, structure.

0.113 Master Status Table

Item	Status	Notes
DEF-ZC105-01 (walk/jump)	Definition	Parameter-free decomposition of δ_1 via the pure-walk rung (Eq. 10)
FIND-ZC105-01 (3 fails)	Negative (struct.)	Mass fails all three Q-pair signatures: not mono-mode, no sign opposition, rational ratio
THM-ZC105-01 (domain)	Near-Formal	Q-pair law holds at mind/force, fails at mass; layer-specific, not universal
GAP-ZC105-01 (first-rung)	Open	Now resists five lens-classes (charge, weight, label, Q-pair, rational); target ZC106

(continues)

(continued)

Item	Status	Notes
PM-ZC105-01 (denom-11)	Scar	Rational hit failed cap-stability, full-precision, structural-home; pattern: extraction artefact (rational-fit)
GAP-ZC104-01 (first-rung)	Sharpened	Inherited; mind-layer Q-pair route excluded here (fifth class)
THM-ZC90-01 (Q-pair law)	Bounded	Domain supplied: mind/force, not mass (THM-ZC105-01)
FIND-ZC91-01 (candidate universal)	Resolved	Not universal; the candidate-law status is closed by the mass boundary

Atomic Registry — ZC105

ID	Description	Status
DEF-ZC105-01	Walk/jump decomposition of the first rung, parameter-free (Eq. 10)	Definition
FIND-ZC105-01	Mass sector fails all three Q-pair signatures (no mono-mode limit; up/down jumps share a sign; rational jump/walk ratio vs THM-ZC91-01)	Negative (struct.)
THM-ZC105-01	The Q-pair law (THM-ZC90-01) is layer-specific: holds at mind/force, fails at the mass sector; not universal	Near-Formal
GAP-ZC105-01	The first-rung effect now resists five lens-classes; real, located, sector-ordered, unexplained; target ZC106	Open
PM-ZC105-01	The denominator-11 rational lead is a rounding artefact (failed cap-stability, full-precision, structural-home)	Scar

Instrument: CRF_MassSector_V2.py (deviation diagnostics). *Inherited verbatim:* DEF-ZC90-01, THM-ZC90-01, THM-ZC91-01, COR-ZC91-02 (ZC90/91); δ_1, δ_2 (ZC104); DEF-MB03/04 (Part F). No value modified. THM-ZC90-01 bounded, not weakened.

0.114 R-Layer Research Note

R-Layer Assignments for ZC105

Atomic unit	R-layer	Cost T	Source
DEF-ZC105-01	R_1	—	a decomposition of the R_1 running deviation
FIND-ZC105-01	R_1	—	a structural test of the R_1 mass residual
THM-ZC105-01	cross	—	a boundary across the mind/force/mass layers
GAP-ZC105-01	R_1	—	the first-rung term at the $R_H = 1$ boundary
PM-ZC105-01	R_0	0	a correction of method (rational-fit)

The mass-sector findings are R_1 (the resolution-running tested against data). THM-ZC105-01 is a *cross-layer* statement — it compares the R_1 mass structure against the mind- and force-layer Q-pair law, and so does not sit at a single R -layer. PM-ZC105-01 is R_0 (a methodological correction). No Born-chain bridge enters.

0.115 Genealogy Map

How ZC105 Bounds the Cross-Layer Law

Branch A — the Q-pair law. ZC89 found the irrational step ratio at the force layer; Part F (DEF-MB03/04) defined the mind layer’s visesa/nibbedha pair; ZC90 unified them (THM-ZC90-01); ZC91 proved the ratio irrational (THM-ZC91-01) and flagged the law as a candidate universal (FIND-ZC91-01); ZC92 built the Barami functional and showed the ratio history-dependent (COR-ZC91-02).

Branch B — the mass residual. ZC95 derived the running form; ZC101 derived $\kappa_R = f(15)$; ZC103 located the residual at the first rung; ZC104 excluded seven candidates and recorded the effect’s shape (sector-ordered, opposite up/down).

Convergence at ZC105. The two branches meet on one question: is the mass first-rung a discontinuous Q-pair “jump”? The walk/jump decomposition (DEF-ZC105-01) tests it under the Q-pair law’s own constraints, and the answer is no on all three signatures (FIND-ZC105-01). The candidate-universal status of FIND-ZC91-01 is thereby resolved: the law has a domain (THM-ZC105-01).

ZC105’s contribution: it gives the Q-pair law a boundary — mind and force, not mass — and removes the mind-layer route to the first-rung effect, leaving GAP-ZC105-01 resisting five classes of explanation.

0.116 Closing Sentence

We carried a structure across the layers to see how far it would reach. At the force layer and the mind layer it is the same structure — two kinds of step, one continuous and one sudden, that cannot be measured against each other, the irrational pairing that ZC90 found and ZC91 proved. We hoped the mass sector would be a third instance, its first rung a sudden jump out of the reference. It is not. The clean sector does not go quiet as a single mode should; the partners do not split by sign; and the ratio that ought to be irrational sits squarely on a fraction. Three signatures, three refusals. The honest reward

is a border drawn where there was only an open guess: the cross-layer law belongs to mind and force, and the mass sector keeps a structure we have not yet learned to read. A law with an edge is worth more than a law without one.

$$\delta_b/\delta_v \notin \mathbb{Q} \text{ (mind, force); } \text{mass: } \frac{\text{jump}}{\text{walk}} = -\frac{27}{11} \in \mathbb{Q}.$$

เรานำโครงสร้างหนึ่งข้ามชั้นต่างๆ เพื่อดูว่ามันจะไปได้ไกลเพียงใด ที่ชั้นแรงและชั้นจิต มันคือโครงสร้างเดียวกัน --- ก้าวสองชนิด ชนิดหนึ่งต่อเนื่องอีกชนิดนับพลัน ที่วัดต่อกันไม่ได้ คือการจับคู่وترรกยะที่ ZC90 พบและ ZC91 พิสูจน์ เราหวังว่าภาคมวลจะเป็นกรณีที่สาม ให้ชั้นแรกของมันเป็นการกระโดดนับพลันออกจากจุดอ้างอิง แต่มันไม่ใช่ ภาคที่สะอาดที่สุดไม่เจียบลงอย่างที่เราคิดควรจะเป็น คู่หูไม่แยกกันด้วยเครื่องหมาย และอัตราส่วนที่ควรเป็นوترรกยะ กลับตกลงบนเศษส่วนพหุดี สามลายเซ็น สามการปฏิเสธ รางวัลที่ซื้อสัตย์คือเส้นพรมแดนที่ถูกขีดขึ้น ณ ที่ซึ่งเดิมมีเพียงการเดาที่เปิดกว้าง --- กฎข้ามชั้นเป็นของชั้นจิตและชั้นแรง และภาคมวลยังเก็บงำโครงสร้างที่เราอ่านไม่ออก กฎที่มีขอบเขต ย่อมมีค่ามากกว่ากฎที่ไร้ขอบเขต.

0.117 ZC106 — The First-Rung Effect Splits

Abstract

ZC104 and ZC105 left the mass-sector first-rung effect (GAP-ZC105-01) located but unexplained, resisting five classes of explanation. This paper re-reads it in the *pure-ladder* convention of ZC16 (THM-MC-01) — the ladder R_H^{nm} with no running — and finds it is not one effect but two.

FIND-ZC106-01 (connective): the lepton first-rung residual *is* ZC16's decade-old GAP-MC-01. In the pure-ladder convention the lepton gen1→2 mass gap is +18% ($3^5 = 243$ vs $m_\mu/m_e = 207$), the value ZC16 recorded (+17%). The ZC104 running-convention residual (−0.734 in ln) and the ZC16 pure-ladder residual (−0.161) differ by *exactly* the running term $n_m \kappa_R \ln 3 = +0.5726$, to machine precision. One gap, seen in two eras.

FIND-ZC106-02 (new): the quark first-rung deviations are not part of GAP-MC-01, which ZC16 verified on leptons only. In the same convention up gen1→2 is −59% and down is +1115% — far larger, opposite-signed, and absent from ZC16.

THM-ZC106-01 records the split: GAP-ZC105-01 is a lepton-piece (old, small) plus a quark-piece (new, large). **PM-ZC106-01** notes that a residual's apparent size is convention-dependent.

0.118 Background: The Same Residual, Two Conventions

A decade before the current mass-sector work, ZC16 (THM-MC-01) derived the charged-lepton mass ratios from a single idea — mass is the entropy *price* a state pays for narrowing from the full possibility space — and got the ratios right to 10–17% with no free parameters, on the bare ladder $m \propto R_H^{nm}$ with $R_H = 15/|H| \in \{1, 3, 5\}$. The leftover 10–17% was recorded honestly as GAP-MC-01 and left open. The recent papers (ZC95 onward) added a *running* correction to that ladder and, in ZC103–ZC105, located a stubborn residual at the *first rung* — the step out of the reference generation — that resisted seven candidate explanations and the mind-layer Q-pair structure besides. This paper connects the two. The key is that ZC103 onward read every sector in the *running* convention (ladder

plus the $n_m \kappa_R$ correction), which moves the baseline; ZC16 read the bare ladder. The terms needed are few: a *rung* (one generation step), the *pure-ladder convention* (the bare $R_H^{n_m}$, no running — ZC16’s frame), and the *running convention* (ladder plus the small $n_m \kappa_R$ correction — ZC104’s frame). Switching to the pure-ladder convention shows the first-rung effect is two things: an old, small lepton residual and a new, large quark one.

Numerical Summary — ZC106 (T-canonical)

Quantity	Value	Status
lepton gen1→2 gap (pure ladder)	+18% (243 vs 207)	GAP-MC-01
ZC16 stated GAP-MC-01 (lepton)	+17%/−10%	ZC16/THM-MC-01
lepton residual, pure vs running	−0.161 vs −0.734	FIND-ZC106-01
running term $n_m \kappa_R \ln 3$	+0.5726 (exact diff)	ZC101
up gen1→2 gap (pure ladder)	−59% (588 vs 243)	FIND-ZC106-02 (new)
down gen1→2 gap (pure ladder)	+1115% (20 vs 243)	FIND-ZC106-02 (new)
ZC16 sector coverage	leptons only	THM-ZC106-01

0.118.1 What ZC16 established, and what it left open

ZC16 (THM-MC-01) — the pure-ladder lepton result

THM-MC-01. Mass is the constraint cost $M_H = n_m \log R_H$, giving the bare ladder $m \propto R_H^{n_m}$ with $R_H = 15/|H|$. On the charged leptons (EQ-MC-05), with *zero* free parameters:

Gen	$ H $	R_H	CRF R_H^5	SM lepton
1st	15	1	1 (baseline)	m_e
2nd	5	3	243	$m_\mu/m_e \approx 207$ (+17%)
3rd	3	5	3125	$m_\tau/m_e \approx 3477$ (−10%)

GAP-MC-01. The 10–17% residual was recorded as open, “real physics that CRF does not yet contain.” Crucially, EQ-MC-05 tested *charged leptons only*; quarks were never placed on this ladder.

0.118.2 What Was Already Known Before ZC106

Pre-ZC106 Status

Known:

- GAP-MC-01: a 10–17% lepton-ratio residual on the bare ladder (ZC16), open for the life of the corpus.
- GAP-ZC105-01: a first-rung mass effect, located, sector-ordered, resisting five classes (charge, weight, label, Q-pair, rational).
- The running constant $\kappa_R = f(15)$ shifts the ladder by $n_m \kappa_R \ln R_H$ per rung (ZC101).

Open:

- Is the ZC104 first-rung residual the same object as ZC16's GAP-MC-01?
- Are the quark first-rung deviations part of that old gap, or new?

ZC106 answers both: the lepton piece is GAP-MC-01; the quark piece is new.

What ZC106 Does NOT Do

- Does not close GAP-MC-01 — it *re-sees* it through the running lens.
- Does not close or explain the quark first-rung — it *isolates* it as new.
- Does not re-open the seven ZC104 exclusions or the ZC105 Q-pair boundary.
- Does not alter $\kappa_R = f(15)$, n_m , the ladder, THM-MC-01, or any result.

Companion Papers — ZC106

Paper / Instrument	What it provides to ZC106
CRF_ZC105_..._v1.tex	GAP-ZC105-01; the five excluded classes
CRF_ZC104_..._v1.tex	δ_1 (first-rung deviation); seven exclusions
CRF_...Part_C.tex	ZC16: THM-MC-01, COR-MC-02, EQ-MC-05, GAP-MC-01
CRF_ZC101_..._v1.tex	$\kappa_R = f(15)$; the running term
CRF_MassSector_V2.py	deviation diagnostics (full-precision masses)

0.118.3 Pre-Work Audit (PWA-1 through PWA-10)

Pre-Work Audit Record — ZC106

Scanned before writing (Upgrade Protocol v1.3):

- ZC16 (THM-MC-01, EQ-MC-05, GAP-MC-01), ZC104 (δ_1), ZC105 (GAP-ZC105-01), ZC101 ($\kappa_R = f(15)$) — **KEY PILLARS**.

PWA-6: macros via `providexcommand`; no collision.

PWA-7–9: not triggered.

PWA-10 (tautology/artefact audit): *triggered and caught* — a mid-session script double-counted the running term and printed a spurious 0.32 mismatch between the ZC16 and ZC104 residuals. Re-derived: the two residuals differ by *exactly* $n_m \kappa_R \ln 3 = +0.5726$, to machine precision. The audited (exact) relation is the one reported; the script artefact is logged in PM-ZC106-01.

PWA-14 (provenance before mechanism): the connection to GAP-MC-01 is checked by matching ZC16's own stated number (+17%), not by a fitted coincidence.

No duplicate atomic units found.

0.119 FIND-ZC106-01: The Lepton First-Rung Residual Is GAP-MC-01

The first-rung effect looked large in ZC104 ($\delta_1^{\text{lepton}} = -0.734$ in $\ln\text{-mass}$). That size is an artefact of the running convention. Read on the bare ladder — ZC16’s frame — the lepton first-rung residual is small, and it is exactly the gap ZC16 already recorded.

FIND-ZC106-01: lepton first-rung residual = GAP-MC-01 (R_1 , numerical)

Statement. The lepton $\text{gen}1 \rightarrow 2$ residual is GAP-MC-01. In the pure-ladder convention,

$$\left. \frac{m_\mu}{m_e} \right|_{\text{CRF}} = 3^{n_m} = 243, \quad \text{observed } 207, \quad \text{gap } +18\%,$$

which is ZC16’s recorded $+17\%$ (EQ-MC-05). The two conventions differ by a fixed, derived shift:

$$\underbrace{\delta_1^{\text{lepton}}}_{\text{running, } -0.734} = \underbrace{\left[\ln \frac{m_\mu}{m_e} - n_m \ln 3 \right]}_{\text{pure ladder, } -0.161} - \underbrace{n_m \kappa_R \ln 3}_{+0.5726} \quad (11)$$

to machine precision. The running term moves the baseline by $+0.5726$; it does not create or remove the residual.

Why this matters. The physical residual — what CRF does not yet explain — is the pure-ladder gap, $+18\%$, not the convention-shifted -0.734 . That gap is GAP-MC-01, open since ZC16. The recent first-rung work, for the leptons, has been re-finding a decade-old result, not a new effect.

Numerical verification:

Quantity	Value	Source
m_μ/m_e pure-ladder (3^5)	243	ZC16/COR-MC-02
m_μ/m_e observed	207	PDG
pure-ladder gap	$+18\%$	$= \text{ZC16's } +17\%$
δ_1^{lepton} (running)	-0.734	ZC104
pure-ladder $\ln$ -residual	-0.161	this paper
difference $= n_m \kappa_R \ln 3$	$+0.5726$	exact (ZC101)

Status: Positive, connective. The lepton first-rung residual is GAP-MC-01 in the running convention.

Physical reading: This is the value of looking back. A residual can wear a different size in a different frame, and the running convention — natural for the recent work — happened to inflate the lepton first-rung gap fourfold in $\ln\text{-mass}$ by moving the baseline. Stripping the running away returns ZC16’s honest $+17\%$, and with it the recognition that the lepton “first-rung effect” is the same $10\text{--}17\%$ that has stood open since the constraint-cost paper. Nothing new is broken; an old gap is correctly re-identified, which is itself a kind of progress — it tells us not to seek a new lepton mechanism where an old one already waits.

0.120 FIND-ZC106-02: The Quark First-Rung Deviations Are New

ZC16's EQ-MC-05 placed only the charged leptons on the ladder; the quarks were never tested there. So whatever the quark first-rung deviations are, they are not part of GAP-MC-01. Read in the same pure-ladder convention, they are far larger than the lepton gap and opposite in sign between up and down.

FIND-ZC106-02: the quark first-rung is large and new (R_1 , numerical)

Statement. In the pure-ladder convention, the quark gen1→2 ratios deviate from $3^{n_m} = 243$ far more than the leptons, and were never in ZC16:

Sector	gen1→2 ratio	ladder 3^5	gap
lepton (μ/e)	207	243	+18% (GAP-MC-01)
up (c/u)	588	243	−59% (new)
down (s/d)	20	243	+1115% (new)

The up ratio over-shoots the ladder ($588 > 243$) and the down ratio massively under-shoots it ($20 \ll 243$): opposite signs, both large, neither resembling the lepton +18%.

Status: New finding. The quark first-rung deviations are not GAP-MC-01; they are a distinct, larger, opposite-signed effect, introduced by ZC103/104 and isolated here.

Physical reading: The split is now plain. The lepton sector carries the small, known residual of ZC16; the quark sector carries something else entirely — an order of magnitude larger on the down side, and of the opposite sign between the doublet partners. Reading the recent work in the running convention had blurred these together, because the convention shift (+0.57 in $\ln$) is comparable to the lepton residual but negligible against the quark ones. In the pure-ladder frame the leptons sit at +18% and the quarks at −59% and +1115%: not one effect with a common cause, but an old gap and a new one sharing a location.

0.121 THM-ZC106-01: The First-Rung Effect Splits

THM-ZC106-01: GAP-ZC105-01 decomposes into an old and a new gap (R_1 , Near-Formal)

Statement. The first-rung mass effect GAP-ZC105-01 is not a single phenomenon. It decomposes into a *lepton-piece*, equal to GAP-MC-01 (ZC16, old, +18%), and a *quark-piece* (new, large, ZC103/104), of different provenance and scale.

Derivation.

Step 1 (lepton = old). FIND-ZC106-01: the lepton first-rung residual equals ZC16's GAP-MC-01, exactly related to ZC104's running-convention value by the derived shift $n_m \kappa_R \ln 3$ (Eq. 11). The lepton piece is +18% in mass and open since ZC16.

Step 2 (quark = new). FIND-ZC106-02: ZC16 (EQ-MC-05) tested charged leptons only; the quark first-rung deviations (−59%, +1115%) are absent from GAP-MC-01 and were introduced by the simulator work (ZC103/104).

Step 3 (distinct provenance and scale). The lepton piece is small, known, and decade-old; the quark piece is large, new, and opposite-signed between up and down. A single common-cause explanation is therefore not indicated; the two pieces should be pursued separately.

Status: Near-Formal decomposition theorem. It refines GAP-ZC105-01 into two targets with different histories, replacing one blurred gap with two sharp ones.

Physical reading: This is the paper’s organizing result, and the reason the look-back was worth it. For five papers the first-rung effect was treated as one thing because one convention made the sectors look comparable. The split says: stop looking for a single mechanism. The leptons need whatever closes GAP-MC-01 — a decade-old, sub-20% correction to the constraint-cost ladder. The quarks need something else, larger and sign-structured, that the corpus has only seen since the simulator was built. Two gaps, clearly separated, are easier to close than one gap that was quietly two.

0.122 Master Status Table

Item	Status	Notes
FIND-ZC106-01 (lepton = old)	Positive	Lepton first-rung residual = GAP-MC-01 (+18%); exact shift $n_m \kappa_R \ln 3 = +0.5726$
FIND-ZC106-02 (quark = new)	Negative/new	Quark gen1→2 gaps −59% (up), +1115% (down); not in ZC16 (leptons only)
THM-ZC106-01 (the split)	Near-Formal	GAP-ZC105-01 = lepton-piece (old, small) + quark-piece (new, large)
PM-ZC106-01 (convention)	Scar	Running convention shifts residual size by +0.57 in ln; read physical residual on pure ladder; pattern: convention mismatch
GAP-MC-01 (ZC16 residual)	Re-seen	The lepton first-rung residual; connected to the running work via FIND-ZC106-01
GAP-ZC105-01 (first-rung)	Split	Decomposed into lepton-piece (= GAP-MC-01) and quark-piece (new); target ZC107
$\kappa_R = f(15)$, THM-MC-01	Stand	Unchanged; used as pillars

Atomic Registry — ZC106

ID	Description	Status
FIND-ZC106-01	Lepton first-rung residual = GAP-MC-01 (+18% pure ladder); exact relation to ZC104 via $n_m \kappa_R \ln 3$ (Eq. 11)	Positive
FIND-ZC106-02	Quark first-rung deviations (−59% up, +1115% down) are new; ZC16 tested leptons only	Negative/new
THM-ZC106-01	GAP-ZC105-01 splits into lepton-piece (= GAP-MC-01, old) and quark-piece (new, large); distinct provenance and scale	Near-Formal
PM-ZC106-01	A residual’s apparent size is convention-dependent (running shift +0.57 in $\ln$); read it on the pure ladder (convention mismatch)	Scar

Instrument: CRF_MassSector_V2.py (full-precision masses). *Inherited verbatim:* THM-MC-01, COR-MC-02, EQ-MC-05, GAP-MC-01 (ZC16); δ_1 (ZC104); $\kappa_R = f(15)$ (ZC101). GAP-MC-01 re-seen, not closed; GAP-ZC105-01 split, not closed. No value modified.

0.123 R-Layer Research Note

R-Layer Assignments for ZC106

Atomic unit	R-layer	Cost T	Source
FIND-ZC106-01	R_1	—	a connection of two R_1 residual readings
FIND-ZC106-02	R_1	—	an R_1 residual isolated as new
THM-ZC106-01	R_1	—	a decomposition of the R_1 first-rung gap
PM-ZC106-01	R_0	0	a convention correction

The findings are R_1 : readings of the mass-sector residual against data, in two conventions (pure ladder vs ladder + running). GAP-MC-01 (the lepton piece) and the quark piece both live at R_1 , at the $R_H = 1$ boundary rung. PM-ZC106-01 is R_0 (a convention correction). No Born-chain bridge enters.

0.124 Genealogy Map

How ZC106 Connects the Old Mass Work to the New

Branch A — the old constraint-cost ladder. ZC15 fixed the generation count; ZC16 (THM-MC-01) derived mass as constraint cost $M_H = n_m \log R_H$, verified the lepton ratios to 10–17% with zero parameters (EQ-MC-05), and recorded the residual as GAP-MC-01 — on charged leptons only.

Branch B — the new running and simulator work. ZC95 added the running correction; ZC101 derived $\kappa_R = f(15)$; ZC103 built the simulator and located the

residual at the first rung; ZC104 excluded seven candidates; ZC105 bounded the Q-pair law away from mass. Throughout, the running convention was used.

Convergence at ZC106. Reading the new work in the old (pure-ladder) convention connects the branches: the lepton first-rung residual is GAP-MC-01 exactly (FIND-ZC106-01, related by $n_m \kappa_R \ln 3$), while the quark deviations — never on ZC16’s ladder — are new and large (FIND-ZC106-02). The first-rung effect splits (THM-ZC106-01).

ZC106’s contribution: it identifies the lepton first-rung effect as a decade-old known gap and isolates the quark first-rung as the genuinely new object, turning one blurred gap into two sharp ones with different histories.

0.125 Closing Sentence

Sometimes the way forward is a look back. We had a single stubborn residual at the first rung, resisting every new idea, and a glance at the oldest mass paper in the corpus showed why the leptons, at least, would not yield: their first-rung gap is the same 10 to 17% that has stood open since mass was first derived as the price of constraint — only wearing a larger number because the running convention had moved its baseline. Strip the convention away and the lepton gap is small and old and named. The quarks are another matter: never placed on that first ladder, they deviate by far more, and in opposite directions, a new thing the simulator found. So the one gap was quietly two. The leptons keep a debt a decade old; the quarks owe something we have only just begun to count.

lepton: + 18% (GAP-MC-01, old); up −59%, down +1115% (new).

บางครั้งทางข้างหน้าคือการเหลียวกลับ เรามีส่วนต่างติดด้านหนึ่งเดียวที่ขึ้นแรก
ด้านทุกความคิดใหม่ และการขายตามองเปเปอร์มวลที่เก่าแก่ที่สุดในคลัง
ก็เผยว่าทำไมอย่างน้อยเลปตอนจึงไม่ยอมอ่อนข้อ --- ช่องว่างชั้นแรกของมันคือ 10 ถึง 17%
เดียวกัน ที่เปิดค้างมาตั้งแต่มวลถูกอนุมานครั้งแรกในฐานะราคาของข้อจำกัด ---
เพียงแต่สวมตัวเลขที่ใหญ่กว่า เพราะ convention ของการไหลย้าย baseline ของมันไป ถอด
convention ออก ช่องว่างเลปตอนก็เล็ก เก่า และมีชื่อ ส่วนควาร์กเป็นอีกเรื่อง ---
ไม่เคยถูกวางบนบันไดแรกนั้น มันเปี่ยมเบนมากกว่ามาก และในทิศตรงข้าม
สิ่งใหม่ที่เครื่องจำลองค้นพบ ดังนั้นช่องว่างหนึ่งเดียว แท้จริงเป็นสอง
เลปตอนแบกหนี้เก่าแก่นับทศวรรษ ส่วนควาร์กเป็นหนี้สิ่งที่เราเพิ่งเริ่มนับ.

Part LXI

Cross-Part Connection Map v17.17 (Part M)

How Part M sits in the edition

Part M is a single continuous arc, not a cluster of separate threads: it takes the mass sector — the one place where CRF had an exact parameter-free *ratio* (EQ-MC-03, Part C) but a standing *residual* against data (GAP-MC-01) — and carries it from “framed problem” (ZC94) to “located, bounded, named residual” (ZC106). The structural move that makes this possible is borrowed wholesale from the electroweak programme of Part I: the same *running bridge* that turned exact R_0 -layer constants into data-matching values (GAP-ZC64-01 $\rightarrow$ COR-ZC70-01) is reused here as a *mass* running (ZC95), and the two-constant boundary it reaches (ZC99) is then traced to an operation-group law (ZC101–102). The mind-layer quantum-pair structure of Part L (DEF-ZC90-01, THM-ZC91-01) is invoked once, as a control: ZC105 proves the mass sector does *not* instantiate that law, bounding it to the mind and force layers.

0.126 Connection Record: Backward Inheritance

Pillars Part M inherits from Parts A–L

From Part C (mass-sector core, ZC14–19):

- **EQ-MC-03** — the parameter-free generation-mass ratio $m_H \propto (15/|H|)^5$. The object every Part M paper runs on; the running bridge corrects it, never replaces it.
- **GAP-MC-01** — the +17%/–10% lepton residual. The frontier Part M opens against (ZC94) and the destination of half the split (ZC106).
- **THM-MC-01, COR-MC-02/03** — the constraint-cost form of the spectrum; carried as the form that survives once PRED-ZC15-01 is read as falsified.

From Part D (ZC20–30): THM-ZC27-01 and FIND-ZC30-02 — compression/grading structure reused in the tower center law (ZC97).

From Part H (ZC50–63): THM-ZC57-01 — carried into the spectral closure argument (ZC98).

From Part I (ZC64–75): the running bridge itself — DEF-ZC64-01/02, GAP-ZC64-01 $\rightarrow$ COR-ZC70-01. Part M’s central method is the mass-sector image of this Part-I result.

From Part L (ZC88–93): DEF-ZC90-01, THM-ZC90-01, THM-ZC91-01, COR-ZC91-02 — the mind-layer quantum-pair law, inherited *as a control* and shown not to extend to mass (ZC105).

0.127 Connection Record: Forward Corrections

Interpretations in Parts A–L that Part M refines

- **PRED-ZC15-01 (the logarithmic mass map)** — read here as *falsified* (ZC94, FIND-ZC94-02): the spectrum is not a closed-form logarithmic map but a constraint-cost form correct up to a scale-running. The falsification is a signpost, not a loss — it is exactly what selects the running-bridge route.
- **GAP-MC-01 (Part C residual)** — not closed, but *re-seen and decomposed* (ZC106, THM-ZC106-01): its lepton piece is the old ZC16 +18% effect (= GAP-MC-01 proper); its quark deviations are genuinely new and were never in ZC16. The single Part-C gap is now two gaps with different ancestry.
- **EQ-MC-03** — sharpened, not corrected: the ratio is exact at the R_0 layer; Part M supplies the missing carry to $R_{\max}$ (the mass running), reducing the data match to two constants κ_R, L_0 (ZC99).

0.128 Connection Record: Phase Misalignments (scars carried)

Phase-Misalignment scars recorded in Part M

Per CRF discipline these are preserved as instructive failure modes, not erased. Each names a *pattern type* so the same trap is recognised next time.

- **PM-ZC102-01** (pattern: *tautology test*) — the “ $\varphi(15) = 2^{d_s}$ ” probe tested whether an identity true by construction carried independent content; it did not.
- **PM-ZC103-01** (pattern: *folded-constant artefact*) — folding L_0 into a per-rung κ manufactures a spurious per-rung measurement; the constant must be carried explicitly.
- **PM-ZC104-01** (pattern: *free-parameter near-miss*) — a freely-fitted R landing near a structural integer is an inverse-fit artefact, not a derivation (the PWA-14 family).
- **PM-ZC104-02** (pattern: *algebraic-identity trap*) — a difference exact to machine precision because it is an identity, not a physical coincidence.
- **PM-ZC105-01** (pattern: *simple-fraction landing*) — a ratio landing on $\sim -27/11$ read as structural before a second independent confirmation (E4 discipline).
- **PM-ZC106-01** (pattern: *convention mismatch*) — the running convention shifts the apparent first-rung residual; convention must be pinned before the residual is attributed.

0.129 Open Gaps at the Close of Part M (real-time, ZC106)

What remains open when the arc stops at ZC106

- **GAP-ZC101-02** — the 0.23% residual on the $\kappa_R = \ln(15)/(15\sqrt{d_s})$ closed form. CONJ-ZC100-01 is promoted HYP $\rightarrow$ DER and GAP-ZC100-01 (the N-assignment) is *closed* by ZC101; this sub-percent residual is what is left.
- **GAP-ZC105-01 (split)** — the first-rung residual decomposes into a *lepton* piece (= GAP-MC-01, the inherited ZC16 effect) and a *quark* piece (new, larger). Both open.
- **GAP-ZC95-01 / GAP-ZC98-01** — the quark common-scale slope (clarified, not closed) and the pinning of L_0, κ (restated and sharpened, not closed).

Arc boundary: Part M stops at ZC106 by decision — a ZC107 quark first-rung paper would duplicate the ZC104 exclusion results. The quantum and lifecycle work that follows (ZC107+) is a separate volume.